
4 APPLICATIONS OF THE DERIVATIVE

4.1 Linear Approximation and Applications

Preliminary Questions

1. True or False? The Linear Approximation says that the vertical change in the graph is approximately equal to the vertical change in the tangent line.

SOLUTION This statement is true. The linear approximation does say that the vertical change in the graph is approximately equal to the vertical change in the tangent line.

2. Estimate $g(1.2) - g(1)$ if $g'(1) = 4$.

SOLUTION Using the Linear Approximation,

$$g(1.2) - g(1) \approx g'(1)(1.2 - 1) = 4(0.2) = 0.8$$

3. Estimate $f(2.1)$ if $f(2) = 1$ and $f'(2) = 3$.

SOLUTION Using the Linearization,

$$f(2.1) \approx f(2) + f'(2)(2.1 - 2) = 1 + 3(0.1) = 1.3$$

4. Complete the following sentence: The Linear Approximation shows that up to a small error, the change in output Δf is directly proportional to _____.

SOLUTION The Linear Approximation tells us that up to a small error, the change in output Δf is directly proportional to the change in input Δx when Δx is small.

Exercises

In Exercises 1–6, use Eq. (1) to estimate $\Delta f = f(3.02) - f(3)$.

1. $f(x) = x^2$

SOLUTION Let $f(x) = x^2$. Then $f'(x) = 2x$ and $\Delta f \approx f'(3)\Delta x = 6(0.02) = 0.12$.

3. $f(x) = x^{-1}$

SOLUTION Let $f(x) = x^{-1}$. Then $f'(x) = -x^{-2}$ and $\Delta f \approx f'(3)\Delta x = -\frac{1}{9}(0.02) = -0.00222$.

5. $f(x) = \sqrt{x+6}$

SOLUTION Let $f(x) = e^{2x}$. Then $f'(x) = 2e^{2x}$ and

$$\Delta f \approx f'(3)\Delta x = 2e^6(0.02) = 0.04e^6 = 16.137$$

7. The cube root of 27 is 3. How much larger is the cube root of 27.2? Estimate using the Linear Approximation.

SOLUTION Let $f(x) = x^{1/3}$, $a = 27$, and $\Delta x = 0.2$. Then $f'(x) = \frac{1}{3}x^{-2/3}$ and $f'(a) = f'(27) = \frac{1}{27}$. The Linear Approximation is

$$\Delta f \approx f'(a)\Delta x = \frac{1}{27}(0.2) = 0.0074074$$

Thus, $\sqrt[3]{27.2}$ is larger than $\sqrt[3]{27}$ by approximately 0.0074074.

In Exercises 9–12, use Eq. (1) to estimate Δf . Use a calculator to compute both the error and the percentage error.

9. $f(x) = \sqrt{1+x}$, $a = 3$, $\Delta x = 0.2$

SOLUTION Let $f(x) = (1+x)^{1/2}$, $a = 3$, and $\Delta x = 0.2$. Then $f'(x) = \frac{1}{2}(1+x)^{-1/2}$, $f'(a) = f'(3) = \frac{1}{4}$ and

$$\Delta f \approx f'(a)\Delta x = \frac{1}{4}(0.2) = 0.05$$

The actual change is

$$\Delta f = f(a + \Delta x) - f(a) = f(3.2) - f(3) = \sqrt{4.2} - 2 \approx 0.049390$$

The error in the Linear Approximation is therefore $|0.049390 - 0.05| = 0.000610$; in percentage terms, the error is

$$\frac{0.000610}{0.049390} \times 100\% \approx 1.24\%$$

11. $f(x) = \frac{1}{1+x^2}$, $a = 3$, $\Delta x = 0.5$

SOLUTION Let $f(x) = \frac{1}{1+x^2}$, $a = 3$, and $\Delta x = 0.5$. Then $f'(x) = -\frac{2x}{(1+x^2)^2}$, $f'(a) = f'(3) = -0.06$ and

$$\Delta f \approx f'(a)\Delta x = -0.06(0.5) = -0.03$$

The actual change is

$$\Delta f = f(a + \Delta x) - f(a) = f(3.5) - f(3) \approx -0.0245283$$

The error in the Linear Approximation is therefore $|-0.0245283 - (-0.03)| = 0.0054717$; in percentage terms, the error is

$$\left| \frac{0.0054717}{-0.0245283} \right| \times 100\% \approx 22.31\%$$

In Exercises 13–20 using Linear Approximation, estimate Δf for a change in x from $x = a$ to $x = b$. Use the estimate to approximate $f(b)$, and find the error using a calculator.

13. $f(x) = \sqrt{x}$, $a = 25$, $b = 26$

SOLUTION Let $f(x) = \sqrt{x}$, $a = 25$, and $\Delta x = 1$. Then $f'(x) = \frac{1}{2}x^{-1/2}$ and $f'(a) = f'(25) = \frac{1}{10}$.

- The Linear Approximation is $\Delta f \approx f'(a)\Delta x = \frac{1}{10}(1) = 0.1$.
- Using the Linear Approximation, $f(b) \approx f(a) + \Delta f = 5 + 0.1 = 5.1$.
- Using a calculator, the error in this estimate is $|\sqrt{26} - 5.1| \approx |5.0990195 - 5.1| = 0.0009805$.

15. $f(x) = \frac{1}{\sqrt{x}}$, $a = 100$, $b = 101$

SOLUTION Let $f(x) = \frac{1}{\sqrt{x}}$, $a = 100$, and $\Delta x = 1$. Then $f'(x) = \frac{d}{dx}(x^{-1/2}) = -\frac{1}{2}x^{-3/2}$ and $f'(a) = -\frac{1}{2}(\frac{1}{1000}) = -0.0005$.

- The Linear Approximation is $\Delta f \approx f'(a)\Delta x = -0.0005(1) = -0.0005$.
- Using the Linear Approximation,

$$f(b) \approx f(a) + \Delta f = \frac{1}{10} + (-0.0005) = 0.0995$$

- Using a calculator, the error in this estimate is

$$\left| \frac{1}{\sqrt{101}} - 0.0995 \right| \approx |0.0995037 - 0.0995| = 3.7 \times 10^{-6}$$

17. $f(x) = x^{1/3}$, $a = 8$, $b = 9$

SOLUTION Let $f(x) = x^{1/3}$, $a = 8$, and $\Delta x = 1$. Then $f'(x) = \frac{1}{3}x^{-2/3}$ and $f'(a) = f'(8) = \frac{1}{12}$.

- The Linear Approximation is $\Delta f \approx f'(a)\Delta x = \frac{1}{12}(1) = 0.083333$.
- Using the Linear Approximation, $f(b) \approx f(a) + \Delta f = 2 + 0.083333 = 2.083333$.
- Using a calculator, the error in this estimate is $|\sqrt[3]{9} - 2.083333| \approx |2.080084 - 2.083333| = 0.003249$.

19. $f(x) = e^x$, $a = 0$, $b = -0.1$

SOLUTION Let $f(x) = e^x$, $a = 0$, and $\Delta x = -0.1$. Then $f'(x) = e^x$ and $f'(a) = f'(0) = 1$.

- The Linear Approximation is $\Delta f \approx f'(a)\Delta x = 1(-0.1) = -0.1$.
- Using the Linear Approximation, $f(b) \approx f(a) + \Delta f = 1 + (-0.1) = 0.9$.
- Using a calculator, the error in this estimate is $|e^{-0.1} - 0.9| \approx |0.904837 - 0.9| = 0.004837$.

In Exercises 21–28, find the linearization at $x = a$ and then use it to approximate $f(b)$.

21. $f(x) = x^4$, $a = 1$, $b = 0.96$

SOLUTION Let $f(x) = x^4$. Then $f'(x) = 4x^3$, $f(a) = 1$, and $f'(a) = 4$. The linearization at $a = 1$ is

$$L(x) = f'(a)(x - a) + f(a) = 4(x - 1) + 1 = 4x - 3$$

and

$$f(b) = f(0.96) \approx L(0.96) = 4(0.96) - 3 = 0.84$$

23. $f(x) = \sin^2 x$, $a = \frac{\pi}{4}$, $b = \frac{1.1\pi}{4}$

SOLUTION Let $f(x) = \sin^2 x$. Then $f'(x) = 2 \sin x \cos x = \sin 2x$, $f(a) = \frac{1}{2}$, and $f'(a) = 1$. The linearization at $a = \frac{\pi}{4}$ is

$$L(x) = f'(a)(x - a) + f(a) = 1\left(x - \frac{\pi}{4}\right) + \frac{1}{2} = x - \frac{\pi}{4} + \frac{1}{2}$$

and

$$f(b) = f\left(\frac{1.1\pi}{4}\right) \approx L\left(\frac{1.1\pi}{4}\right) = \frac{1.1\pi}{4} - \frac{\pi}{4} + \frac{1}{2} = \frac{0.1\pi}{4} + \frac{1}{2} \approx 0.578540$$

25. $f(x) = (1 + x)^{-1/2}$, $a = 0$, $b = 0.08$

SOLUTION Let $f(x) = (1 + x)^{-1/2}$. Then $f'(x) = -\frac{1}{2}(1 + x)^{-3/2}$, $f(a) = 1$, and $f'(a) = -\frac{1}{2}$. The linearization at $a = 0$ is

$$L(x) = f'(a)(x - a) + f(a) = -\frac{1}{2}x + 1$$

and

$$f(b) = f(0.08) \approx L(0.08) = -\frac{1}{2}(0.08) + 1 = 0.96$$

27. $f(x) = e^{\sqrt{x}}$, $a = 1$, $b = 0.85$

SOLUTION Let $f(x) = e^{\sqrt{x}}$. Then

$$f'(x) = \frac{1}{2\sqrt{x}}e^{\sqrt{x}}, \quad f(a) = e, \quad \text{and} \quad f'(a) = \frac{1}{2}e$$

so the linearization of $f(x)$ at $a = 1$ is

$$L(x) = f'(a)(x - a) + f(a) = \frac{1}{2}e(x - 1) + e = \frac{1}{2}e(x + 1)$$

and

$$f(b) = f(0.85) \approx L(0.85) = \frac{1}{2}e(0.85 + 1) = 0.925e \approx 2.514411$$

In Exercises 29–32, estimate Δy using differentials [Eq. (3)].

29. $y = \cos x$, $a = \frac{\pi}{6}$, $dx = 0.014$

SOLUTION Let $f(x) = \cos x$. Then $f'(x) = -\sin x$ and

$$\Delta y \approx dy = f'(a)dx = -\sin\left(\frac{\pi}{6}\right)(0.014) = -0.007$$

31. $y = \frac{10 - x^2}{2 + x^2}$, $a = 1$, $dx = 0.01$

SOLUTION Let $f(x) = \frac{10 - x^2}{2 + x^2}$. Then

$$f'(x) = \frac{(2 + x^2)(-2x) - (10 - x^2)(2x)}{(2 + x^2)^2} = -\frac{24x}{(2 + x^2)^2}$$

and

$$\Delta y \approx dy = f'(a)dx = -\frac{24}{9}(0.01) = -0.026667$$

33. Estimate $f(4.03)$ for $f(x)$ as in Figure 9.

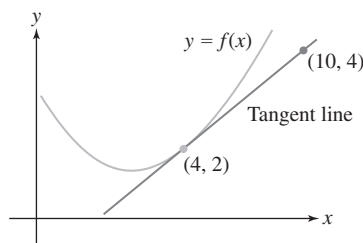


FIGURE 9

SOLUTION Using the Linear Approximation, $f(4.03) \approx f(4) + f'(4)(0.03)$. From the figure, we find that $f(4) = 2$ and

$$f'(4) = \frac{4 - 2}{10 - 4} = \frac{1}{3}$$

Thus,

$$f(4.03) \approx 2 + \frac{1}{3}(0.03) = 2.01$$

35. Which is larger: $\sqrt{2.1} - \sqrt{2}$ or $\sqrt{9.1} - \sqrt{9}$? Explain using the Linear Approximation.

SOLUTION Let $f(x) = \sqrt{x}$, and $\Delta x = 0.1$. Then $f'(x) = \frac{1}{2}x^{-1/2}$ and the Linear Approximation at $x = a$ gives

$$\Delta f = \sqrt{a + 0.1} - \sqrt{a} \approx f'(a)(0.1) = \frac{1}{2}a^{-1/2}(0.1) = \frac{0.05}{\sqrt{a}}$$

We see that Δf decreases as a increases. In particular

$$\sqrt{2.1} - \sqrt{2} \approx \frac{0.05}{\sqrt{2}} \quad \text{is larger than} \quad \sqrt{9.1} - \sqrt{9} \approx \frac{0.05}{3}$$

37. Box office revenue at a multiplex cinema in Paris is $R(p) = 3600p - 10p^3$ euros per showing when the ticket price is p euros. Calculate $R(p)$ for $p = 9$ and use the Linear Approximation to estimate ΔR if p is raised or lowered by 0.5 euro.

SOLUTION Let $R(p) = 3600p - 10p^3$. Then $R(9) = 3600(9) - 10(9)^3 = 25110$ euros. Moreover, $R'(p) = 3600 - 30p^2$, so by the Linear Approximation,

$$\Delta R \approx R'(9)\Delta p = 1170\Delta p$$

If p is raised by 0.5 euros, then $\Delta R \approx 585$ euros; on the other hand, if p is lowered by 0.5 euros, then $\Delta R \approx -585$ euros.

39. A thin silver wire has length $L = 18$ cm when the temperature is $T = 30^\circ\text{C}$. Estimate ΔL when T decreases to 25°C if the coefficient of thermal expansion is $k = 1.9 \times 10^{-5}^\circ\text{C}^{-1}$ (see Example 3).

SOLUTION We have

$$\frac{dL}{dT} = kL = (1.9 \times 10^{-5})(18) = 3.42 \times 10^{-4} \text{ cm}/^\circ\text{C}$$

The change in temperature is $\Delta T = -5^\circ\text{C}$, so by the Linear Approximation, the change in length is approximately

$$\Delta L \approx 3.42 \times 10^{-4} \Delta T = (3.42 \times 10^{-4})(-5) = -0.00171 \text{ cm}$$

At $T = 25^\circ\text{C}$, the length of the wire is approximately 17.99829 cm.

41. The atmospheric pressure at altitude h (kilometers) for $11 \leq h \leq 25$ is approximately

$$P(h) = 128e^{-0.157h} \text{ kilopascals}$$

- (a) Estimate ΔP at $h = 20$ when $\Delta h = 0.5$.
 (b) Compute the actual change, and compute the percentage error in the Linear Approximation.

SOLUTION

- (a) Let $P(h) = 128e^{-0.157h}$. Then $P'(h) = -20.096e^{-0.157h}$. Using the Linear Approximation,

$$\Delta P \approx P'(h)\Delta h = P'(20)(0.5) = -0.434906 \text{ kilopascals}$$

- (b) The actual change in pressure is

$$P(20.5) - P(20) = -0.418274 \text{ kilopascals}$$

The percentage error in the Linear Approximation is

$$\frac{-0.434906 - (-0.418274)}{-0.418274} \times 100\% \approx 3.98\%$$

- 43.** Newton's Law of Gravitation shows that if a person weighs w pounds on the surface of the earth, then his or her weight at distance x from the center of the earth is

$$W(x) = \frac{wR^2}{x^2} \quad (\text{for } x \geq R)$$

where $R = 3960$ miles is the radius of the earth (Figure 10).

- (a) Show that the weight lost at altitude h miles above the earth's surface is approximately $\Delta W \approx -(0.0005w)h$. *Hint:* Use the Linear Approximation with $dx = h$.
 (b) Estimate the weight lost by a 200-lb football player flying in a jet at an altitude of 7 miles.

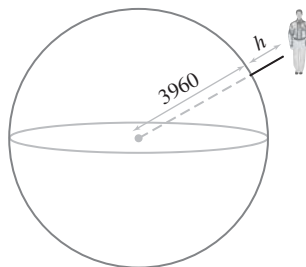


FIGURE 10 The distance to the center of the earth is $3960 + h$ miles.

SOLUTION

- (a) Using the Linear Approximation

$$\Delta W \approx W'(R)\Delta x = -\frac{2wR^2}{R^3}h = -\frac{2wh}{R} \approx -0.0005wh$$

- (b) Substitute $w = 200$ and $h = 7$ into the result from part (a) to obtain

$$\Delta W \approx -0.0005(200)(7) = -0.7 \text{ pounds}$$

- 45.** A stone tossed vertically into the air with initial velocity v cm/s reaches a maximum height of $h = v^2/1960$ cm.

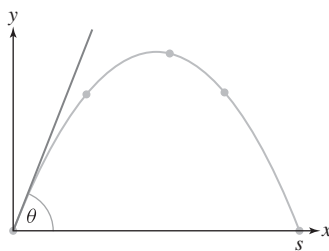
- (a) Estimate Δh if $v = 700$ cm/s and $\Delta v = 1$ cm/s.
 (b) Estimate Δh if $v = 1000$ cm/s and $\Delta v = 1$ cm/s.
 (c) In general, does a 1-cm/s increase in v lead to a greater change in h at low or high initial velocities? Explain.

SOLUTION A stone tossed vertically with initial velocity v cm/s attains a maximum height of $h(v) = v^2/1960$ cm. Thus, $h'(v) = v/980$.

- (a) If $v = 700$ and $\Delta v = 1$, then $\Delta h \approx h'(v)\Delta v = \frac{1}{980}(700)(1) \approx 0.71$ cm.
 (b) If $v = 1000$ and $\Delta v = 1$, then $\Delta h \approx h'(v)\Delta v = \frac{1}{980}(1000)(1) = 1.02$ cm.
 (c) A 1 centimeter per second increase in initial velocity v increases the maximum height by approximately $v/980$ cm. Accordingly, there is a bigger effect at higher velocities.

In Exercises 47 and 48, use the following fact derived from Newton's Laws: An object released at an angle θ with initial velocity v ft/s travels a horizontal distance

$$s = \frac{1}{32}v^2 \sin 2\theta \text{ ft (Figure 11)}$$

FIGURE 11 Trajectory of an object released at an angle θ .

47. A player located 18.1 ft from the basket launches a successful jump shot from a height of 10 ft (level with the rim of the basket), at an angle $\theta = 34^\circ$ and initial velocity $v = 25$ ft/s.

- (a) Show that $\Delta s \approx 0.255\Delta\theta$ ft for a small change of $\Delta\theta$.
 (b) Is it likely that the shot would have been successful if the angle had been off by 2° ?
 (c) Estimate Δs if $\theta = 34^\circ$, $v = 25$ ft/s, and $\Delta v = 2$.

SOLUTION Using Newton's laws and the given initial velocity of $v = 25$ ft/s, the shot travels $s = \frac{1}{32}v^2 \sin 2t = \frac{625}{32} \sin 2t$ ft, where t is in radians.

- (a) If $\theta = 34^\circ$ (i.e., $t = \frac{17}{90}\pi$), then

$$\Delta s \approx s'(t)\Delta t = \frac{625}{16} \cos\left(\frac{17}{45}\pi\right)\Delta t = \frac{625}{16} \cos\left(\frac{17}{45}\pi\right)\Delta\theta \cdot \frac{\pi}{180} \approx 0.255\Delta\theta$$

- (b) If $\Delta\theta = 2^\circ$, this gives $\Delta s \approx 0.51$ ft, in which case the shot would not have been successful, having been off half a foot.

- (c) Using Newton's laws and the fixed angle of $\theta = 34^\circ = \frac{17}{90}\pi$, the shot travels

$$s = \frac{1}{32}v^2 \sin \frac{17}{45}\pi$$

With $v = 25$ ft/s and $\Delta v = 2$ ft/s, we find

$$\Delta s \approx s'(v)\Delta v = \frac{1}{16}(25) \sin \frac{17\pi}{45} \cdot 2 = 2.897 \text{ ft}$$

49. The radius of a spherical ball is measured at $r = 25$ cm. Estimate the maximum error in the volume and surface area if r is accurate to within 0.5 cm.

SOLUTION The volume and surface area of the sphere are given by $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$, respectively. If $r = 25$ and $\Delta r = \pm 0.5$, then

$$\Delta V \approx V'(25)\Delta r = 4\pi(25)^2(0.5) \approx 3927 \text{ cm}^3$$

and

$$\Delta S \approx S'(25)\Delta r = 8\pi(25)(0.5) \approx 314.2 \text{ cm}^2$$

51. The volume (in liters) and pressure P (in atmospheres) of a certain gas satisfy $PV = 24$. A measurement yields $V = 4$ with a possible error of ± 0.3 L. Compute P and estimate the maximum error in this computation.

SOLUTION Given $PV = 24$ and $V = 4$, it follows that $P = 6$ atmospheres. Solving $PV = 24$ for P yields $P = 24V^{-1}$. Thus, $P' = -24V^{-2}$ and

$$\Delta P \approx P'(4)\Delta V = -24(4)^{-2}(\pm 0.3) = \pm 0.45 \text{ atmospheres}$$

53. Approximate $f(2)$ if the linearization of $f(x)$ at $a = 2$ is $L(x) = 2x + 4$?

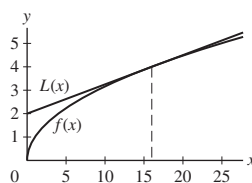
SOLUTION $f(2) \approx L(2) = 2(2) + 4 = 8$.

55. Estimate $\sqrt{16.2}$ using the linearization $L(x)$ of $f(x) = \sqrt{x}$ at $a = 16$. Plot f and L on the same set of axes and determine whether the estimate is too large or too small.

SOLUTION Let $f(x) = x^{1/2}$, $a = 16$, and $\Delta x = 0.2$. Then $f'(x) = \frac{1}{2}x^{-1/2}$ and $f'(a) = f'(16) = \frac{1}{8}$. The linearization to $f(x)$ is

$$L(x) = f'(a)(x - a) + f(a) = \frac{1}{8}(x - 16) + 4 = \frac{1}{8}x + 2$$

Thus, we have $\sqrt{16.2} \approx L(16.2) = 4.025$. Graphs of $f(x)$ and $L(x)$ are shown below. Because the graph of $L(x)$ lies above the graph of $f(x)$, we expect that the estimate from the Linear Approximation is too large.



In Exercises 57–65, approximate using linearization and use a calculator to compute the percentage error.

57. $\frac{1}{\sqrt{17}}$

SOLUTION Let $f(x) = x^{-1/2}$, $a = 16$, and $\Delta x = 1$. Then $f'(x) = -\frac{1}{2}x^{-3/2}$, $f'(a) = f'(16) = -\frac{1}{128}$ and the linearization to $f(x)$ is

$$L(x) = f'(a)(x - a) + f(a) = -\frac{1}{128}(x - 16) + \frac{1}{4} = -\frac{1}{128}x + \frac{3}{8}$$

Thus, we have $\frac{1}{\sqrt{17}} \approx L(17) \approx 0.24219$. The percentage error in this estimate is

$$\left| \frac{\frac{1}{\sqrt{17}} - 0.24219}{\frac{1}{\sqrt{17}}} \right| \times 100\% \approx 0.14\%$$

59. $\frac{1}{(10.03)^2}$

SOLUTION Let $f(x) = x^{-2}$, $a = 10$ and $\Delta x = 0.03$. Then $f'(x) = -2x^{-3}$, $f'(a) = f'(10) = -0.002$ and the linearization to $f(x)$ is

$$L(x) = f'(a)(x - a) + f(a) = -0.002(x - 10) + 0.01 = -0.002x + 0.03$$

Thus, we have

$$\frac{1}{(10.03)^2} \approx L(10.03) = -0.002(10.03) + 0.03 = 0.00994$$

The percentage error in this estimate is

$$\left| \frac{\frac{1}{(10.03)^2} - 0.00994}{\frac{1}{(10.03)^2}} \right| \times 100\% \approx 0.0027\%$$

61. $(64.1)^{1/3}$

SOLUTION Let $f(x) = x^{1/3}$, $a = 64$, and $\Delta x = 0.1$. Then $f'(x) = \frac{1}{3}x^{-2/3}$, $f'(a) = f'(64) = \frac{1}{48}$ and the linearization to $f(x)$ is

$$L(x) = f'(a)(x - a) + f(a) = \frac{1}{48}(x - 64) + 4 = \frac{1}{48}x + \frac{8}{3}$$

Thus, we have $(64.1)^{1/3} \approx L(64.1) \approx 4.002083$. The percentage error in this estimate is

$$\left| \frac{(64.1)^{1/3} - 4.002083}{(64.1)^{1/3}} \right| \times 100\% \approx 0.000019\%$$

63. $\cos^{-1}(0.52)$

SOLUTION Let $f(x) = \cos^{-1} x$ and $a = 0.5$. Then

$$f'(x) = -\frac{1}{\sqrt{1-x^2}}, \quad f'(a) = f'(0.5) = -\frac{2\sqrt{3}}{3}$$

and the linearization to $f(x)$ is

$$L(x) = f'(a)(x - a) + f(a) = -\frac{2\sqrt{3}}{3}(x - 0.5) + \frac{\pi}{3}$$

Thus, we have $\cos^{-1}(0.52) \approx L(0.52) = 1.024104$. The percentage error in this estimate is

$$\left| \frac{\cos^{-1}(0.52) - 1.024104}{\cos^{-1}(0.52)} \right| \times 100\% \approx 0.015\%$$

65. $e^{-0.012}$

SOLUTION Let $f(x) = e^x$ and $a = 0$. Then $f'(x) = e^x$, $f'(a) = f'(0) = 1$ and the linearization to $f(x)$ is

$$L(x) = f'(a)(x - a) + f(a) = 1(x - 0) + 1 = x + 1$$

Thus, we have $e^{-0.012} \approx L(-0.012) = 1 - 0.012 = 0.988$. The percentage error in this estimate is

$$\left| \frac{e^{-0.012} - 0.988}{e^{-0.012}} \right| \times 100\% \approx 0.0073\%$$

67. Show that the Linear Approximation to $f(x) = \sqrt{x}$ at $x = 9$ yields the estimate $\sqrt{9+h} - 3 \approx \frac{1}{6}h$. Set $K = 0.01$ and show that $|f''(x)| \leq K$ for $x \geq 9$. Then verify numerically that the error E satisfies Eq. (5) for $h = 10^{-n}$, for $1 \leq n \leq 4$.

SOLUTION Let $f(x) = \sqrt{x}$. Then $f(9) = 3$, $f'(x) = \frac{1}{2}x^{-1/2}$ and $f'(9) = \frac{1}{6}$. Therefore, by the Linear Approximation,

$$f(9+h) - f(9) = \sqrt{9+h} - 3 \approx \frac{1}{6}h$$

Moreover, $f''(x) = -\frac{1}{4}x^{-3/2}$, so $|f''(x)| = \frac{1}{4}x^{-3/2}$. Because this is a decreasing function, it follows that for $x \geq 9$,

$$K = \max |f''(x)| \leq |f''(9)| = \frac{1}{108} < 0.01$$

From the following table, we see that for $h = 10^{-n}$, $1 \leq n \leq 4$, $E \leq \frac{1}{2}Kh^2$.

h	$E = \sqrt{9+h} - 3 - \frac{1}{6}h $	$\frac{1}{2}Kh^2$
10^{-1}	4.604×10^{-5}	5.00×10^{-5}
10^{-2}	4.627×10^{-7}	5.00×10^{-7}
10^{-3}	4.629×10^{-9}	5.00×10^{-9}
10^{-4}	4.627×10^{-11}	5.00×10^{-11}

Further Insights and Challenges

69. Compute dy/dx at the point $P = (2, 1)$ on the curve $y^3 + 3xy = 7$ and show that the linearization at P is $L(x) = -\frac{1}{3}x + \frac{5}{3}$. Use $L(x)$ to estimate the y -coordinate of the point on the curve where $x = 2.1$.

SOLUTION Differentiating both sides of the equation $y^3 + 3xy = 7$ with respect to x yields

$$3y^2 \frac{dy}{dx} + 3x \frac{dy}{dx} + 3y = 0$$

so

$$\frac{dy}{dx} = -\frac{y}{y^2 + x}$$

Thus,

$$\left. \frac{dy}{dx} \right|_{(2,1)} = -\frac{1}{1^2 + 2} = -\frac{1}{3}$$

and the linearization at $P = (2, 1)$ is

$$L(x) = 1 - \frac{1}{3}(x - 2) = -\frac{1}{3}x + \frac{5}{3}$$

Finally, when $x = 2.1$, we estimate that the y -coordinate of the point on the curve is

$$y \approx L(2.1) = -\frac{1}{3}(2.1) + \frac{5}{3} = 0.967$$

71. Apply the method of Exercise 69 to $P = (-1, 2)$ on $y^4 + 7xy = 2$ to estimate the solution of $y^4 - 7.7y = 2$ near $y = 2$.

SOLUTION Differentiating both sides of the equation $y^4 + 7xy = 2$ with respect to x yields

$$4y^3 \frac{dy}{dx} + 7x \frac{dy}{dx} + 7y = 0$$

so

$$\frac{dy}{dx} = -\frac{7y}{4y^3 + 7x}$$

Thus,

$$\left. \frac{dy}{dx} \right|_{(-1,2)} = -\frac{7(2)}{4(2)^3 + 7(-1)} = -\frac{14}{25}$$

and the linearization at $P = (-1, 2)$ is

$$L(x) = 2 - \frac{14}{25}(x + 1) = -\frac{14}{25}x + \frac{36}{25}$$

Finally, the equation $y^4 - 7.7y = 2$ corresponds to $x = -1.1$, so we estimate the solution of this equation near $y = 2$ is

$$y \approx L(-1.1) = -\frac{14}{25}(-1.1) + \frac{36}{25} = 2.056$$

73. Let $\Delta f = f(5 + h) - f(5)$, where $f(x) = x^2$. Verify directly that $E = |\Delta f - f'(5)h|$ satisfies (5) with $K = 2$.

SOLUTION Let $f(x) = x^2$. Then

$$\Delta f = f(5 + h) - f(5) = (5 + h)^2 - 5^2 = h^2 + 10h$$

and

$$E = |\Delta f - f'(5)h| = |h^2 + 10h - 10h| = h^2 \leq 2h^2$$

4.2 Extreme Values

Preliminary Questions

1. What is the definition of a critical point?

SOLUTION A critical point is a value of the independent variable x in the domain of a function f at which either $f'(x) = 0$ or $f'(x)$ does not exist.

In Questions 2 and 3, which is the correct conclusion, (a) or (b)?

2. If f is not continuous on $[0, 1]$, then

- (a) f has no extreme values on $[0, 1]$.
- (b) f might not have any extreme values on $[0, 1]$.

SOLUTION The correct response is **(b)**: f might not have any extreme values on $[0, 1]$. Although $[0, 1]$ is closed, because f is not continuous, the function is not guaranteed to have any extreme values on $[0, 1]$.

3. If f is continuous but has no critical points in $[0, 1]$, then

- (a) f has no min or max on $[0, 1]$.
- (b) Either $f(0)$ or $f(1)$ is the minimum value on $[0, 1]$.

SOLUTION The correct response is **(b)**: either $f(0)$ or $f(1)$ is the minimum value on $[0, 1]$. Remember that extreme values occur either at critical points or endpoints. If a continuous function on a closed interval has no critical points, the extreme values must occur at the endpoints.

4. For each statement, indicate whether it is true or false. If false, correct the statement or explain why it is false.

- (a) If $f'(c) = 0$, then $f(c)$ is either a local minimum or a local maximum.
- (b) If $f(c)$ is the absolute maximum of f on an interval I , then $f'(c) = 0$.
- (c) If f is differentiable and $f(c)$ is a local minimum of f , then $f'(c) = 0$.
- (d) If there is one local minimum of f on an interval I , then it is the absolute minimum on I .

SOLUTION

(a) This statement is false. The condition $f'(c) = 0$ is not sufficient to guarantee that $f(c)$ is either a local minimum or a local maximum. For example, consider the function $f(x) = x^3$. Then $f'(0) = 0$, but $f(0)$ is neither a local minimum nor a local maximum.

(b) This statement is false. If $f(c)$ is the absolute maximum of f on an interval I , it could be that $f'(c)$ does not exist or that $f'(c)$ is any real number if c happens to be an endpoint of the interval I . For example, consider the function $f(x) = x^2$ on the interval $[0, 2]$. The absolute maximum value of f on $[0, 2]$ is $f(2) = 4$, but $f'(2) = 4 \neq 0$.

(c) This statement is true.

(d) This statement is false. If there is one local minimum of f on an interval I , then the absolute minimum might occur at an endpoint of the interval. For example, consider the function $f(x) = x^3 - x^2$ on the interval $[-1, 1]$. The function f has only one local minimum on this interval, at $x = 2/3$, but the absolute minimum occurs at the endpoint $x = -1$.

Exercises

- The following refer to Figure 18.
 - What are the critical points of f on $[0, 8]$?
 - What are the maximum and minimum values of f on $[0, 8]$?
 - What are the local maximum and minimum values of f , and where do they occur?
 - Find a closed interval on which both the minimum and maximum values of f occur at critical points.
 - Find an open interval on which f has neither a minimum nor a maximum value.
 - Find an open interval on which f has a maximum value but no minimum value.

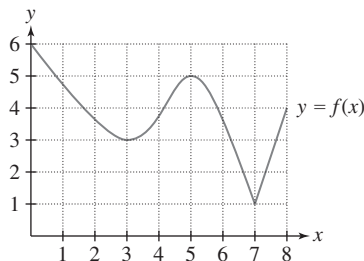


FIGURE 18

SOLUTION

- The function f has three critical points on the interval $[0, 8]$: at $x = 3$, $x = 5$, and $x = 7$. Two of these, $x = 3$ and $x = 5$, are where the derivative is zero and one; $x = 7$ is where the derivative does not exist.
- The maximum value of f on $[0, 8]$ is 6; the function takes this value at $x = 0$. The minimum value is 1; the function takes this value at $x = 7$.
- The function f achieves a local maximum of 5 at $x = 5$. On the other hand, the function f achieves a local minimum value of 3 at $x = 3$ and a local minimum value of 1 at $x = 7$.
- Answers may vary. One example is the interval $[4, 8]$. Another is $[2, 6]$.
- Answers may vary. One example is $(0, 2)$.
- Answers may vary. One example is $(4, 6)$.

In Exercises 3–20, find all critical points of the function.

3. $f(x) = x^2 - 2x + 4$

SOLUTION Let $f(x) = x^2 - 2x + 4$. Then $f'(x) = 2x - 2 = 0$ implies that $x = 1$ is the lone critical point of f .

5. $f(x) = x^3 - \frac{9}{2}x^2 - 54x + 2$

SOLUTION Let $f(x) = x^3 - \frac{9}{2}x^2 - 54x + 2$. Then $f'(x) = 3x^2 - 9x - 54 = 3(x + 3)(x - 6) = 0$ implies that $x = -3$ and $x = 6$ are the critical points of f .

7. $f(x) = x^{-1} - x^{-2}$

SOLUTION Let $f(x) = x^{-1} - x^{-2}$. Then

$$f'(x) = -x^{-2} + 2x^{-3} = \frac{2 - x}{x^3} = 0$$

implies that $x = 2$ is the only critical point of f . Though f' does not exist at $x = 0$, this is not a critical point of f because $x = 0$ is not in the domain of f .

9. $f(x) = \frac{x}{x^2 + 1}$

SOLUTION Let $f(x) = \frac{x}{x^2 + 1}$. Then $f'(x) = \frac{1 - x^2}{(x^2 + 1)^2} = 0$ implies that $x = \pm 1$ are the critical points of f .

11. $f(t) = t - 4\sqrt{t+1}$

SOLUTION Let $f(t) = t - 4\sqrt{t+1}$. Then

$$f'(t) = 1 - \frac{2}{\sqrt{t+1}} = 0$$

implies that $t = 3$ is a critical point of f . Because f' does not exist at $t = -1$, this is another critical point of f .

13. $f(x) = xe^{2x}$

SOLUTION Let $f(x) = xe^{2x}$. Then $f'(x) = (2x + 1)e^{2x} = 0$ implies that $x = -\frac{1}{2}$ is the only critical point of f .

15. $g(\theta) = \sin^2 \theta$

SOLUTION Let $g(\theta) = \sin^2 \theta$. Then $g'(\theta) = 2 \sin \theta \cos \theta = \sin 2\theta = 0$ implies that

$$\theta = \frac{n\pi}{2}$$

is a critical value of g for all integer values of n .

17. $f(x) = x \ln x$

SOLUTION Let $f(x) = x \ln x$. Then $f'(x) = 1 + \ln x = 0$ implies that $x = e^{-1} = \frac{1}{e}$ is the only critical point of f .

19. $f(x) = \sin^{-1} x - 2x$

SOLUTION Let $f(x) = \sin^{-1} x - 2x$. Then

$$f'(x) = \frac{1}{\sqrt{1-x^2}} - 2 = 0$$

implies that $x = \pm \frac{\sqrt{3}}{2}$ are the critical points of f .

21. Let $f(x) = 2x^2 - 8x + 7$.

(a) Find the critical point c of f and compute $f(c)$.

(b) Find the extreme values of f on $[0, 5]$.

(c) Find the extreme values of f on $[-4, 1]$.

SOLUTION

(a) Let $f(x) = 2x^2 - 8x + 7$. Then $f'(x) = 4x - 8 = 0$ implies that $c = 2$ is the critical point of f . At the critical point, $f(2) = -1$.

(b) First note that the critical point $c = 2$ lies in the interval $[0, 5]$. Comparing the value of f at the endpoints of the interval $[0, 5]$, $f(0) = 7$ and $f(5) = 17$, with the value of f at the critical point, $f(2) = -1$, we see that the maximum value of f on $[0, 5]$ is 17 and the minimum value is -1 .

(c) First note that the critical point $c = 2$ does not lie in the interval $[-4, 1]$. Comparing the value of f at the endpoints of the interval $[-4, 1]$, $f(-4) = 71$ and $f(1) = 1$, we see that the maximum value of f on $[-4, 1]$ is 71 and the minimum value is 1.

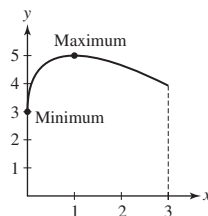
23. Find the critical points of $f(x) = \sin x + \cos x$ and determine the extreme values on $[0, \frac{\pi}{2}]$.

SOLUTION

- Let $f(x) = \sin x + \cos x$. Then on the interval $[0, \frac{\pi}{2}]$, we have $f'(x) = \cos x - \sin x = 0$ at $x = \frac{\pi}{4}$, the only critical point of f in this interval.
- Since $f(\frac{\pi}{4}) = \sqrt{2}$ and $f(0) = f(\frac{\pi}{2}) = 0$, the maximum value of f on $[0, \frac{\pi}{2}]$ is $\sqrt{2}$, while the minimum value is 0.

25. **[GU]** Plot $f(x) = 4\sqrt{x} - 2x + 3$ on $[0, 3]$ and indicate where it appears that the minimum and maximum occur. Then determine the minimum and maximum using calculus.

SOLUTION From the graph of $f(x) = 4\sqrt{x} - 2x + 3$, which is shown below, the minimum value of f on the interval $[0, 3]$ appears to occur at $x = 0$ while the maximum value appears to occur at $x \approx 1$.



To determine the minimum and maximum values using calculus, first note that the domain of f is the set $\{x|x \geq 0\}$. Now,

$$f'(x) = \frac{2}{\sqrt{x}} - 2$$

We see that $f'(x) = 0$ when $x = 1$, and, on the domain of f , $f'(x)$ does not exist when $x = 0$. Thus, $x = 0$ and $x = 1$ are the critical points of f . With $f(0) = 3$, $f(1) = 5$, and $f(3) = 4\sqrt{3} - 3 \approx 3.9282$, it follows that the minimum value of f on the interval $[0, 3]$ is $f(0) = 3$, and the maximum value is $f(1) = 5$.

In Exercises 27–60, find the minimum and maximum values of the function on the given interval by comparing values at the critical points and endpoints.

27. $y = 2x^2 + 4x + 5$, $[-2, 2]$

SOLUTION Let $f(x) = 2x^2 + 4x + 5$. Then $f'(x) = 4x + 4 = 0$ implies that $x = -1$ is the only critical point of f . The minimum of f on the interval $[-2, 2]$ is $f(-1) = 3$, whereas its maximum is $f(2) = 21$. (Note: $f(-2) = 5$.)

29. $y = 6t - t^2$, $[0, 5]$

SOLUTION Let $f(t) = 6t - t^2$. Then $f'(t) = 6 - 2t = 0$ implies that $t = 3$ is the only critical point of f . The minimum of f on the interval $[0, 5]$ is $f(0) = 0$, whereas the maximum is $f(3) = 9$. (Note: $f(5) = 5$.)

31. $y = x^3 - 6x^2 + 8$, $[1, 6]$

SOLUTION Let $f(x) = x^3 - 6x^2 + 8$. Then $f'(x) = 3x^2 - 12x = 3x(x - 4) = 0$ implies that $x = 0$ and $x = 4$ are the critical points of f . The minimum of f on the interval $[1, 6]$ is $f(4) = -24$, whereas the maximum is $f(6) = 8$. (Note: $f(1) = 3$ and the critical point $x = 0$ is not on the interval $[1, 6]$.)

33. $y = x^3 - 6x^2 + 8$, $[1, 3]$

SOLUTION Let $f(x) = x^3 - 6x^2 + 8$. Then $f'(x) = 3x^2 - 12x = 3x(x - 4) = 0$ implies that $x = 0$ and $x = 4$ are the critical points of f . The minimum of f on the interval $[1, 3]$ is $f(3) = -19$, whereas the maximum is $f(1) = 3$. (Note: Neither of the critical points are on the interval $[1, 3]$.)

35. $y = 2t^3 + 3t^2$, $[1, 2]$

SOLUTION Let $f(t) = 2t^3 + 3t^2$. Then $f'(t) = 6t^2 + 6t = 6t(t + 1) = 0$ implies that $t = 0$ and $t = -1$ are the critical points of f . The minimum of f on the interval $[1, 2]$ is $f(1) = 5$, whereas the maximum is $f(2) = 28$. (Note: Neither of the critical points are on the interval $[1, 2]$.)

37. $y = z^5 - 80z$, $[-3, 3]$

SOLUTION Let $f(z) = z^5 - 80z$. Then $f'(z) = 5z^4 - 80 = 5(z^4 - 16) = 5(z^2 + 4)(z + 2)(z - 2) = 0$ implies that $z = \pm 2$ are the critical points of f . The minimum value of f on the interval $[-3, 3]$ is $f(2) = -128$, whereas the maximum is $f(-2) = 128$. (Note: $f(-3) = 3$ and $f(3) = -3$.)

39. $y = \frac{x^2 + 1}{x - 4}$, $[5, 6]$

SOLUTION Let $f(x) = \frac{x^2 + 1}{x - 4}$. Then

$$f'(x) = \frac{(x - 4) \cdot 2x - (x^2 + 1) \cdot 1}{(x - 4)^2} = \frac{x^2 - 8x - 1}{(x - 4)^2} = 0$$

implies $x = 4 \pm \sqrt{17}$ are critical points of f . $x = 4$ is not a critical point because $x = 4$ is not in the domain of f . On the interval $[5, 6]$, the minimum of f is $f(6) = \frac{37}{2} = 18.5$, whereas the maximum of f is $f(5) = 26$. (Note: The critical points $x = 4 \pm \sqrt{17}$ are not on the interval $[5, 6]$.)

41. $y = x - \frac{4x}{x + 1}$, $[0, 3]$

SOLUTION Let $f(x) = x - \frac{4x}{x + 1}$. Then

$$f'(x) = 1 - \frac{4}{(x + 1)^2} = \frac{(x - 1)(x + 3)}{(x + 1)^2} = 0$$

implies that $x = 1$ and $x = -3$ are critical points of f . $x = -1$ is not a critical point because $x = -1$ is not in the domain of f . The minimum of f on the interval $[0, 3]$ is $f(1) = -1$, whereas the maximum is $f(0) = f(3) = 0$. (Note: The critical point $x = -3$ is not on the interval $[0, 3]$.)

43. $y = (2 + x)\sqrt{2 + (2 - x)^2}$, $[0, 2]$

SOLUTION Let $f(x) = (2 + x)\sqrt{2 + (2 - x)^2}$. Then

$$f'(x) = \sqrt{2 + (2 - x)^2} - (2 + x)(2 + (2 - x)^2)^{-1/2}(2 - x) = \frac{2(x - 1)^2}{\sqrt{2 + (2 - x)^2}} = 0$$

implies that $x = 1$ is the critical point of f . On the interval $[0, 2]$, the minimum is $f(0) = 2\sqrt{6} \approx 4.9$ and the maximum is $f(2) = 4\sqrt{2} \approx 5.66$. (Note: $f(1) = 3\sqrt{3} \approx 5.2$.)

45. $y = \sqrt{x + x^2} - 2\sqrt{x}$, $[0, 4]$

SOLUTION Let $f(x) = \sqrt{x + x^2} - 2\sqrt{x}$. Then

$$f'(x) = \frac{1}{2}(x + x^2)^{-1/2}(1 + 2x) - x^{-1/2} = \frac{1 + 2x - 2\sqrt{1 + x}}{2\sqrt{x}\sqrt{1 + x}} = 0$$

implies that $x = 0$ and $x = \frac{\sqrt{3}}{2}$ are the critical points of f . Neither $x = -1$ nor $x = -\frac{\sqrt{3}}{2}$ is a critical point because neither is in the domain of f . On the interval $[0, 4]$, the minimum of f is $f\left(\frac{\sqrt{3}}{2}\right) \approx -0.589980$ and the maximum is $f(4) \approx 0.472136$. (Note: $f(0) = 0$.)

47. $y = \sin x \cos x, \quad [0, \frac{\pi}{2}]$

SOLUTION Let $f(x) = \sin x \cos x = \frac{1}{2} \sin 2x$. On the interval $[0, \frac{\pi}{2}]$, $f'(x) = \cos 2x = 0$ when $x = \frac{\pi}{4}$. The minimum of f on this interval is $f(0) = f(\frac{\pi}{2}) = 0$, whereas the maximum is $f(\frac{\pi}{4}) = \frac{1}{2}$.

49. $y = \sqrt{2}\theta - \sec \theta, \quad [0, \frac{\pi}{3}]$

SOLUTION Let $f(\theta) = \sqrt{2}\theta - \sec \theta$. On the interval $[0, \frac{\pi}{3}]$, $f'(\theta) = \sqrt{2} - \sec \theta \tan \theta = 0$ at $\theta = \frac{\pi}{4}$. The minimum value of f on this interval is $f(0) = -1$, whereas the maximum value over this interval is $f(\frac{\pi}{4}) = \sqrt{2}(\frac{\pi}{4} - 1) \approx -0.303493$. (Note: $f(\frac{\pi}{3}) = \sqrt{2}\frac{\pi}{3} - 2 \approx -0.519039$.)

51. $y = x^3 + x^2 - x, \quad [-2, 2]$

SOLUTION Let $f(x) = x^3 + x^2 - x$. Then $f'(x) = 3x^2 + 2x - 1 = (3x - 1)(x + 1) = 0$ implies that $x = 1/3$ and $x = -1$ are critical points of f . The minimum of f on the interval $[-2, 2]$ is $f(-2) = -2$, whereas the maximum is $f(2) = 10$. (Note: $f(-1) = 1$ and $f(1/3) = -5/27$.)

53. $y = \theta - 2 \sin \theta, \quad [0, 2\pi]$

SOLUTION Let $g(\theta) = \theta - 2 \sin \theta$. On the interval $[0, 2\pi]$, $g'(\theta) = 1 - 2 \cos \theta = 0$ at $\theta = \frac{\pi}{3}$ and $\theta = \frac{5}{3}\pi$. The minimum of g on this interval is $g(\frac{\pi}{3}) = \frac{\pi}{3} - \sqrt{3} \approx -0.685$ and the maximum is $g(\frac{5}{3}\pi) = \frac{5}{3}\pi + \sqrt{3} \approx 6.968$. (Note: $g(0) = 0$ and $g(2\pi) = 2\pi \approx 6.283$.)

55. $y = \tan x - 2x, \quad [0, 1]$

SOLUTION Let $f(x) = \tan x - 2x$. Then on the interval $[0, 1]$, $f'(x) = \sec^2 x - 2 = 0$ at $x = \frac{\pi}{4}$. The minimum of f is $f(\frac{\pi}{4}) = 1 - \frac{\pi}{2} \approx -0.570796$ and the maximum is $f(0) = 0$. (Note: $f(1) = \tan 1 - 2 \approx -0.442592$.)

57. $y = \frac{\ln x}{x}, \quad [1, 3]$

SOLUTION Let $f(x) = \frac{\ln x}{x}$. Then, on the interval $[1, 3]$,

$$f'(x) = \frac{1 - \ln x}{x^2} = 0$$

at $x = e$. The minimum of f on this interval is $f(1) = 0$ and the maximum is $f(e) = e^{-1} \approx 0.367879$. (Note: $f(3) = \frac{1}{3} \ln 3 \approx 0.366204$.)

59. $y = 3e^x - e^{2x}, \quad [-\frac{1}{2}, 1]$

SOLUTION Let $f(x) = 3e^x - e^{2x}$. Then, on the interval $[-\frac{1}{2}, 1]$, $f'(x) = 3e^x - 2e^{2x} = e^x(3 - 2e^x) = 0$ at $x = \ln(3/2)$. The minimum of f on this interval is $f(1) = 3e - e^2 \approx 0.765789$ and the maximum is $f(\ln(3/2)) = 2.25$. (Note: $f(-1/2) = 3e^{-1/2} - e^{-1} \approx 1.451713$.)

61.  Plot $f(x) = \frac{2+x^2}{x}$ on $(0, 5)$ and use the graph to explain why there is a minimum value, but no maximum value, of f on $(0, 5)$. Use calculus to find the minimum value.

SOLUTION The figure below shows the graph of $f(x) = \frac{2+x^2}{x}$ on $(0, 5)$. As $x \rightarrow 0^+$, $f(x) \rightarrow \infty$. Therefore, f has no absolute maximum value on $(0, 5)$ but the graph suggests there is an absolute minimum at $x \approx 1.4$. To determine the absolute minimum value of f on $(0, 5)$, first rewrite

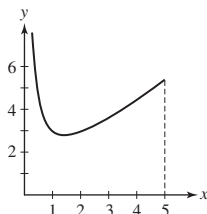
$$f(x) = \frac{2+x^2}{x} = \frac{2}{x} + x = 2x^{-1} + x$$

Then

$$f'(x) = -2x^{-2} + 1 = -\frac{2}{x^2} + 1$$

On the interval $(0, 5)$, the derivative is defined and is equal to zero when $x = \sqrt{2}$. Therefore, the absolute minimum value of f on $(0, 5)$ is

$$f(\sqrt{2}) = \frac{2 + (\sqrt{2})^2}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$



63. Let $f(\theta) = 2 \sin 2\theta + \sin 4\theta$.

(a) Show that θ is a critical point if $\cos 4\theta = -\cos 2\theta$.

(b) Show, using a unit circle, that $\cos \theta_1 = -\cos \theta_2$ if and only if $\theta_1 = \pi \pm \theta_2 + 2\pi k$ for an integer k .

(c) Show that $\cos 4\theta = -\cos 2\theta$ if and only if $\theta = \frac{\pi}{2} + \pi k$ or $\theta = \frac{\pi}{6} + \frac{\pi}{3}k$.

(d) Find the six critical points of f on $[0, 2\pi]$ and find the extreme values of f on this interval.

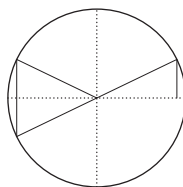
(e)  Check your results against a graph of f .

SOLUTION $f(\theta) = 2 \sin 2\theta + \sin 4\theta$ is differentiable at all θ , so the way to find the critical points is to find all points such that $f'(\theta) = 0$.

(a) $f'(\theta) = 4 \cos 2\theta + 4 \cos 4\theta$. If $f'(\theta) = 0$, then $4 \cos 4\theta = -4 \cos 2\theta$, so $\cos 4\theta = -\cos 2\theta$.

(b) Given the point $(\cos \theta, \sin \theta)$ at angle θ on the unit circle, there are two points with x coordinate $-\cos \theta$. The graphic shows these two points, which are:

- The point $(\cos(\theta + \pi), \sin(\theta + \pi))$ on the opposite end of the unit circle.
- The point $(\cos(\pi - \theta), \sin(\pi - \theta))$ obtained by reflecting through the y axis.



If we include all angles representing these points on the circle, we find that $\cos \theta_1 = -\cos \theta_2$ if and only if $\theta_1 = (\pi + \theta_2) + 2\pi k$ or $\theta_1 = (\pi - \theta_2) + 2\pi k$ for integers k .

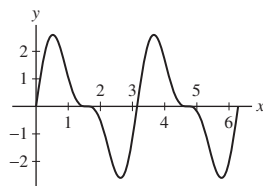
(c) Using (b), we recognize that $\cos 4\theta = -\cos 2\theta$ if $4\theta = 2\theta + \pi + 2\pi k$ or $4\theta = \pi - 2\theta + 2\pi k$. Solving for θ , we obtain $\theta = \frac{\pi}{2} + k\pi$ or $\theta = \frac{\pi}{6} + \frac{\pi}{3}k$.

(d) To find all θ , $0 \leq \theta < 2\pi$ indicated by (c), we use the following table:

k	0	1	2	3	4	5
$\frac{\pi}{2} + k\pi$	$\frac{\pi}{2}$	$\frac{3\pi}{2}$				
$\frac{\pi}{6} + \frac{\pi}{3}k$	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$

The critical points in the range $[0, 2\pi]$ are $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2},$ and $\frac{11\pi}{6}$. On this interval, the maximum value is $f(\frac{\pi}{6}) = f(\frac{7\pi}{6}) = \frac{3\sqrt{3}}{2}$ and the minimum value is $f(\frac{5\pi}{6}) = f(\frac{11\pi}{6}) = -\frac{3\sqrt{3}}{2}$.

(e) The graph of $f(\theta) = 2 \sin 2\theta + \sin 4\theta$ is shown here:



We can see that there are six flat points on the graph between 0 and 2π , as predicted. There are 4 local extrema, and two points at $(\frac{\pi}{2}, 0)$ and $(\frac{3\pi}{2}, 0)$ where the graph has neither a local maximum nor a local minimum.

In Exercises 65–68, find the critical points and the extreme values on $[0, 4]$. In Exercises 67 and 68, refer to Figure 21.

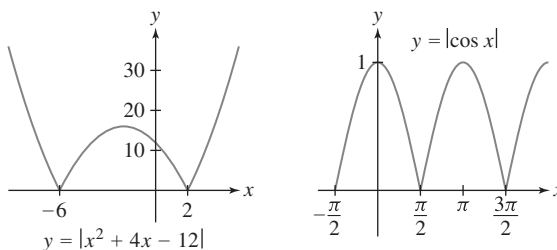
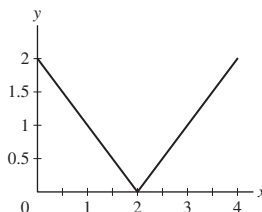


FIGURE 21

65. $y = |x - 2|$

SOLUTION Let $f(x) = |x - 2|$. For $x < 2$, we have $f'(x) = -1$. For $x > 2$, we have $f'(x) = 1$. Now as $x \rightarrow 2^-$, we have $\frac{f(x) - f(2)}{x - 2} = \frac{(2 - x) - 0}{x - 2} \rightarrow -1$; whereas as $x \rightarrow 2^+$, we have $\frac{f(x) - f(2)}{x - 2} = \frac{(x - 2) - 0}{x - 2} \rightarrow 1$. Therefore, $f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$ does not exist and the lone critical point of f is $x = 2$. Alternately, we examine the graph of $f(x) = |x - 2|$ shown below.

To find the extremum, we check the values of f at the critical point and the endpoints. $f(0) = 2$, $f(3) = 1$, and $f(2) = 0$. f takes its minimum value of 0 at $x = 2$, and its maximum of 2 at $x = 0$.



67. $y = |x^2 + 4x - 12|$

SOLUTION Let $f(x) = |x^2 + 4x - 12| = |(x + 6)(x - 2)|$. From the graph of f in Figure 21, we see that $f'(x)$ does not exist at $x = -6$ and at $x = 2$, so these are critical points of f . There is also a critical point between $x = -6$ and $x = 2$ at which $f'(x) = 0$. For $-6 < x < 2$, $f(x) = -x^2 - 4x + 12$, so $f'(x) = -2x - 4 = 0$ when $x = -2$. On the interval $[0, 3]$ the minimum value of f is $f(2) = 0$ and the maximum value is $f(0) = 12$. (Note: $f(3) = 9$ and the critical points $x = -6$ and $x = -2$ are not in the interval.)

In Exercises 69–72, verify Rolle's Theorem for the given interval by checking $f(a) = f(b)$ and then finding a value c in (a, b) such that $f'(c) = 0$.

69. $f(x) = x + x^{-1}$, $[\frac{1}{2}, 2]$

SOLUTION Because f is continuous on $[\frac{1}{2}, 2]$, differentiable on $(\frac{1}{2}, 2)$ and

$$f\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{1}{\frac{1}{2}} = \frac{5}{2} = 2 + \frac{1}{2} = f(2)$$

we may conclude from Rolle's Theorem that there exists a $c \in (\frac{1}{2}, 2)$ at which $f'(c) = 0$. Here, $f'(x) = 1 - x^{-2} = \frac{x^2 - 1}{x^2}$, so we may take $c = 1$.

71. $f(x) = \frac{x^2}{8x - 15}$, $[3, 5]$

SOLUTION Because f is continuous on $[3, 5]$, differentiable on $(3, 5)$ and $f(3) = f(5) = 1$, we may conclude from Rolle's Theorem that there exists a $c \in (3, 5)$ at which $f'(c) = 0$. Here,

$$f'(x) = \frac{(8x - 15)(2x) - 8x^2}{(8x - 15)^2} = \frac{2x(4x - 15)}{(8x - 15)^2}$$

so we may take $c = \frac{15}{4}$.

73. Prove that $f(x) = x^5 + 2x^3 + 4x - 12$ has precisely one real root.

SOLUTION Let's first establish that $f(x) = x^5 + 2x^3 + 4x - 12$ has at least one root. Because f is a polynomial, it is continuous for all x . Moreover, $f(0) = -12 < 0$ and $f(2) = 44 > 0$. Therefore, by the Intermediate Value Theorem, there exists a $c \in (0, 2)$ such that $f(c) = 0$.

Next, we prove that this is the only root. We will use proof by contradiction. Suppose $f(x) = x^5 + 2x^3 + 4x - 12$ has two real roots, $x = a$ and $x = b$. Then $f(a) = f(b) = 0$ and Rolle's Theorem guarantees that there exists a $c \in (a, b)$ at which $f'(c) = 0$. However, $f'(x) = 5x^4 + 6x^2 + 4 \geq 4$ for all x , so there is no $c \in (a, b)$ at which $f'(c) = 0$. Based on this contradiction, we conclude that $f(x) = x^5 + 2x^3 + 4x - 12$ cannot have more than one real root. Finally, f must have precisely one real root.

75. Prove that $f(x) = x^4 + 5x^3 + 4x$ has no root c satisfying $c > 0$. *Hint:* Note that $x = 0$ is a root and apply Rolle's Theorem.

SOLUTION We will proceed by contradiction. Note that $f(0) = 0$ and suppose that there exists a $c > 0$ such that $f(c) = 0$. Then $f(0) = f(c) = 0$ and Rolle's Theorem guarantees that there exists a $d \in (0, c)$ such that $f'(d) = 0$. However, $f'(x) = 4x^3 + 15x^2 + 4 > 4$ for all $x > 0$, so there is no $d \in (0, c)$ such that $f'(d) = 0$. Based on this contradiction, we conclude that $f(x) = x^4 + 5x^3 + 4x$ has no root c satisfying $c > 0$.

77. The position of a mass oscillating at the end of a spring is $s(t) = A \sin \omega t$, where A is the amplitude and ω is the angular frequency. Show that the speed $|v(t)|$ is at a maximum when the acceleration $a(t)$ is zero and that $|a(t)|$ is at a maximum when $v(t)$ is zero.

SOLUTION Let $s(t) = A \sin \omega t$. Then

$$v(t) = \frac{ds}{dt} = A\omega \cos \omega t$$

and

$$a(t) = \frac{dv}{dt} = -A\omega^2 \sin \omega t$$

Thus, the speed

$$|v(t)| = |A\omega \cos \omega t|$$

is a maximum when $|\cos \omega t| = 1$, which is precisely when $\sin \omega t = 0$; that is, the speed $|v(t)|$ is at a maximum when the acceleration $a(t)$ is zero. Similarly,

$$|a(t)| = |A\omega^2 \sin \omega t|$$

is a maximum when $|\sin \omega t| = 1$, which is precisely when $\cos \omega t = 0$; that is, $|a(t)|$ is at a maximum when $v(t)$ is zero.

79. CAS Antibiotic Levels A study shows that the concentration $C(t)$ (in micrograms per milliliter) of antibiotic in a patient's blood serum after t hours is $C(t) = 120(e^{-0.2t} - e^{-bt})$, where $b \geq 1$ is a constant that depends on the particular combination of antibiotic agents used. Solve numerically for the value of b (to two decimal places) for which maximum concentration occurs at $t = 1$ h. You may assume that the maximum occurs at a critical point, as suggested by Figure 22.

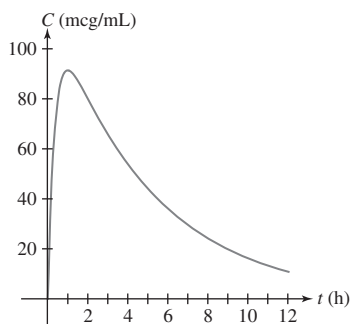


FIGURE 22 Graph of $C(t) = 120(e^{-0.2t} - e^{-bt})$ with b chosen so that the maximum occurs at $t = 1$ h.

SOLUTION Answer is $b = 2.86$. The max of $C(t)$ occurs at $t = \ln(5b)/(b - 0.2)$ so we solve $\ln(5b)/(b - 0.2) = 1$ numerically.

81. In 1919, physicist Alfred Betz argued that the maximum efficiency of a wind turbine is around 59%. If wind enters a turbine with speed v_1 and exits with speed v_2 , then the power extracted is the difference in kinetic energy per unit time:

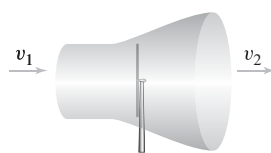
$$P = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 \quad \text{watts}$$

where m is the mass of wind flowing through the rotor per unit time (Figure 23). Betz assumed that $m = \rho A(v_1 + v_2)/2$, where ρ is the density of air and A is the area swept out by the rotor. Wind flowing undisturbed through the same area A would have mass per unit time $\rho A v_1$ and power $P_0 = \frac{1}{2}\rho A v_1^3$. The fraction of power extracted by the turbine is $F = P/P_0$.

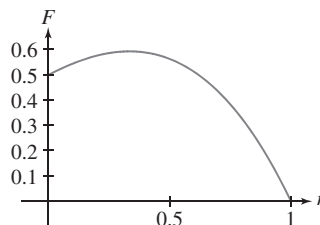
(a) Show that F depends only on the ratio $r = v_2/v_1$ and is equal to $F(r) = \frac{1}{2}(1 - r^2)(1 + r)$, where $0 \leq r \leq 1$.

(b) Show that the maximum value of F , called the **Betz Limit**, is $16/27 \approx 0.59$.

(c)  Explain why Betz's formula for F is not meaningful for r close to zero. *Hint:* How much wind would pass through the turbine if v_2 were zero? Is this realistic?



(A) Wind flowing through a turbine.



(B) F is the fraction of energy extracted by the turbine as a function of $r = v_2/v_1$.

FIGURE 23

SOLUTION

(a) We note that

$$\begin{aligned}
 F &= \frac{P}{P_0} = \frac{\frac{1}{2} \frac{\rho A (v_1 + v_2)}{2} (v_1^2 - v_2^2)}{\frac{1}{2} \rho A v_1^3} \\
 &= \frac{1}{2} \frac{v_1^2 - v_2^2}{v_1^3} \cdot \frac{v_1 + v_2}{v_1} \\
 &= \frac{1}{2} \left(1 - \frac{v_2^2}{v_1^2} \right) \left(1 + \frac{v_2}{v_1} \right) \\
 &= \frac{1}{2} (1 - r^2)(1 + r)
 \end{aligned}$$

(b) Based on part (a),

$$F'(r) = \frac{1}{2}(1 - r^2) - r(1 + r) = -\frac{3}{2}r^2 - r + \frac{1}{2}$$

The roots of this quadratic are $r = -1$ and $r = \frac{1}{3}$. Now, $F(0) = \frac{1}{2}$, $F(1) = 0$ and

$$F\left(\frac{1}{3}\right) = \frac{1}{2} \cdot \frac{8}{9} \cdot \frac{4}{3} = \frac{16}{27} \approx 0.59.$$

Thus, the Betz Limit is $16/27 \approx 0.59$.

(c) If v_2 were 0, then no air would be passing through the turbine, which is not realistic.

83. The response of a circuit or other oscillatory system to an input of frequency ω (“omega”) is described by the function

$$\phi(\omega) = \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4D^2\omega^2}}$$

Both ω_0 (the natural frequency of the system) and D (the damping factor) are positive constants. The graph of ϕ is called a **resonance curve**, and the positive frequency $\omega_r > 0$, where ϕ takes its maximum value, if it exists, is called the **resonant frequency**. Show that $\omega_r = \sqrt{\omega_0^2 - 2D^2}$ if $0 < D < \omega_0/\sqrt{2}$ and that no resonant frequency exists otherwise (Figure 25).

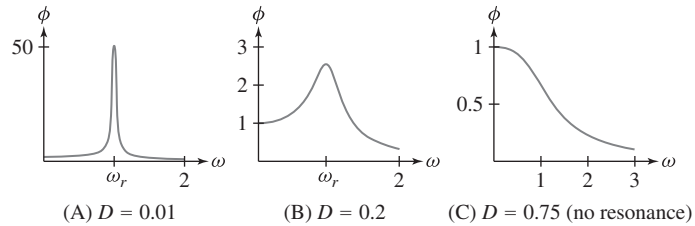


FIGURE 25 Resonance curves with $\omega_0 = 1$.

SOLUTION Let $\phi(\omega) = ((\omega_0^2 - \omega^2)^2 + 4D^2\omega^2)^{-1/2}$. Then

$$\phi'(\omega) = \frac{2\omega((\omega_0^2 - \omega^2) - 2D^2)}{((\omega_0^2 - \omega^2)^2 + 4D^2\omega^2)^{3/2}}$$

and the nonnegative critical points are $\omega = 0$ and $\omega = \sqrt{\omega_0^2 - 2D^2}$. The latter critical point is positive if and only if $\omega_0^2 - 2D^2 > 0$, and since we are given $D > 0$, this is equivalent to $0 < D < \omega_0/\sqrt{2}$.

Define $\omega_r = \sqrt{\omega_0^2 - 2D^2}$. Now, $\phi(0) = 1/\omega_0^2$ and $\phi(\omega) \rightarrow 0$ as $\omega \rightarrow \infty$. Finally,

$$\phi(\omega_r) = \frac{1}{2D\sqrt{\omega_0^2 - D^2}}$$

which, for $0 < D < \omega_0/\sqrt{2}$, is larger than $1/\omega_0^2$. Hence, the point $\omega = \sqrt{\omega_0^2 - 2D^2}$, if defined, is a local maximum.

85. Find the maximum of $y = x^a - x^b$ on $[0, 1]$, where $0 < a < b$. In particular, find the maximum of $y = x^5 - x^{10}$ on $[0, 1]$.

SOLUTION

- Let $f(x) = x^a - x^b$. Then $f'(x) = ax^{a-1} - bx^{b-1}$. Since $a < b$, $f'(x) = x^{a-1}(a - bx^{b-a}) = 0$ implies critical points $x = 0$ and $x = \left(\frac{a}{b}\right)^{1/(b-a)}$, which is in the interval $[0, 1]$ as $a < b$ implies $\frac{a}{b} < 1$ and consequently $x = \left(\frac{a}{b}\right)^{1/(b-a)} < 1$. Also, $f(0) = f(1) = 0$ and $a < b$ implies $x^a > x^b$ on the interval $[0, 1]$, which gives $f(x) > 0$ and thus the maximum value of f on $[0, 1]$ is

$$f\left(\left(\frac{a}{b}\right)^{1/(b-a)}\right) = \left(\frac{a}{b}\right)^{a/(b-a)} - \left(\frac{a}{b}\right)^{b/(b-a)}$$

- Let $f(x) = x^5 - x^{10}$. Then by part (a), the maximum value of f on $[0, 1]$ is

$$f\left(\left(\frac{1}{2}\right)^{1/5}\right) = \left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

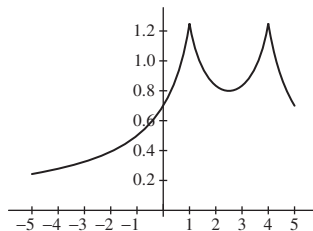
In Exercises 86–88, plot the function using a graphing utility and find its critical points and extreme values on $[-5, 5]$.

87. GU $y = \frac{1}{1 + |x - 1|} + \frac{1}{1 + |x - 4|}$

SOLUTION Let

$$f(x) = \frac{1}{1 + |x - 1|} + \frac{1}{1 + |x - 4|}$$

The plot follows:



We can see on the plot that the critical points of $f(x)$ lie at the cusps at $x = 1$ and $x = 4$ and at the location of the horizontal tangent line at $x = \frac{5}{2}$. With $f(-5) = \frac{17}{70}$, $f(1) = f(4) = \frac{5}{4}$, $f(\frac{5}{2}) = \frac{4}{5}$ and $f(5) = \frac{7}{10}$, it follows that the maximum value of f on $[-5, 5]$ is $f(1) = f(4) = \frac{5}{4}$ and the minimum value is $f(-5) = \frac{17}{70}$.

89. (a) Use implicit differentiation to find the critical points on the curve $27x^2 = (x^2 + y^2)^3$.

(b) GU Plot the curve and the horizontal tangent lines on the same set of axes.

SOLUTION

(a) Differentiating both sides of the equation $27x^2 = (x^2 + y^2)^3$ with respect to x yields

$$54x = 3(x^2 + y^2)^2 \left(2x + 2y \frac{dy}{dx}\right)$$

Solving for dy/dx we obtain

$$\frac{dy}{dx} = \frac{27x - 3x(x^2 + y^2)^2}{3y(x^2 + y^2)^2} = \frac{x(9 - (x^2 + y^2)^2)}{y(x^2 + y^2)^2}$$

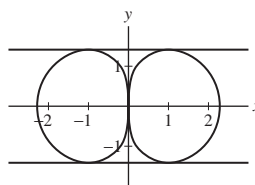
Thus, the derivative is zero when $x^2 + y^2 = 3$. Substituting into the equation for the curve, this yields $x^2 = 1$, or $x = \pm 1$. There are therefore four points at which the derivative is zero:

$$(-1, -\sqrt{2}), (-1, \sqrt{2}), (1, -\sqrt{2}), (1, \sqrt{2})$$

There are also critical points where the derivative does not exist. This occurs when $y = 0$ and gives the following points with vertical tangents:

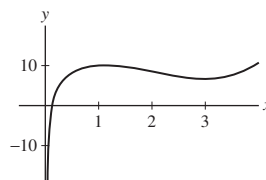
$$(0, 0), (\pm\sqrt[4]{27}, 0)$$

(b) The curve $27x^2 = (x^2 + y^2)^3$ and its horizontal tangents are plotted below.



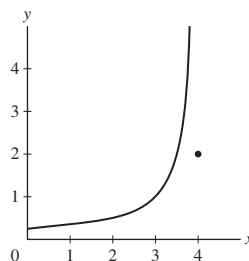
91. Sketch the graph of a continuous function on $(0, 4)$ having a local minimum but no absolute minimum.

SOLUTION Here is the graph of a function f on $(0, 4)$ with a local minimum value [between $x = 2$ and $x = 4$] but no absolute minimum [since $f(x) \rightarrow -\infty$ as $x \rightarrow 0^+$].



93. Sketch the graph of a function f on $[0, 4]$ with a discontinuity such that f has an absolute minimum but no absolute maximum.

SOLUTION Here is the graph of a function f on $[0, 4]$ that (a) has a discontinuity [at $x = 4$] and (b) has an absolute minimum [at $x = 0$] but no absolute maximum [since $f(x) \rightarrow \infty$ as $x \rightarrow 4^-$].



Further Insights and Challenges

95. Show that the extreme values of $f(x) = a \sin x + b \cos x$ are $\pm \sqrt{a^2 + b^2}$.

SOLUTION If $f(x) = a \sin x + b \cos x$, then $f'(x) = a \cos x - b \sin x$, so that $f'(x) = 0$ implies $a \cos x - b \sin x = 0$. This implies $\tan x = \frac{a}{b}$. Then,

$$\sin x = \frac{\pm a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \cos x = \frac{\pm b}{\sqrt{a^2 + b^2}}$$

Therefore

$$f(x) = a \sin x + b \cos x = a \frac{\pm a}{\sqrt{a^2 + b^2}} + b \frac{\pm b}{\sqrt{a^2 + b^2}} = \pm \frac{a^2 + b^2}{\sqrt{a^2 + b^2}} = \pm \sqrt{a^2 + b^2}$$

97. Show that if the quadratic polynomial $f(x) = x^2 + rx + s$ takes on both positive and negative values, then its minimum value occurs at the midpoint between the two roots.

SOLUTION Let $f(x) = x^2 + rx + s$ and suppose that f takes on both positive and negative values. This will guarantee that f has two real roots. By the quadratic formula, the roots of f are

$$x = \frac{-r \pm \sqrt{r^2 - 4s}}{2}$$

Observe that the midpoint between these roots is

$$\frac{1}{2} \left(\frac{-r + \sqrt{r^2 - 4s}}{2} + \frac{-r - \sqrt{r^2 - 4s}}{2} \right) = -\frac{r}{2}$$

Next, $f'(x) = 2x + r = 0$ when $x = -\frac{r}{2}$ and, because the graph of $f(x)$ is an upward opening parabola, it follows that $f(-\frac{r}{2})$ is a minimum. Thus, f takes on its minimum value at the midpoint between the two roots.

99. A cubic polynomial may have a local min and max, or it may have neither (Figure 28). Find conditions on the coefficients a and b of

$$f(x) = \frac{1}{3}x^3 + \frac{1}{2}ax^2 + bx + c$$

that ensure f has neither a local min nor a local max. *Hint:* Apply Exercise 96 to $f'(x)$.

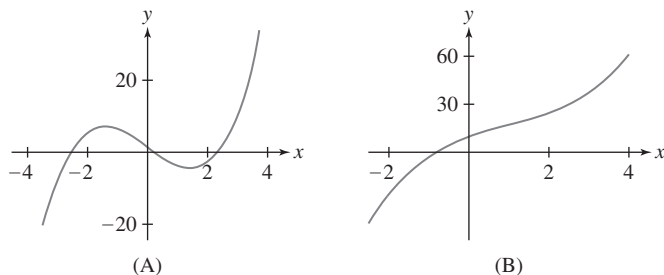


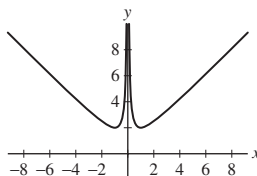
FIGURE 28 Cubic polynomials.

SOLUTION Let $f(x) = \frac{1}{3}x^3 + \frac{1}{2}ax^2 + bx + c$. Using Exercise 96, we have $g(x) = f'(x) = x^2 + ax + b > 0$ for all x provided $b > \frac{1}{4}a^2$, in which case f has no critical points and hence no local extrema. (Actually $b \geq \frac{1}{4}a^2$ will suffice, since in this case [as we'll see in a later section] f has an inflection point but no local extrema.)

101.  Prove that if f is continuous and $f(a)$ and $f(b)$ are local minima where $a < b$, then there exists a value c between a and b such that $f(c)$ is a local maximum. (*Hint:* Apply Theorem 1 to the interval $[a, b]$.) Show that continuity is a necessary hypothesis by sketching the graph of a function (necessarily discontinuous) with two local minima but no local maximum.

SOLUTION

- Let $f(x)$ be a continuous function with $f(a)$ and $f(b)$ local minima on the interval $[a, b]$. By Theorem 1, $f(x)$ must take on both a minimum and a maximum on $[a, b]$. Since local minima occur at $f(a)$ and $f(b)$, the maximum must occur at some other point in the interval, call it c , where $f(c)$ is a local maximum.
- The function graphed here is discontinuous at $x = 0$.



4.3 The Mean Value Theorem and Monotonicity

Preliminary Questions

1. For which value of m is the following statement correct? If $f(2) = 3$ and $f(4) = 9$, and f is differentiable, then f has a tangent line of slope m .

SOLUTION The Mean Value Theorem guarantees that the function has a tangent line with slope equal to

$$\frac{f(4) - f(2)}{4 - 2} = \frac{9 - 3}{4 - 2} = 3$$

Hence, $m = 3$ makes the statement correct.

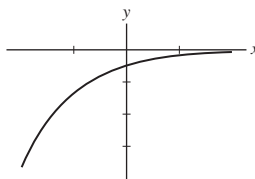
2. Assume f is differentiable. Which of the following statements does *not* follow from the MVT?

- (a) If f has a secant line of slope 0, then f has a tangent line of slope 0.
- (b) If $f(5) < f(9)$, then $f'(c) > 0$ for some $c \in (5, 9)$.
- (c) If f has a tangent line of slope 0, then f has a secant line of slope 0.
- (d) If $f'(x) > 0$ for all x , then every secant line has positive slope.

SOLUTION Conclusion (c) does not follow from the Mean Value Theorem. As a counterexample, consider the function $f(x) = x^3$. Note that $f'(0) = 0$, but no secant line has zero slope.

3. Can a function with the real numbers as its domain that takes on only negative values have a positive derivative? If so, sketch an example.

SOLUTION Yes. The figure below displays a function that takes on only negative values but has a positive derivative.



4. For f with derivative as in Figure 14:

- (a) Is $f(c)$ a local minimum or maximum?
 (b) Is f a decreasing function?

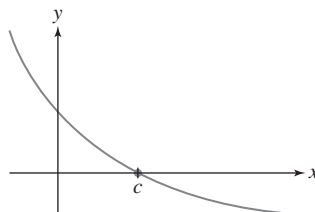


FIGURE 14 Graph of derivative f' .

SOLUTION

(a) To the left of $x = c$, the derivative is positive, so f is increasing; to the right of $x = c$, the derivative is negative, so f is decreasing. Consequently, $f(c)$ must be a local maximum.

(b) No. The derivative is a decreasing function, but as noted in part (a), $f(x)$ is increasing for $x < c$ and decreasing for $x > c$.

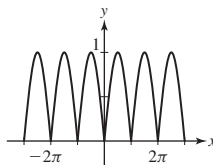
5. Which of the six standard trigonometric functions have infinitely many local minima and infinitely many local maxima but no absolute maximum and no absolute minimum over their whole domain?

SOLUTION The trigonometric functions $\sec x$ and $\csc x$ have infinitely many local minima and infinitely many local maxima but no absolute maximum and no absolute minimum over their whole domain.

6. Compose the absolute value with a familiar function to define a function f that

- has infinitely many local maxima, all of which occur where $f' = 0$, and
- has infinitely many local minima, all of which occur where f' is undefined.

SOLUTION The figure below shows the graph of $f(x) = |\sin x|$. We see that f has infinitely many local maxima, all of which occur where $f' = 0$, and has infinitely many local minima, all of which occur where f' is undefined. The function $f(x) = |\cos x|$ also satisfies these conditions.



Exercises

In Exercises 1–8, find a point c satisfying the conclusion of the MVT for the given function and interval.

1. $y = x^{-1}$, $[2, 8]$

SOLUTION Let $f(x) = x^{-1}$, $a = 2$, $b = 8$. Then $f'(x) = -x^{-2}$, and by the MVT, there exists a $c \in (2, 8)$ such that

$$-\frac{1}{c^2} = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{\frac{1}{8} - \frac{1}{2}}{8 - 2} = -\frac{1}{16}$$

Thus $c^2 = 16$ and $c = \pm 4$. Choose $c = 4 \in (2, 8)$.

3. $y = \cos x - \sin x$, $[0, 2\pi]$

SOLUTION Let $f(x) = \cos x - \sin x$, $a = 0$, $b = 2\pi$. Then $f'(x) = -\sin x - \cos x$, and by the MVT, there exists a $c \in (0, 2\pi)$ such that

$$-\sin c - \cos c = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{1 - 1}{2\pi - 0} = 0$$

Thus $-\sin c = \cos c$. Choose either $c = \frac{3\pi}{4}$ or $c = \frac{7\pi}{4} \in (0, 2\pi)$.

5. $y = x^3$, $[-4, 5]$

SOLUTION Let $f(x) = x^3$, $a = -4$, $b = 5$. Then $f'(x) = 3x^2$, and by the MVT, there exists a $c \in (-4, 5)$ such that

$$3c^2 = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{189}{9} = 21$$

Solving for c yields $c^2 = 7$, so $c = \pm\sqrt{7}$. Both of these values are in the interval $[-4, 5]$, so either value can be chosen.

7. $y = e^{-2x}$, $[0, 3]$

SOLUTION Let $f(x) = e^{-2x}$, $a = 0$, $b = 3$. Then $f'(x) = -2e^{-2x}$, and by the MVT, there exists a $c \in (0, 3)$ such that

$$-2e^{-2c} = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{e^{-6} - 1}{3 - 0} = \frac{e^{-6} - 1}{3}$$

Solving for c yields

$$c = -\frac{1}{2} \ln \frac{1 - e^{-6}}{4} \approx 0.6944 \in (0, 3)$$

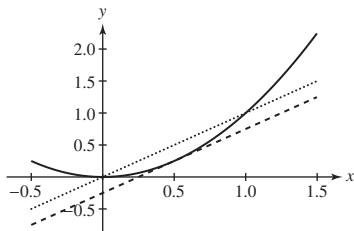
In Exercises 9–12, find a point c satisfying the conclusion of the MVT for the given function and interval. Then draw the graph of the function, the secant line between the endpoints of the graph and the tangent line at $(c, f(c))$, to see that the secant and tangent lines are, in fact, parallel.

9. $y = x^2$, $[0, 1]$

SOLUTION Let $f(x) = x^2$, $a = 0$, and $b = 1$. Then $f'(x) = 2x$. Because f is continuous on the closed interval $[0, 1]$ and differentiable on the open interval $(0, 1)$, by the MVT, there exists a $c \in (0, 1)$ such that

$$2c = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{1 - 0}{1 - 0} = 1$$

Solving for c yields $c = \frac{1}{2}$. The figure below displays the graph of $y = x^2$ as the solid curve, the secant line between the points $(0, 0)$ and $(1, 1)$ as the dotted line, and the tangent line at $x = \frac{1}{2}$ as the dashed line. It is clear that the secant line and the tangent line are parallel.

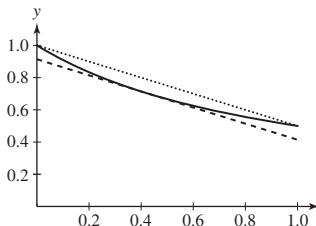


11. $y = e^x$, $[0, 1]$

SOLUTION Let $f(x) = e^x$, $a = 0$, and $b = 1$. Then $f'(x) = e^x$. Because f is continuous on the closed interval $[0, 1]$ and differentiable on the open interval $(0, 1)$, by the MVT, there exists a $c \in (0, 1)$ such that

$$e^c = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{e - 1}{1 - 0} = e - 1$$

Solving for c yields $c = \ln(e - 1)$. The figure below displays the graph of $y = e^x$ as the solid curve, the secant line between the points $(0, 1)$ and $(1, e)$ as the dotted line, and the tangent line at $x = \ln(e - 1)$ as the dashed line. It is clear that the secant line and the tangent line are parallel.

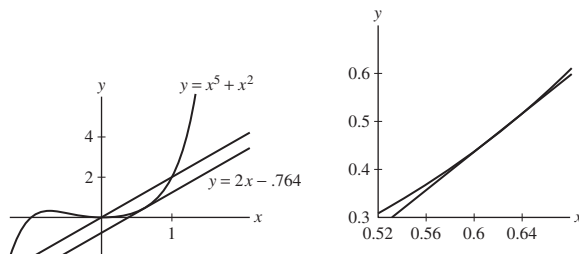


13. [GU] Let $f(x) = x^5 + x^2$. The secant line between $(0, 0)$ and $(1, 2)$ has slope 2 (check this), so by the MVT, $f'(c) = 2$ for some $c \in (0, 1)$. Plot f and the secant line on the same axes. Then plot $y = 2x + b$ for different values of b until the line becomes tangent to the graph of f . Zoom in on the point of tangency to estimate the x -coordinate c of the point of tangency.

SOLUTION Let $f(x) = x^5 + x^2$. The slope of the secant line between $x = 0$ and $x = 1$ is

$$\frac{f(1) - f(0)}{1 - 0} = \frac{2 - 0}{1} = 2$$

A plot of f , the secant line between $x = 0$ and $x = 1$, and the line $y = 2x - 0.764$ is shown below at the left. The line $y = 2x - 0.764$ appears to be tangent to the graph of $y = f(x)$. Zooming in on the point of tangency (see below at the right), it appears that the x -coordinate of the point of tangency is approximately 0.62.



15. Determine the intervals on which $f'(x)$ is positive and negative, assuming that Figure 15 is the graph of f .

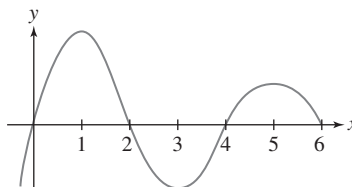


FIGURE 15

SOLUTION The derivative of f is positive on the intervals $(0, 1)$ and $(3, 5)$ where f is increasing; it is negative on the intervals $(1, 3)$ and $(5, 6)$ where f is decreasing.

17. State whether $f(2)$ and $f(4)$ are local minima or local maxima, assuming that Figure 15 is the graph of f' .

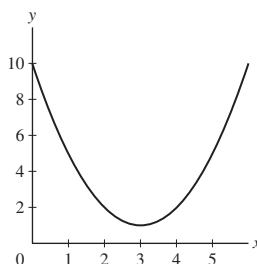
SOLUTION

- f' makes a transition from positive to negative at $x = 2$, so $f(2)$ is a local maximum.
- f' makes a transition from negative to positive at $x = 4$, so $f(4)$ is a local minimum.

In Exercises 19–22, sketch the graph of a function f whose derivative f' has the given description.

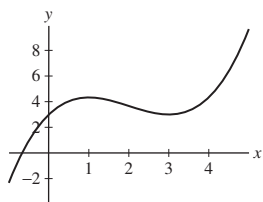
19. $f'(x) > 0$ for $x > 3$ and $f'(x) < 0$ for $x < 3$

SOLUTION Here is the graph of a function f for which $f'(x) > 0$ for $x > 3$ and $f'(x) < 0$ for $x < 3$.



21. $f'(x)$ is negative on $(1, 3)$ and positive everywhere else.

SOLUTION Here is the graph of a function f for which $f'(x)$ is negative on $(1, 3)$ and positive elsewhere.



In Exercises 23–26, find all critical points of f and use the First Derivative Test to determine whether they are local minima or maxima.

23. $f(x) = 4 + 6x - x^2$

SOLUTION Let $f(x) = 4 + 6x - x^2$. Then $f'(x) = 6 - 2x = 0$ implies that $x = 3$ is the only critical point of f . As x increases through 3, $f'(x)$ makes the sign transition $+, -$. Therefore, $f(3) = 13$ is a local maximum.

25. $f(x) = \frac{x^2}{x+1}$

SOLUTION Let $f(x) = \frac{x^2}{x+1}$. Then

$$f'(x) = \frac{x(x+2)}{(x+1)^2} = 0$$

implies that $x = 0$ and $x = -2$ are critical points. Note that $x = -1$ is not a critical point because it is not in the domain of f . As x increases through -2 , $f'(x)$ makes the sign transition $+, -$ so $f(-2) = -4$ is a local maximum. As x increases through 0, $f'(x)$ makes the sign transition $-, +$ so $f(0) = 0$ is a local minimum.

In Exercises 27–58, find the critical points and the intervals on which the function is increasing or decreasing. Use the First Derivative Test to determine whether the critical point yields a local min or max (or neither).

SOLUTION Here is a table legend for Exercises 27–48.

SYMBOL	MEANING
$-$	The entity is negative on the given interval.
0	The entity is zero at the specified point.
$+$	The entity is positive on the given interval.
U	The entity is undefined at the specified point.
$\nearrow$	f is increasing on the given interval.
$\searrow$	f is decreasing on the given interval.
M	f has a local maximum at the specified point.
m	f has a local minimum at the specified point.
$\neg$	There is no local extremum here.

27. $y = -x^2 + 7x - 17$

SOLUTION Let $f(x) = -x^2 + 7x - 17$. Then $f'(x) = 7 - 2x = 0$ yields the critical point $c = \frac{7}{2}$.

x	$(-\infty, \frac{7}{2})$	$7/2$	$(\frac{7}{2}, \infty)$
f'	$+$	0	$-$
f	$\nearrow$	M	$\searrow$

29. $y = x^3 - 12x^2$

SOLUTION Let $f(x) = x^3 - 12x^2$. Then $f'(x) = 3x^2 - 24x = 3x(x - 8) = 0$ yields critical points $c = 0, 8$.

x	$(-\infty, 0)$	0	$(0, 8)$	8	$(8, \infty)$
f'	$+$	0	$-$	0	$+$
f	$\nearrow$	M	$\searrow$	m	$\nearrow$

31. $y = 3x^4 + 8x^3 - 6x^2 - 24x$

SOLUTION Let $f(x) = 3x^4 + 8x^3 - 6x^2 - 24x$. Then

$$\begin{aligned} f'(x) &= 12x^3 + 24x^2 - 12x - 24 \\ &= 12x^2(x+2) - 12(x+2) = 12(x+2)(x^2-1) \\ &= 12(x-1)(x+1)(x+2) = 0 \end{aligned}$$

yields critical points $c = -2, -1, 1$.

x	$(-\infty, -2)$	-2	$(-2, -1)$	-1	$(-1, 1)$	1	$(1, \infty)$
f'	$-$	0	$+$	0	$-$	0	$+$
f	$\searrow$	m	$\nearrow$	M	$\searrow$	m	$\nearrow$

33. $y = \frac{1}{3}x^3 + \frac{3}{2}x^2 + 2x + 4$

SOLUTION Let $f(x) = \frac{1}{3}x^3 + \frac{3}{2}x^2 + 2x + 4$. Then $f'(x) = x^2 + 3x + 2 = (x+1)(x+2) = 0$ yields critical points $c = -2, -1$.

x	$(-\infty, -2)$	-2	$(-2, -1)$	-1	$(-1, \infty)$
f'	$+$	0	$-$	0	$+$
f	$\nearrow$	M	$\searrow$	m	$\nearrow$

35. $y = x^5 + x^3 + 1$

SOLUTION Let $f(x) = x^5 + x^3 + 1$. Then $f'(x) = 5x^4 + 3x^2 = x^2(5x^2 + 3)$ yields a single critical point: $c = 0$.

x	$(-\infty, 0)$	0	$(0, \infty)$
f'	$+$	0	$+$
f	$\nearrow$	$\neg$	$\nearrow$

37. $y = x^4 - 4x^{3/2} \quad (x > 0)$

SOLUTION Let $f(x) = x^4 - 4x^{3/2}$ for $x > 0$. Then $f'(x) = 4x^3 - 6x^{1/2} = 2x^{1/2}(2x^{5/2} - 3) = 0$, which gives us the critical point $c = (\frac{3}{2})^{2/5}$. (Note: $c = 0$ is not in the interval under consideration.)

x	$(0, (\frac{3}{2})^{2/5})$	$\frac{3}{2}^{2/5}$	$((\frac{3}{2})^{2/5}, \infty)$
f'	$-$	0	$+$
f	$\searrow$	m	$\nearrow$

39. $y = x + x^{-1}$

SOLUTION Let $f(x) = x + x^{-1}$. Then

$$f'(x) = 1 - x^{-2} = 1 - \frac{1}{x^2} = \frac{x^2 - 1}{x^2} = 0$$

yields the critical points $c = \pm 1$. $c = 0$ is not a critical point because it is not in the domain of f .

x	$(-\infty, -1)$	-1	$(-1, 0)$	0	$(0, 1)$	1	$(1, \infty)$
f'	$+$	0	$-$	∞	$-$	0	$+$
f	$\nearrow$	M	$\searrow$	$\neg$	$\searrow$	m	$\nearrow$

41. $y = \frac{1}{x^2 + 1}$

SOLUTION Let $f(x) = (x^2 + 1)^{-1}$. Then $f'(x) = -2x(x^2 + 1)^{-2} = 0$ yields critical point $c = 0$.

x	$(-\infty, 0)$	0	$(0, \infty)$
f'	+	0	-
f	$\nearrow$	M	$\searrow$

43. $y = \frac{x^3}{x^2 + 1}$

SOLUTION Let $f(x) = \frac{x^3}{x^2 + 1}$. Then

$$f'(x) = \frac{(x^2 + 1)(3x^2) - x^3(2x)}{(x^2 + 1)^2} = \frac{x^2(x^2 + 3)}{(x^2 + 1)^2} = 0$$

yields the single critical point $c = 0$.

x	$(-\infty, 0)$	0	$(0, \infty)$
f'	+	0	+
f	$\nearrow$	$\cap$	$\nearrow$

45. $y = \theta + \sin \theta + \cos \theta, \quad [0, 2\pi]$

SOLUTION Let $f(\theta) = \theta + \sin \theta + \cos \theta$. Then $f'(\theta) = 1 + \cos \theta - \sin \theta = 0$ yields the critical points $c = \frac{\pi}{2}$ and $c = \pi$.

θ	$(0, \frac{\pi}{2})$	$\frac{\pi}{2}$	$(\frac{\pi}{2}, \pi)$	π	$(\pi, 2\pi)$
f'	+	0	-	0	+
f	$\nearrow$	M	$\searrow$	m	$\nearrow$

47. $y = \sin^2 \theta + \sin \theta, \quad [0, 2\pi]$

SOLUTION Let $f(\theta) = \sin^2 \theta + \sin \theta$. Then $f'(\theta) = 2 \sin \theta \cos \theta + \cos \theta = \cos \theta(2 \sin \theta + 1) = 0$ yields the critical points $c = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}$, and $\frac{11\pi}{6}$.

θ	$(0, \frac{\pi}{2})$	$\frac{\pi}{2}$	$(\frac{\pi}{2}, \frac{7\pi}{6})$	$\frac{7\pi}{6}$	$(\frac{7\pi}{6}, \frac{3\pi}{2})$	$\frac{3\pi}{2}$	$(\frac{3\pi}{2}, \frac{11\pi}{6})$	$\frac{11\pi}{6}$	$(\frac{11\pi}{6}, 2\pi)$
f'	+	0	-	0	+	0	-	0	+
f	$\nearrow$	M	$\searrow$	m	$\nearrow$	M	$\searrow$	m	$\nearrow$

49. $y = x + e^{-x}$

SOLUTION Let $f(x) = x + e^{-x}$. Then $f'(x) = 1 - e^{-x}$, which yields $c = 0$ as the only critical point.

x	$(-\infty, 0)$	0	$(0, \infty)$
f'	-	0	+
f	$\searrow$	m	$\nearrow$

51. $y = e^{-x} \cos x, \quad [-\frac{\pi}{2}, \frac{\pi}{2}]$

SOLUTION Let $f(x) = e^{-x} \cos x$. Then

$$f'(x) = -e^{-x} \sin x - e^{-x} \cos x = -e^{-x}(\sin x + \cos x),$$

which yields $c = -\frac{\pi}{4}$ as the only critical point on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

x	$[-\frac{\pi}{2}, -\frac{\pi}{4})$	$-\frac{\pi}{4}$	$(-\frac{\pi}{4}, \frac{\pi}{2}]$
f'	+	0	-
f	$\nearrow$	M	$\searrow$

53. $y = \tan^{-1} x - \frac{1}{2}x$

SOLUTION Let $f(x) = \tan^{-1} x - \frac{1}{2}x$. Then

$$f'(x) = \frac{1}{1+x^2} - \frac{1}{2},$$

which yields $c = \pm 1$ as critical points.

x	$(-\infty, -1)$	-1	$(-1, 1)$	1	$(1, \infty)$
f'	$-$	0	$+$	0	$-$
f	$\searrow$	m	$\nearrow$	M	$\searrow$

55. $y = x - \ln x^2$

SOLUTION Let $f(x) = x - \ln(x^2)$. Then

$$f'(x) = 1 - \frac{1}{x^2}(2x) = 1 - \frac{2}{x} = \frac{x-2}{x} = 0$$

yields the critical point $c = 2$. $c = 0$ is not a critical point because it is not in the domain of f .

x	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, \infty)$
f'	$+$	∞	$-$	0	$+$
f	$\nearrow$	$\neg$	$\searrow$	m	$\nearrow$

57. $y = x^{1/3}$

SOLUTION Let $f(x) = x^{1/3}$. Then $f'(x) = \frac{1}{3}x^{-2/3}$, which yields $c = 0$ as the only critical point.

x	$(-\infty, 0)$	0	$(0, \infty)$
f'	$+$	∞	$+$
f	$\nearrow$	$\neg$	$\nearrow$

59. Find the maximum value of $f(x) = x^{-x}$ for $x > 0$.

SOLUTION Let $f(x) = x^{-x}$. To find the derivative of f , first rewrite the function as $f(x) = e^{-x \ln x}$. Then

$$f'(x) = e^{-x \ln x} \left(-\ln x + (-x) \frac{1}{x} \right) = -x^{-x}(1 + \ln x) = 0$$

yields the critical point $c = e^{-1}$. For $0 < x < e^{-1}$, $f'(x) > 0$ and f is increasing, while for $x > e^{-1}$, $f'(x) < 0$ and f is decreasing. Thus, f achieves its maximum value at $x = e^{-1}$, and that maximum value is

$$f(e^{-1}) = (e^{-1})^{-e^{-1}} = e^{1/e}$$

61. Show that $f(x) = x^3 - 2x^2 + 2x$ is an increasing function. *Hint:* Find the minimum value of f' .

SOLUTION Let $f(x) = x^3 - 2x^2 + 2x$. For all x , we have

$$f'(x) = 3x^2 - 4x + 2 = 3 \left(x - \frac{2}{3} \right)^2 + \frac{2}{3} \geq \frac{2}{3} > 0$$

Since $f'(x) > 0$ for all x , the function f is everywhere increasing.

63. Ron's toll pass recorded him entering the tollway at mile 0 at 12:17 PM. He exited at mile 115 at 1:52 PM, and soon thereafter he was pulled over by the state police. "The speed limit on the tollway is 65 miles per hour," the trooper told Ron. "You exceeded that by more than five miles per hour this afternoon." "No way!" responded Ron. "I glance at the speedometer frequently, and not once did it read over 65!" How did the trooper use the Mean Value Theorem to support her claim that Ron must have gone more than 70 miles per hour at some point?

SOLUTION Ron traveled 115 miles in 95 minutes, so his average velocity was

$$\frac{115 \text{ miles}}{95 \text{ minutes}} \cdot \frac{60 \text{ minutes}}{1 \text{ hour}} \approx 72.63 \text{ miles/hour}$$

The Mean Value theorem guarantees that if the average velocity was 72.63 miles/hour, then at some time during the trip the instantaneous velocity was 72.63 miles/hour. At that time, Ron's speed exceeded 70 miles/hour.

65. Determine where $f(x) = (1,000 - x)^2 + x^2$ is decreasing. Use this to decide which is larger: $800^2 + 200^2$ or $600^2 + 400^2$.

SOLUTION If $f(x) = (1000 - x)^2 + x^2$, then $f'(x) = -2(1000 - x) + 2x = 4x - 2000$. $f'(x) < 0$ as long as $x < 500$. Therefore, $800^2 + 200^2 = f(200) > f(400) = 600^2 + 400^2$.

67. Which values of c satisfy the conclusion of the MVT on the interval $[a, b]$ if f is a linear function?

SOLUTION Let $f(x) = px + q$, where p and q are constants. Then the slope of every secant line and tangent line of f is p . Accordingly, considering the interval $[a, b]$, every point $c \in (a, b)$ satisfies $f'(c) = p = \frac{f(b) - f(a)}{b - a}$, the conclusion of the MVT.

69. Suppose that $f(0) = 2$ and $f'(x) \leq 3$ for $x > 0$. Apply the MVT to the interval $[0, 4]$ to prove that $f(4) \leq 14$. Prove more generally that $f(x) \leq 2 + 3x$ for all $x > 0$.

SOLUTION The MVT, applied to the interval $[0, 4]$, guarantees that there exists a $c \in (0, 4)$ such that

$$f'(c) = \frac{f(4) - f(0)}{4 - 0} \quad \text{or} \quad f(4) - f(0) = 4f'(c)$$

Because $c > 0$, $f'(c) \leq 3$, so $f(4) - f(0) \leq 12$. Finally, $f(4) \leq f(0) + 12 = 14$.

More generally, let $x > 0$. The MVT, applied to the interval $[0, x]$, guarantees there exists a $c \in (0, x)$ such that

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \quad \text{or} \quad f(x) - f(0) = f'(c)x$$

Because $c > 0$, $f'(c) \leq 3$, so $f(x) - f(0) \leq 3x$. Finally, $f(x) \leq f(0) + 3x = 3x + 2$.

71. Show that if $f(2) = 5$ and $f'(x) \geq 10$ for $x > 2$, then $f(x) \geq 10x - 15$ for all $x > 2$.

SOLUTION Let $x > 2$. The MVT, applied to the interval $[2, x]$, guarantees there exists a $c \in (2, x)$ such that

$$f'(c) = \frac{f(x) - f(2)}{x - 2} \quad \text{or} \quad f(x) - f(2) = (x - 2)f'(c)$$

Because $f'(x) \geq 10$, it follows that $f(x) - f(2) \geq 10(x - 2)$, or $f(x) \geq f(2) + 10(x - 2) = 10x - 15$.

Further Insights and Challenges

73. Prove that if $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for $x \geq 0$, then $f(x) \leq g(x)$ for all $x \geq 0$. *Hint:* Show that the function given by $y = f(x) - g(x)$ is nonincreasing.

SOLUTION Let $h(x) = f(x) - g(x)$. By the sum rule, $h'(x) = f'(x) - g'(x)$. Since $f'(x) \leq g'(x)$ for all $x \geq 0$, $h'(x) \leq 0$ for all $x \geq 0$. This implies that h is nonincreasing. Since $h(0) = f(0) - g(0) = 0$, $h(x) \leq 0$ for all $x \geq 0$ (as h is nonincreasing, it cannot climb above zero). Hence $f(x) - g(x) \leq 0$ for all $x \geq 0$, and so $f(x) \leq g(x)$ for $x \geq 0$.

75. Use Exercises 73 and 74 to prove the following assertions for all $x \geq 0$ (each assertion follows from the previous one):

(a) $\cos x \geq 1 - \frac{1}{2}x^2$

(b) $\sin x \geq x - \frac{1}{6}x^3$

(c) $\cos x \leq 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

Can you guess the next inequality in the series?

SOLUTION

(a) Let $g(x) = \cos x$ and $f(x) = 1 - \frac{1}{2}x^2$. Then $f(0) = g(0) = 1$ and $g'(x) = -\sin x \geq -x = f'(x)$ for $x \geq 0$ by Exercise 74. Now apply Exercise 73 to conclude that $\cos x \geq 1 - \frac{1}{2}x^2$ for $x \geq 0$.

(b) Let $g(x) = \sin x$ and $f(x) = x - \frac{1}{6}x^3$. Then $f(0) = g(0) = 0$ and $g'(x) = \cos x \geq 1 - \frac{1}{2}x^2 = f'(x)$ for $x \geq 0$ by part (a). Now apply Exercise 73 to conclude that $\sin x \geq x - \frac{1}{6}x^3$ for $x \geq 0$.

(c) Let $g(x) = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$ and $f(x) = \cos x$. Then $f(0) = g(0) = 1$ and $g'(x) = -x + \frac{1}{6}x^3 \geq -\sin x = f'(x)$ for $x \geq 0$ by part (b). Now apply Exercise 73 to conclude that $\cos x \leq 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$ for $x \geq 0$.

The next inequality in the series is $\sin x \leq x - \frac{1}{6}x^3 + \frac{1}{120}x^5$, valid for $x \geq 0$.

77. Assume that f'' exists and $f''(x) = 0$ for all x . Prove that $f(x) = mx + b$, where $m = f'(0)$ and $b = f(0)$.

SOLUTION

- Let $f''(x) = 0$ for all x . Then $f'(x) = \text{constant}$ for all x . Since $f'(0) = m$, we conclude that $f'(x) = m$ for all x .
- Let $g(x) = f(x) - mx$. Then $g'(x) = f'(x) - m = m - m = 0$ which implies that $g(x) = \text{constant}$ for all x and consequently $f(x) - mx = \text{constant}$ for all x . Rearranging the statement, $f(x) = mx + \text{constant}$. Since $f(0) = b$, we conclude that $f(x) = mx + b$ for all x .

79. Suppose that $f(x)$ satisfies the following equation (an example of a **differential equation**):

$$f''(x) = -f(x)$$

- (a) Show that $f(x)^2 + f'(x)^2 = f(0)^2 + f'(0)^2$ for all x . *Hint:* Show that the function on the left has zero derivative.
 (b) Verify that $\sin x$ and $\cos x$ satisfy Eq. (1), and deduce that $\sin^2 x + \cos^2 x = 1$.

SOLUTION

- (a) Let $g(x) = f(x)^2 + f'(x)^2$. Then

$$g'(x) = 2f(x)f'(x) + 2f'(x)f''(x) = 2f(x)f'(x) + 2f'(x)(-f(x)) = 0$$

where we have used the fact that $f''(x) = -f(x)$. Because $g'(0) = 0$ for all x , $g(x) = f(x)^2 + f'(x)^2$ must be a constant function. In other words, $f(x)^2 + f'(x)^2 = C$ for some constant C . To determine the value of C , we can substitute any number for x . In particular, for this problem, we want to substitute $x = 0$ and find $C = f(0)^2 + f'(0)^2$. Hence,

$$f(x)^2 + f'(x)^2 = f(0)^2 + f'(0)^2.$$

- (b) Let $f(x) = \sin x$. Then $f'(x) = \cos x$ and $f''(x) = -\sin x$, so $f''(x) = -f(x)$. Next, let $f(x) = \cos x$. Then $f'(x) = -\sin x$, $f''(x) = -\cos x$, and we again have $f''(x) = -f(x)$. Finally, if we take $f(x) = \sin x$, the result from part (a) guarantees that

$$\sin^2 x + \cos^2 x = \sin^2 0 + \cos^2 0 = 0 + 1 = 1$$

81. Use Exercise 80 to prove $f(x) = \sin x$ is the unique solution of Eq. (1) such that $f(0) = 0$ and $f'(0) = 1$; and $g(x) = \cos x$ is the unique solution such that $g(0) = 1$ and $g'(0) = 0$. This result can be used to develop all the properties of the trigonometric functions “analytically”—that is, without reference to triangles.

SOLUTION In part (b) of Exercise 79, it was shown that $f(x) = \sin x$ satisfies Eq. (1), and we can directly calculate that $f(0) = \sin 0 = 0$ and $f'(0) = \cos 0 = 1$. Suppose there is another function, call it $F(x)$, that satisfies Eq. (1) with the same initial conditions: $F(0) = 0$ and $F'(0) = 1$. By Exercise 80, it follows that $F(x) = \sin x$ for all x . Hence, $f(x) = \sin x$ is the unique solution of Eq. (1) satisfying $f(0) = 0$ and $f'(0) = 1$. The proof that $g(x) = \cos x$ is the unique solution of Eq. (1) satisfying $g(0) = 1$ and $g'(0) = 0$ is carried out in a similar manner.

4.4 The Second Derivative and Concavity

Preliminary Questions

1. If f is concave up, then f' is (choose one)
 (a) increasing (b) decreasing

SOLUTION The correct response is (a): increasing. If the function is concave up, then f'' is positive. Since f'' is the derivative of f' , it follows that the derivative of f' is positive and f' must therefore be increasing.

2. What conclusion can you draw if $f'(c) = 0$ and $f''(c) < 0$?

SOLUTION If $f'(c) = 0$ and $f''(c) < 0$, then $f(c)$ is a local maximum.

3. True or false? If $f(c)$ is a local min, then $f''(c)$ must be positive.

SOLUTION False. $f''(c)$ could be zero.

4. True or false? If $f''(c) = 0$, then f has an inflection point at $x = c$.

SOLUTION This statement is false. The condition $f''(c) = 0$ is not sufficient to conclude that f has an inflection point at $x = c$. For example, consider the function $f(x) = x^4$. Then $f''(x) = 12x^2$, so that $f''(0) = 0$. However, $f''(x)$ is positive for any nonzero value of x . This means that f is concave up on $(-\infty, 0)$ and on $(0, \infty)$. Because the concavity of f did not change at $x = 0$, f does not have an inflection point at $x = 0$.

5. The function $f(x) = \frac{x^2+1}{x}$ is concave down for $x < 0$ and concave up for $x > 0$. Is there an inflection point at $x = 0$? Explain.

SOLUTION No, the function $f(x) = \frac{x^2+1}{x}$ does not have an inflection point at $x = 0$ because $x = 0$ is not in the domain of f . Thus, there is no point on the graph at $x = 0$.

6. Can a function have an inflection point at a critical point? Explain.

SOLUTION Yes, a function can have an inflection point at a critical point. Consider the function $f(x) = x^3$. Then $f'(x) = 3x^2$ and $f''(x) = 6x$. It follows that $f'(0) = 0$, so $x = 0$ is a critical point of f . Moreover, $f''(x) < 0$ for $x < 0$ and $f''(x) > 0$ for $x > 0$. This means that f is concave down for $x < 0$ and concave up for $x > 0$, so that f has an inflection point at $x = 0$.

Exercises

1. Match the graphs in Figure 13 with the description:

(a) $f''(x) < 0$ for all x .

(b) $f''(x)$ goes from + to −.

(c) $f''(x) > 0$ for all x .

(d) $f''(x)$ goes from − to +.

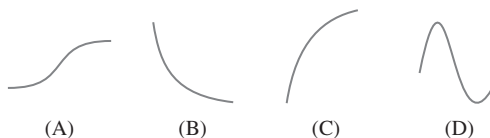


FIGURE 13

SOLUTION

(a) In C, we have $f''(x) < 0$ for all x .

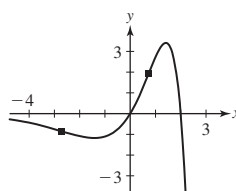
(b) In A, $f''(x)$ goes from + to −.

(c) In B, we have $f''(x) > 0$ for all x .

(d) In D, $f''(x)$ goes from − to +.

3. Plot $f(x) = (2x - x^2)e^x$ and indicate on the graph where it appears that inflection points occur. Then find the inflection points using calculus.

SOLUTION The figure below shows the graph of $f(x) = (2x - x^2)e^x$ with the approximate location of inflection points indicated by a square.



To determine the inflection points, first calculate

$$f'(x) = (2 - 2x)e^x + (2x - x^2)e^x = (2 - x^2)e^x$$

and

$$f''(x) = -2xe^x + (2 - x^2)e^x = (2 - 2x - x^2)e^x$$

The second derivative exists for all x and is equal to zero when

$$x = \frac{2 \pm \sqrt{4 - 4(-1)(2)}}{-2} = \frac{2 \pm 2\sqrt{3}}{-2} = -1 \pm \sqrt{3}$$

Now, f is concave down for $x < -1 - \sqrt{3}$ and for $x > -1 + \sqrt{3}$ because $f''(x) < 0$ there. Moreover, f is concave up for $-1 - \sqrt{3} < x < -1 + \sqrt{3}$ because $f''(x) > 0$ there. Concavity changes at $x = -1 - \sqrt{3}$ and at $x = -1 + \sqrt{3}$, so f has inflection points at both $x = -1 - \sqrt{3}$ and $x = -1 + \sqrt{3}$. The inflection points are

$$(-1 - \sqrt{3}, f(-1 - \sqrt{3})) = (-1 - \sqrt{3}, (-6 - 4\sqrt{3})e^{-1 - \sqrt{3}}) \quad \text{and} \quad (-1 + \sqrt{3}, f(-1 + \sqrt{3})) = (-1 + \sqrt{3}, (-6 + 4\sqrt{3})e^{-1 + \sqrt{3}})$$

In Exercises 5–24, determine the intervals on which the function is concave up or down and find the points of inflection.

5. $y = x^2 - 4x + 3$

SOLUTION Let $f(x) = x^2 - 4x + 3$. Then $f'(x) = 2x - 4$ and $f''(x) = 2 > 0$ for all x . Therefore, f is concave up everywhere, and there are no points of inflection.

7. $y = 10x^3 - x^5$

SOLUTION Let $f(x) = 10x^3 - x^5$. Then $f'(x) = 30x^2 - 5x^4$ and $f''(x) = 60x - 20x^3 = 20x(3 - x^2)$. Now, f is concave up for $x < -\sqrt{3}$ and for $0 < x < \sqrt{3}$ since $f''(x) > 0$ there. Moreover, f is concave down for $-\sqrt{3} < x < 0$ and for $x > \sqrt{3}$ since $f''(x) < 0$ there. Finally, because $f''(x)$ changes sign at $x = 0$ and at $x = \pm\sqrt{3}$, $f(x)$ has a point of inflection at $x = 0$ and at $x = \pm\sqrt{3}$. The points of inflection are

$$(0, f(0)) = (0, 0), \quad (\sqrt{3}, f(\sqrt{3})) = (\sqrt{3}, 21\sqrt{3}), \quad \text{and} \quad (-\sqrt{3}, f(-\sqrt{3})) = (-\sqrt{3}, -21\sqrt{3})$$

9. $y = \theta - 2 \sin \theta, \quad [0, 2\pi]$

SOLUTION Let $f(\theta) = \theta - 2 \sin \theta$. Then $f'(\theta) = 1 - 2 \cos \theta$ and $f''(\theta) = 2 \sin \theta$. Now, f is concave up for $0 < \theta < \pi$ since $f''(\theta) > 0$ there. Moreover, f is concave down for $\pi < \theta < 2\pi$ since $f''(\theta) < 0$ there. Finally, because $f''(\theta)$ changes sign at $\theta = \pi$, $f(\theta)$ has a point of inflection at $\theta = \pi$. The point of inflection is $(\pi, f(\pi)) = (\pi, \pi)$.

11. $y = x(x - 8\sqrt{x}) \quad (x \geq 0)$

SOLUTION Let $f(x) = x(x - 8\sqrt{x}) = x^2 - 8x^{3/2}$. Then $f'(x) = 2x - 12x^{1/2}$ and $f''(x) = 2 - 6x^{-1/2}$. Now, f is concave down for $0 < x < 9$ since $f''(x) < 0$ there. Moreover, f is concave up for $x > 9$ since $f''(x) > 0$ there. Finally, because $f''(x)$ changes sign at $x = 9$, $f(x)$ has a point of inflection at $x = 9$. The point of inflection is $(9, f(9)) = (9, -135)$.

13. $y = (x - 2)(1 - x^3)$

SOLUTION Let $f(x) = (x - 2)(1 - x^3) = x - x^4 - 2 + 2x^3$. Then $f'(x) = 1 - 4x^3 + 6x^2$ and $f''(x) = 12x - 12x^2 = 12x(1 - x) = 0$ at $x = 0$ and $x = 1$. Now, f is concave up on $(0, 1)$ since $f''(x) > 0$ there. Moreover, f is concave down on $(-\infty, 0) \cup (1, \infty)$ since $f''(x) < 0$ there. Finally, because $f''(x)$ changes sign at both $x = 0$ and $x = 1$, $f(x)$ has a point of inflection at both $x = 0$ and $x = 1$. The points of inflections are $(0, f(0)) = (0, -2)$ and $(1, f(1)) = (1, 0)$.

15. $y = \frac{1}{x^2 + 3}$

SOLUTION Let $f(x) = \frac{1}{x^2 + 3}$. Then $f'(x) = -\frac{2x}{(x^2 + 3)^2}$ and

$$f''(x) = -\frac{2(x^2 + 3)^2 - 8x^2(x^2 + 3)}{(x^2 + 3)^4} = \frac{6x^2 - 6}{(x^2 + 3)^3}$$

Now, f is concave up for $|x| > 1$ since $f''(x) > 0$ there. Moreover, f is concave down for $|x| < 1$ since $f''(x) < 0$ there. Finally, because $f''(x)$ changes sign at both $x = -1$ and $x = 1$, $f(x)$ has a point of inflection at both $x = -1$ and $x = 1$. The points of inflection are

$$(-1, f(-1)) = \left(-1, \frac{1}{4}\right) \quad \text{and} \quad (1, f(1)) = \left(1, \frac{1}{4}\right)$$

17. $f(x) = \frac{x^3}{1+x}$

SOLUTION Let $f(x) = \frac{x^3}{1+x}$. Then

$$f'(x) = \frac{(1+x)(3x^2) - x^3(1)}{(1+x)^2} = \frac{3x^2 + 2x^3}{(1+x)^2}$$

and

$$f''(x) = \frac{(1+x)^2(6x + 6x^2) - (3x^2 + 2x^3)(2)(1+x)}{(1+x)^4} = \frac{2x(x^2 + 3x + 3)}{(1+x)^3}$$

Now, f is concave up for $x < -1$ and for $x > 0$ since $f''(x) > 0$ there. Moreover, f is concave down for $-1 < x < 0$ since $f''(x) < 0$ there. Finally, because $f''(x)$ changes sign at $x = 0$, $f(x)$ has a point of inflection at $x = 0$. The point of inflection is $(0, f(0)) = (0, 0)$. Though the sign $f''(x)$ does change sign at $x = -1$, there is no point of inflection at $x = -1$ because $x = -1$ is not in the domain of f .

19. $y = xe^{-3x}$

SOLUTION Let $f(x) = xe^{-3x}$. Then $f'(x) = -3xe^{-3x} + e^{-3x} = (1 - 3x)e^{-3x}$ and $f''(x) = -3(1 - 3x)e^{-3x} - 3e^{-3x} = (9x - 6)e^{-3x}$. Now, f is concave down for $x < \frac{2}{3}$ since $f''(x) < 0$ there. Moreover, f is concave up for $x > \frac{2}{3}$ since $f''(x) > 0$ there. Finally, because $f''(x)$ changes sign at $x = \frac{2}{3}$, $f(x)$ has a point of inflection at $x = \frac{2}{3}$. The point of inflection is

$$\left(\frac{2}{3}, f\left(\frac{2}{3}\right)\right) = \left(\frac{2}{3}, \frac{2}{3}e^{-2}\right)$$

21. $y = 2x^2 + \ln x \quad (x > 0)$

SOLUTION Let $f(x) = 2x^2 + \ln x$. Then $f'(x) = 4x + x^{-1}$ and $f''(x) = 4 - x^{-2}$. Now, f is concave down for $x < \frac{1}{2}$ since $f''(x) < 0$ there. Moreover, f is concave up for $x > \frac{1}{2}$ since $f''(x) > 0$ there. Finally, because $f''(x)$ changes sign at $x = \frac{1}{2}$, f has a point of inflection at $x = \frac{1}{2}$. The point of inflection is

$$\left(\frac{1}{2}, f\left(\frac{1}{2}\right)\right) = \left(\frac{1}{2}, \frac{1}{2} + \ln \frac{1}{2}\right) = \left(\frac{1}{2}, \frac{1}{2} - \ln 2\right)$$

23. $f(t) = te^{-t^2}$

SOLUTION Let $f(t) = te^{-t^2}$. Then

$$f'(t) = e^{-t^2} + te^{-t^2}(-2t) = (1 - 2t^2)e^{-t^2}$$

and

$$f''(t) = -4te^{-t^2} + (1 - 2t^2)e^{-t^2}(-2t) = 2t(2t^2 - 3)e^{-t^2}$$

Now, f is concave up for $-\sqrt{6}/2 < t < 0$ and for $t > \sqrt{6}/2$ since $f''(t) > 0$ there. Moreover, f is concave down for $t < -\sqrt{6}/2$ and for $0 < t < \sqrt{6}/2$ since $f''(t) < 0$ there. Finally, because $f''(t)$ changes sign at $t = \pm\sqrt{6}/2$ and at $t = 0$, $f(t)$ has points of inflection at $t = -\sqrt{6}/2$, $t = 0$, and $t = \sqrt{6}/2$. The points of inflection are

$$\left(-\frac{\sqrt{6}}{2}, f\left(-\frac{\sqrt{6}}{2}\right)\right) = \left(-\frac{\sqrt{6}}{2}, -\frac{\sqrt{6}}{2}e^{-3/2}\right) \quad (0, f(0)) = (0, 0), \quad \text{and} \quad \left(\frac{\sqrt{6}}{2}, f\left(\frac{\sqrt{6}}{2}\right)\right) = \left(\frac{\sqrt{6}}{2}, \frac{\sqrt{6}}{2}e^{-3/2}\right)$$

25. The position of an ambulance in kilometers on a straight road over a period of 4 hours is given by the graph in Figure 15.

(a) Describe the motion of the ambulance.

(b) Explain what the fact that this graph is concave up tells us about the speed of the ambulance.

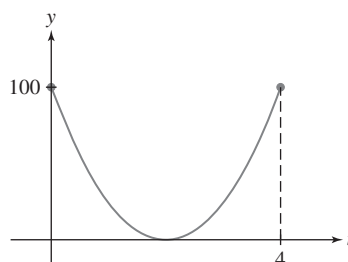


FIGURE 15

SOLUTION

(a) Suppose, for point of reference, the straight road along which the ambulance is driving travels north and south, with north being the positive direction. The ambulance travels south for 100 kilometers, turns around, and travels 100 kilometers north, returning to its original position.

(b) The fact that the graph is concave up indicates that the velocity of the ambulance is always increasing. On the first part of its trip, the ambulance is traveling south, so the velocity of the ambulance is negative and increasing, which means the speed of the ambulance is decreasing. On the second part of the trip, the ambulance is traveling north, so the velocity of the ambulance is positive and increasing, which means the speed of the ambulance is increasing.

27.  The growth of a sunflower during the first 100 days after sprouting is modeled well by the *logistic curve* $y = h(t)$ shown in Figure 17. Estimate the growth rate at the point of inflection and explain its significance. Then make a rough sketch of the first and second derivatives of h .

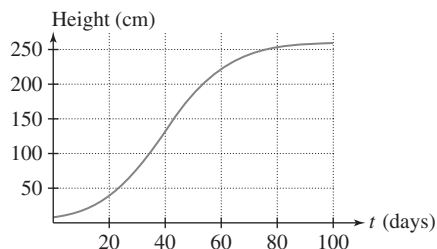
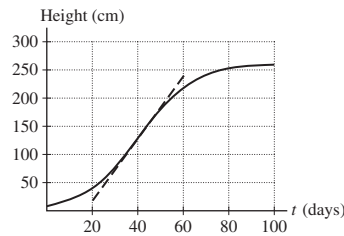


FIGURE 17

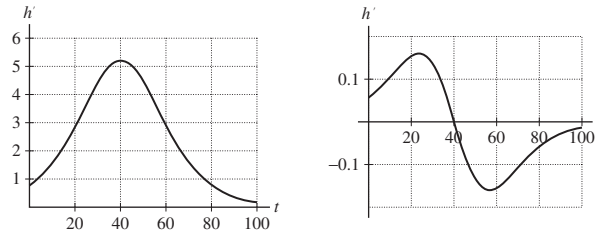
SOLUTION The point of inflection in Figure 17 appears to occur at $t = 40$ days. The graph below shows the logistic curve with an approximate tangent line drawn at $t = 40$. The approximate tangent line passes roughly through the points (20, 20) and (60, 240). The growth rate at the point of inflection is thus

$$\frac{240 - 20}{60 - 20} = \frac{220}{40} = 5.5 \text{ cm/day}$$

Because the logistic curve changes from concave up to concave down at $t = 40$, the growth rate at this point is the maximum growth rate for the sunflower plant.



Sketches of the first and second derivative of $h(t)$ are shown below at the left and at the right, respectively.



29. Repeat Exercise 28 but assume that Figure 18 is the graph of the *derivative* f' .

SOLUTION Points of inflection occur when $f''(x)$ changes sign. Consequently, points of inflection occur when $f'(x)$ changes from increasing to decreasing or from decreasing to increasing. In Figure 18, this occurs at $x = b$ and at $x = e$; therefore, $f(x)$ has an inflection point at $x = b$ and another at $x = e$. The function $f(x)$ will be concave down when $f''(x) < 0$ or when $f'(x)$ is decreasing. Thus, $f(x)$ is concave down for $b < x < e$.

31. Figure 19 shows the *derivative* f' on $[0, 1.2]$. Locate the points of inflection of f and the points where the local minima and maxima occur. Determine the intervals on which f has the following properties:

- | | |
|----------------|------------------|
| (a) Increasing | (b) Decreasing |
| (c) Concave up | (d) Concave down |

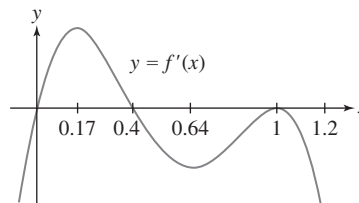


FIGURE 19

SOLUTION Recall that the graph is that of f' , *not* f . The inflection points of f occur where f' changes from increasing to decreasing or vice versa because it is at these points that the sign of f'' changes. From the graph we conclude that f has points of inflection at $x = 0.17$, $x = 0.64$, and $x = 1$. The local extrema of f occur where f' changes sign. This occurs at $x = 0.4$. Because the sign of f' changes from $+$ to $-$, $f(0.4)$ is a local maximum. There are no local minima.

- (a) f is increasing when f' is positive. Hence, f is increasing on $(0, 0.4)$.
 (b) f is decreasing when f' is negative. Hence, f is decreasing on $(0.4, 1) \cup (1, 1.2)$.
 (c) Now f is concave up where f' is increasing. This occurs on $(0, 0.17) \cup (0.64, 1)$.
 (d) Moreover, f is concave down where f' is decreasing. This occurs on $(0.17, 0.64) \cup (1, 1.2)$.

In Exercises 33–46, find the critical points and apply the Second Derivative Test (or state that it fails).

33. $f(x) = x^3 - 12x^2 + 45x$

SOLUTION Let $f(x) = x^3 - 12x^2 + 45x$. Then $f'(x) = 3x^2 - 24x + 45 = 3(x - 3)(x - 5)$, and the critical points are $x = 3$ and $x = 5$. Moreover, $f''(x) = 6x - 24$, so $f''(3) = -6 < 0$ and $f''(5) = 6 > 0$. Therefore, by the Second Derivative Test, $f(3) = 54$ is a local maximum, and $f(5) = 50$ is a local minimum.

35. $f(x) = 3x^4 - 8x^3 + 6x^2$

SOLUTION Let $f(x) = 3x^4 - 8x^3 + 6x^2$. Then $f'(x) = 12x^3 - 24x^2 + 12x = 12x(x - 1)^2 = 0$ at $x = 0, 1$ and $f''(x) = 36x^2 - 48x + 12$. Thus, $f''(0) > 0$, which implies $f(0)$ is a local minimum; however, $f''(1) = 0$, which is inconclusive.

37. $f(x) = \frac{x^2 - 8x}{x + 1}$

SOLUTION Let $f(x) = \frac{x^2 - 8x}{x + 1}$. Then

$$f'(x) = \frac{x^2 + 2x - 8}{(x + 1)^2} \quad \text{and} \quad f''(x) = \frac{2(x + 1)^2 - 2(x^2 + 2x - 8)}{(x + 1)^3}$$

Thus, the critical points are $x = -4$ and $x = 2$. Moreover, $f''(-4) < 0$ and $f''(2) > 0$. Therefore, by the second derivative test, $f(-4) = -16$ is a local maximum and $f(2) = -4$ is a local minimum.

39. $y = 6x^{3/2} - 4x^{1/2}$

SOLUTION Let $f(x) = 6x^{3/2} - 4x^{1/2}$. Then $f'(x) = 9x^{1/2} - 2x^{-1/2} = x^{-1/2}(9x - 2)$, so there are two critical points: $x = 0$ and $x = \frac{2}{9}$. Now,

$$f''(x) = \frac{9}{2}x^{-1/2} + x^{-3/2} = \frac{1}{2}x^{-3/2}(9x + 2)$$

Thus, $f''\left(\frac{2}{9}\right) > 0$, which implies $f\left(\frac{2}{9}\right)$ is a local minimum. $f''(x)$ is undefined at $x = 0$, so the Second Derivative Test cannot be applied there.

41. $f(x) = \sin^2 x + \cos x$, $[0, \pi]$

SOLUTION Let $f(x) = \sin^2 x + \cos x$. Then $f'(x) = 2 \sin x \cos x - \sin x = \sin x(2 \cos x - 1)$. On the interval $[0, \pi]$, $f'(x) = 0$ at $x = 0$, $x = \frac{\pi}{3}$ and $x = \pi$. Now,

$$f''(x) = 2 \cos^2 x - 2 \sin^2 x - \cos x$$

Thus, $f''(0) > 0$, so $f(0)$ is a local minimum. On the other hand, $f''\left(\frac{\pi}{3}\right) < 0$, so $f\left(\frac{\pi}{3}\right)$ is a local maximum. Finally, $f''(\pi) > 0$, so $f(\pi)$ is a local minimum.

43. $f(x) = xe^{-x^2}$

SOLUTION Let $f(x) = xe^{-x^2}$. Then $f'(x) = -2x^2e^{-x^2} + e^{-x^2} = (1 - 2x^2)e^{-x^2}$, so there are two critical points: $x = \pm \frac{\sqrt{2}}{2}$. Now,

$$f''(x) = (4x^3 - 2x)e^{-x^2} - 4xe^{-x^2} = (4x^3 - 6x)e^{-x^2}$$

Thus, $f''\left(\frac{\sqrt{2}}{2}\right) < 0$, so $f\left(\frac{\sqrt{2}}{2}\right)$ is a local maximum. On the other hand, $f''\left(-\frac{\sqrt{2}}{2}\right) > 0$, so $f\left(-\frac{\sqrt{2}}{2}\right)$ is a local minimum.

45. $f(x) = x^3 \ln x$ ($x > 0$)

SOLUTION Let $f(x) = x^3 \ln x$. Then $f'(x) = x^2 + 3x^2 \ln x = x^2(1 + 3 \ln x)$, so there is only one critical point: $x = e^{-1/3}$. Now,

$$f''(x) = 3x + 2x(1 + 3 \ln x) = x(5 + 6 \ln x)$$

Thus, $f''\left(e^{-1/3}\right) > 0$, so $f\left(e^{-1/3}\right)$ is a local minimum.

In Exercises 47–62, find the intervals on which f is concave up or down, the points of inflection, the critical points, and the local minima and maxima.

SOLUTION Here is a table legend for Exercises 47–59.

SYMBOL	MEANING
–	The entity is negative on the given interval.
0	The entity is zero at the specified point.
+	The entity is positive on the given interval.
U	The entity is undefined at the specified point.
↗	The function (f , g , etc.) is increasing on the given interval.
↘	The function (f , g , etc.) is decreasing on the given interval.
∩	The function (f , g , etc.) is concave up on the given interval.
∪	The function (f , g , etc.) is concave down on the given interval.
M	The function (f , g , etc.) has a local maximum at the specified point.
m	The function (f , g , etc.) has a local minimum at the specified point.
I	The function (f , g , etc.) has an inflection point here.
↯	There is no local extremum or inflection point here.

47. $f(x) = x^3 - 2x^2 + x$

SOLUTION Let $f(x) = x^3 - 2x^2 + x$.

- Then $f'(x) = 3x^2 - 4x + 1 = (x-1)(3x-1) = 0$ yields $x = 1$ and $x = \frac{1}{3}$ as candidates for extrema.
- Moreover, $f''(x) = 6x - 4 = 0$ gives a candidate for a point of inflection at $x = \frac{2}{3}$.

x	$(-\infty, \frac{1}{3})$	$\frac{1}{3}$	$(\frac{1}{3}, 1)$	1	$(1, \infty)$
f'	+	0	-	0	+
f	$\nearrow$	M	$\searrow$	m	$\nearrow$

x	$(-\infty, \frac{2}{3})$	$\frac{2}{3}$	$(\frac{2}{3}, \infty)$
f''	-	0	+
f		I	

49. $f(t) = t^2 - t^3$

SOLUTION Let $f(t) = t^2 - t^3$.

- Then $f'(t) = 2t - 3t^2 = t(2 - 3t) = 0$ yields $t = 0$ and $t = \frac{2}{3}$ as candidates for extrema.
- Moreover, $f''(t) = 2 - 6t = 0$ gives a candidate for a point of inflection at $t = \frac{1}{3}$.

t	$(-\infty, 0)$	0	$(0, \frac{2}{3})$	$\frac{2}{3}$	$(\frac{2}{3}, \infty)$
f'	-	0	+	0	-
f	$\searrow$	m	$\nearrow$	M	$\searrow$

t	$(-\infty, \frac{1}{3})$	$\frac{1}{3}$	$(\frac{1}{3}, \infty)$
f''	+	0	-
f	$\smile$	I	$\frown$

51. $f(x) = x^2 - 8x^{1/2} \quad (x \geq 0)$

SOLUTION Let $f(x) = x^2 - 8x^{1/2}$. Note that the domain of f is $x \geq 0$.

- Then $f'(x) = 2x - 4x^{-1/2} = x^{-1/2}(2x^{3/2} - 4) = 0$ yields $x = 0$ and $x = (2)^{2/3}$ as candidates for extrema.
- Moreover, $f''(x) = 2 + 2x^{-3/2} > 0$ for all $x \geq 0$, which means there are no inflection points.

x	0	$(0, (2)^{2/3})$	$(2)^{2/3}$	$((2)^{2/3}, \infty)$
f'	U	-	0	+
f	M	$\searrow$	m	$\nearrow$

53. $f(x) = \frac{x}{x^2 + 27}$

SOLUTION Let $f(x) = \frac{x}{x^2 + 27}$.

- Then $f'(x) = \frac{27 - x^2}{(x^2 + 27)^2} = 0$ yields $x = \pm 3\sqrt{3}$ as candidates for extrema.
- Moreover, $f''(x) = \frac{-2x(x^2 + 27)^2 - (27 - x^2)(2)(x^2 + 27)(2x)}{(x^2 + 27)^4} = \frac{2x(x^2 - 81)}{(x^2 + 27)^3} = 0$ gives candidates for a point of inflection at $x = 0$ and at $x = \pm 9$.

x	$(-\infty, -3\sqrt{3})$	$-3\sqrt{3}$	$(-3\sqrt{3}, 3\sqrt{3})$	$3\sqrt{3}$	$(3\sqrt{3}, \infty)$
f'	-	0	+	0	-
f	$\searrow$	m	$\nearrow$	M	$\searrow$

x	$(-\infty, -9)$	-9	$(-9, 0)$	0	$(0, 9)$	9	$(9, \infty)$
f''	-	0	+	0	-	0	+
f	$\frown$	I	$\smile$	I	$\frown$	I	$\smile$

55. $f(x) = x^{5/3} - x$

SOLUTION Let $f(x) = x^{5/3} - x$.

- Then $f'(x) = \frac{5}{3}x^{2/3} - 1 = 0$ yields $x = \pm \left(\frac{3}{5}\right)^{3/2}$ as a candidate for an extremum.
- Moreover, $f''(x) = \frac{10}{9}x^{-1/3}$, which yields $x = 0$ as a candidate for a point of inflection.

x	$\left(-\infty, -\left(\frac{3}{5}\right)^{3/2}\right)$	$-\left(\frac{3}{5}\right)^{3/2}$	$\left(-\left(\frac{3}{5}\right)^{3/2}, \left(\frac{3}{5}\right)^{3/2}\right)$	$\left(\frac{3}{5}\right)^{3/2}$	$\left(\left(\frac{3}{5}\right)^{3/2}, \infty\right)$
f'	+	0	-	0	+
f	$\nearrow$	M	$\searrow$	m	$\nearrow$

x	$(-\infty, 0)$	0	$(0, \infty)$
f''	-	U	+
f		I	

57. $f(\theta) = \theta + \sin \theta$, $[0, 2\pi]$

SOLUTION Let $f(\theta) = \theta + \sin \theta$ on $[0, 2\pi]$.

- Then $f'(\theta) = 1 + \cos \theta = 0$ yields $\theta = \pi$ as a candidate for an extremum.
- Moreover, $f''(\theta) = -\sin \theta = 0$ gives candidates for a point of inflection at $\theta = 0$, at $\theta = \pi$, and at $\theta = 2\pi$.

θ	$(0, \pi)$	π	$(\pi, 2\pi)$
f'	+	0	+
f	$\nearrow$	$\neg$	$\nearrow$

θ	0	$(0, \pi)$	π	$(\pi, 2\pi)$	2π
f''	0	-	0	+	0
f	$\neg$	$\frown$	I	$\smile$	$\neg$

59. $f(x) = \tan x$, $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

SOLUTION Let $f(x) = \tan x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

- Then $f'(x) = \sec^2 x \geq 1 > 0$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.
- Moreover, $f''(x) = 2 \sec x \cdot \sec x \tan x = 2 \sec^2 x \tan x = 0$ gives a candidate for a point of inflection at $x = 0$.

x	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
f'	+
f	$\nearrow$

x	$\left(-\frac{\pi}{2}, 0\right)$	0	$\left(0, \frac{\pi}{2}\right)$
f''	-	0	+
f	$\frown$	I	$\smile$

61. $y = (x^2 - 2)e^{-x}$ ($x > 0$)

SOLUTION Let $f(x) = (x^2 - 2)e^{-x}$.

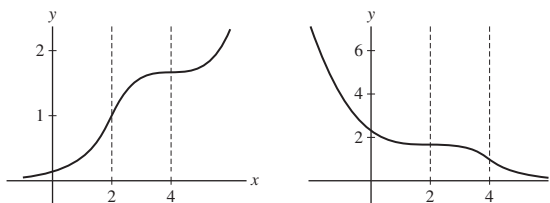
- Then $f'(x) = -(x^2 - 2x - 2)e^{-x} = 0$ gives $x = 1 + \sqrt{3}$ as a candidate for an extrema.
- Moreover, $f''(x) = (x^2 - 4x)e^{-x} = 0$ gives $x = 4$ as a candidate for a point of inflection.

x	$(0, 1 + \sqrt{3})$	$1 + \sqrt{3}$	$(1 + \sqrt{3}, \infty)$
f'	+	0	-
f	$\nearrow$	M	$\searrow$

x	$(0, 4)$	4	$(4, \infty)$
f''	-	0	+
f	$\frown$	I	$\smile$

63. Sketch the graph of an increasing function such that $f''(x)$ changes from + to - at $x = 2$ and from - to + at $x = 4$. Do the same for a decreasing function.

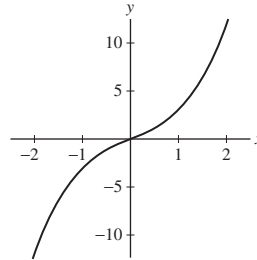
SOLUTION The graph shown below at the left is an increasing function which changes from concave up to concave down at $x = 2$ and from concave down to concave up at $x = 4$. The graph shown below at the right is a decreasing function which changes from concave up to concave down at $x = 2$ and from concave down to concave up at $x = 4$.



In Exercises 64–66, sketch the graph of a function f satisfying all of the given conditions.

65. (i) $f'(x) > 0$ for all x , and
 (ii) $f''(x) < 0$ for $x < 0$ and $f''(x) > 0$ for $x > 0$

SOLUTION Here is the graph of a function $f(x)$ satisfying (i) $f'(x) > 0$ for all x and (ii) $f''(x) < 0$ for $x < 0$ and $f''(x) > 0$ for $x > 0$.



67.  An infectious flu spreads slowly at the beginning of an epidemic. The infection process accelerates until a majority of the susceptible individuals are infected, at which point the process slows down.

- (a) If $R(t)$ is the number of individuals infected at time t , describe the concavity of the graph of R near the beginning and end of the epidemic.
 (b) Describe the status of the epidemic on the day that R has a point of inflection.

SOLUTION

- (a) Near the beginning of the epidemic, the graph of R is concave up. Near the epidemic's end, R is concave down.
 (b) “Epidemic subsiding: number of new cases declining.”

69.  Water is pumped into a sphere of radius R at a variable rate in such a way that the water level rises at a constant rate (Figure 20). Let $V(t)$ be the volume of water in the tank at time t . Sketch the graph V (approximately, but with the correct concavity). Where does the point of inflection occur?

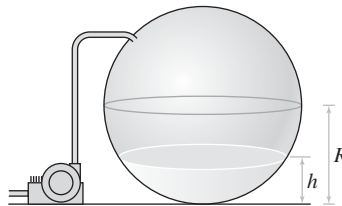
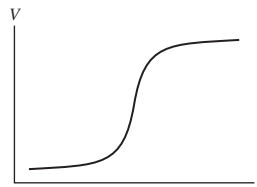


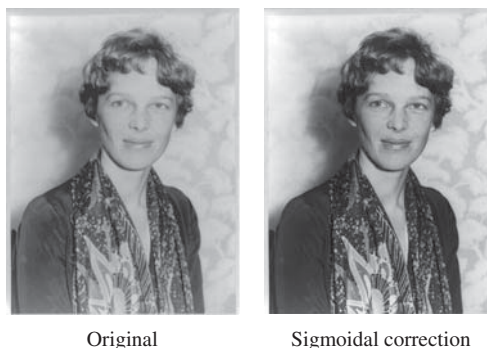
FIGURE 20

SOLUTION Because water is entering the sphere in such a way that the water level rises at a constant rate, we expect the volume to increase more slowly near the bottom and top of the sphere where the sphere is not as “wide” and to increase more rapidly near the middle of the sphere. The graph of $V(t)$ should therefore start concave up and change to concave down when the sphere is half full; that is, the point of inflection should occur when the water level is equal to the radius of the sphere. A possible graph of $V(t)$ is shown below.



71. **Image Processing** The intensity of a pixel in a digital image is measured by a number u between 0 and 1. Often, images can be enhanced by rescaling intensities, as in the images of Amelia Earhart in Figure 21. When rescaling, pixels of intensity u are displayed with intensity $g(u)$ for a suitable function g . One common choice is the **sigmoidal correction**, defined for constants a, b by

$$g(u) = \frac{f(u) - f(0)}{f(1) - f(0)}, \quad \text{where } f(u) = (1 + e^{b(a-u)})^{-1}$$



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and Photographs Division

FIGURE 21

Figure 22 shows that $g(u)$ reduces the intensity of low-intensity pixels [where $g(u) < u$] and increases the intensity of high-intensity pixels.

- (a) Verify that $f'(u) > 0$ and use this to show that $g(u)$ increases from 0 to 1 for $0 \leq u \leq 1$.
 (b) Where does $g(u)$ have a point of inflection?

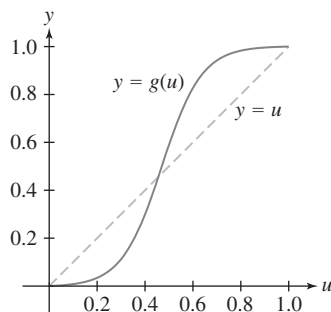


FIGURE 22 Sigmoidal correction with $a = 0.47$, $b = 12$.

SOLUTION

- (a) With $f(u) = (1 + e^{b(a-u)})^{-1}$, it follows that

$$f'(u) = -(1 + e^{b(a-u)})^{-2} \cdot -be^{b(a-u)} = \frac{be^{b(a-u)}}{(1 + e^{b(a-u)})^2} > 0$$

for all u . Next, observe that

$$g(0) = \frac{f(0) - f(0)}{f(1) - f(0)} = 0, \quad g(1) = \frac{f(1) - f(0)}{f(1) - f(0)} = 1,$$

and

$$g'(u) = \frac{1}{f(1) - f(0)} f'(u) > 0$$

for all u . Thus, $g(u)$ increases from 0 to 1 for $0 \leq u \leq 1$.

- (b) Working from part (a), we find

$$f''(u) = \frac{b^2 e^{b(a-u)} (2e^{b(a-u)} - 1)}{(1 + e^{b(a-u)})^3}$$

Because

$$g''(u) = \frac{1}{f(1) - f(0)} f''(u)$$

it follows that $g(u)$ has a point of inflection when

$$2e^{b(a-u)} - 1 = 0 \quad \text{or} \quad u = a + \frac{1}{b} \ln 2$$

Further Insights and Challenges

In Exercises 73–75, assume that f is differentiable.

73. Proof of the Second Derivative Test Let c be a critical point such that $f''(c) > 0$ [the case $f''(c) < 0$ is similar].

(a) Show that $f''(c) = \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c)}{h}$.

(b) Use (a) to show that there exists an open interval (a, b) containing c such that $f'(x) < 0$ if $a < x < c$ and $f'(x) > 0$ if $c < x < b$. Conclude that $f(c)$ is a local minimum.

SOLUTION

(a) Because c is a critical point, either $f'(c) = 0$ or $f'(c)$ does not exist; however, $f''(c)$ exists, so $f'(c)$ must also exist. Therefore, $f'(c) = 0$. Now, from the definition of the derivative, we have

$$f''(c) = \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c)}{h} = \lim_{h \rightarrow 0} \frac{f'(c+h)}{h}$$

(b) We are given that $f''(c) > 0$. By part (a), it follows that

$$\lim_{h \rightarrow 0} \frac{f'(c+h)}{h} > 0$$

in other words, for sufficiently small h ,

$$\frac{f'(c+h)}{h} > 0$$

Now, if h is sufficiently small but negative, then $f'(c+h)$ must also be negative (so that the ratio $f'(c+h)/h$ will be positive) and $c+h < c$. On the other hand, if h is sufficiently small but positive, then $f'(c+h)$ must also be positive and $c+h > c$. Thus, there exists an open interval (a, b) containing c such that $f'(x) < 0$ for $a < x < c$ and $f'(x) > 0$ for $c < x < b$. Finally, because $f'(x)$ changes from negative to positive at $x = c$, $f(c)$ must be a local minimum.

75.  Assume that f'' exists and let c be a point of inflection of f .

(a) Use the method of Exercise 74 to prove that the tangent line at $x = c$ crosses the graph (Figure 23). *Hint:* Show that $G(x)$ changes sign at $x = c$.

(b)  Verify this conclusion for $f(x) = \frac{x}{3x^2 + 1}$ by graphing f and the tangent line at each inflection point on the same set of axes.

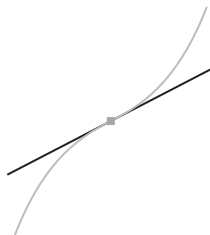


FIGURE 23 Tangent line crosses graph at point of inflection.

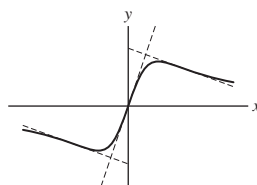
SOLUTION

(a) Let $G(x) = f(x) - f'(c)(x - c) - f(c)$. Then, as in Exercise 74, $G(c) = G'(c) = 0$ and $G''(x) = f''(x)$. If $f''(x)$ changes from positive to negative at $x = c$, then so does $G''(x)$ and $G'(x)$ is increasing for $x < c$ and decreasing for $x > c$. This means that $G'(x) < 0$ for $x < c$ and $G'(x) < 0$ for $x > c$. This in turn implies that $G(x)$ is decreasing, so $G(x) > 0$ for $x < c$ but $G(x) < 0$ for $x > c$. On the other hand, if $f''(x)$ changes from negative to positive at $x = c$, then so does $G''(x)$ and $G'(x)$ is decreasing for $x < c$ and increasing for $x > c$. Thus, $G'(x) > 0$ for $x < c$ and $G'(x) > 0$ for $x > c$. This in turn implies that $G(x)$ is increasing, so $G(x) < 0$ for $x < c$ and $G(x) > 0$ for $x > c$. In either case, $G(x)$ changes sign at $x = c$, and the tangent line at $x = c$ crosses the graph of the function.

(b) Let $f(x) = \frac{x}{3x^2 + 1}$. Then

$$f'(x) = \frac{1 - 3x^2}{(3x^2 + 1)^2} \quad \text{and} \quad f''(x) = \frac{-18x(1 - x^2)}{(3x^2 + 1)^3}.$$

Therefore $f(x)$ has a point of inflection at $x = 0$ and at $x = \pm 1$. The figure below shows the graph of $y = f(x)$ and its tangent lines at each of the points of inflection. It is clear that each tangent line crosses the graph of $f(x)$ at the inflection point.



77. Let f be a polynomial of degree $n \geq 2$. Show that f has at least one point of inflection if n is odd. Then give an example to show that f need not have a point of inflection if n is even.

SOLUTION Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial of degree n . Then $f'(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \cdots + 2 a_2 x + a_1$ and $f''(x) = n(n-1) a_n x^{n-2} + (n-1)(n-2) a_{n-1} x^{n-3} + \cdots + 6 a_3 x + 2 a_2$. If $n \geq 3$ and is odd, then $n-2$ is also odd and $f''(x)$ is a polynomial of odd degree. Therefore $f''(x)$ must take on both positive and negative values. It follows that $f''(x)$ has at least one root c such that $f''(x)$ changes sign at c . The function $f(x)$ will then have a point of inflection at $x = c$. On the other hand, the functions $f(x) = x^2$, x^4 and x^8 are polynomials of even degree that do not have any points of inflection.

4.5 L'Hôpital's Rule

Preliminary Questions

1. What is wrong with applying L'Hôpital's Rule to $\lim_{x \rightarrow 0} \frac{x^2 - 2x}{3x - 2}$?

SOLUTION As $x \rightarrow 0$,

$$\frac{x^2 - 2x}{3x - 2}$$

is not of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, so L'Hôpital's Rule cannot be used.

2. Does L'Hôpital's Rule apply to $\lim_{x \rightarrow a} f(x)g(x)$ if $f(x)$ and $g(x)$ both approach ∞ as $x \rightarrow a$?

SOLUTION No. L'Hôpital's Rule only applies to limits of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

3. What is wrong with saying, "To apply L'Hôpital's Rule to $\lim_{x \rightarrow 0} \frac{\ln(1-x)}{x}$, use the Quotient Rule to differentiate $\frac{\ln(1-x)}{x}$ and then take the limit"?

SOLUTION To apply L'Hôpital's Rule, you do not apply the Quotient Rule. Rather, you differentiate the numerator and the denominator separately and then take the limit of the rational function obtained in this way.

4. What is wrong with applying L'Hôpital's Rule to $\lim_{x \rightarrow 0^+} x^{\frac{1}{x}}$?

SOLUTION The function $f(x) = x^{1/x}$ is not indeterminate as $x \rightarrow 0^+$. The function is of the form 0^∞ as $x \rightarrow 0^+$, which corresponds to a limit with the value 0.

5. What property of the function $f(x) = e^x$ allows us to say $\lim_{x \rightarrow a} e^{g(x)} = e^{\lim_{x \rightarrow a} g(x)}$?

SOLUTION The fact that the function $f(x) = e^x$ is continuous allows us to say $\lim_{x \rightarrow a} e^{g(x)} = e^{\lim_{x \rightarrow a} g(x)}$.

Exercises

In Exercises 1–10, use L'Hôpital's Rule to evaluate the limit, or state that L'Hôpital's Rule does not apply.

1. $\lim_{x \rightarrow 3} \frac{2x^2 - 5x - 3}{x - 4}$

SOLUTION Because the quotient is not indeterminate at $x = 3$,

$$\left. \frac{2x^2 - 5x - 3}{x - 4} \right|_{x=3} = \frac{18 - 15 - 3}{3 - 4} = \frac{0}{-1}$$

L'Hôpital's Rule does not apply.

3. $\lim_{x \rightarrow 4} \frac{x^3 - 64}{x^2 + 16}$

SOLUTION Because the quotient is not indeterminate at $x = 4$,

$$\left. \frac{x^3 - 64}{x^2 + 16} \right|_{x=4} = \frac{64 - 64}{16 + 16} = \frac{0}{32}$$

L'Hôpital's Rule does not apply.

5. $\lim_{x \rightarrow 9} \frac{x^{1/2} + x - 6}{x^{3/2} - 27}$

SOLUTION Because the quotient is not indeterminate at $x = 9$,

$$\left. \frac{x^{1/2} + x - 6}{x^{3/2} - 27} \right|_{x=9} = \frac{3 + 9 - 6}{27 - 27} = \frac{6}{0}$$

L'Hôpital's Rule does not apply.

7. $\lim_{x \rightarrow 0} \frac{\sin 4x}{x^2 + 3x + 1}$

SOLUTION Because the quotient is not indeterminate at $x = 0$,

$$\left. \frac{\sin 4x}{x^2 + 3x + 1} \right|_{x=0} = \frac{0}{0 + 0 + 1} = \frac{0}{1}$$

L'Hôpital's Rule does not apply.

9. $\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{\sin 5x}$

SOLUTION The functions $\cos 2x - 1$ and $\sin 5x$ are differentiable, but the quotient is indeterminate at $x = 0$,

$$\left. \frac{\cos 2x - 1}{\sin 5x} \right|_{x=0} = \frac{1 - 1}{0} = \frac{0}{0}$$

so L'Hôpital's Rule applies. We find

$$\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{\sin 5x} = \lim_{x \rightarrow 0} \frac{-2 \sin 2x}{5 \cos 5x} = \frac{0}{5} = 0$$

In Exercises 11–16, use L'Hôpital's Rule to evaluate the limit.

11. $\lim_{x \rightarrow \infty} \frac{9x + 4}{3 - 2x}$

SOLUTION As $x \rightarrow \infty$, the quotient $\frac{9x + 4}{3 - 2x}$ is of the form $\frac{\infty}{\infty}$, so L'Hôpital's Rule applies. We find

$$\lim_{x \rightarrow \infty} \frac{9x + 4}{3 - 2x} = \lim_{x \rightarrow \infty} \frac{9}{-2} = -\frac{9}{2}$$

13. $\lim_{x \rightarrow \infty} \frac{\ln x}{x^{1/2}}$

SOLUTION As $x \rightarrow \infty$, the quotient $\frac{\ln x}{x^{1/2}}$ is of the form $\frac{\infty}{\infty}$, so L'Hôpital's Rule applies. We find

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^{1/2}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2}x^{-1/2}} = \lim_{x \rightarrow \infty} \frac{1}{2x^{1/2}} = 0$$

15. $\lim_{x \rightarrow -\infty} \frac{\ln(x^4 + 1)}{x}$

SOLUTION As $x \rightarrow \infty$, the quotient $\frac{\ln(x^4 + 1)}{x}$ is of the form $\frac{\infty}{\infty}$, so L'Hôpital's Rule applies. Here, we use L'Hôpital's Rule twice to find

$$\lim_{x \rightarrow \infty} \frac{\ln(x^4 + 1)}{x} = \lim_{x \rightarrow \infty} \frac{\frac{4x^3}{x^4 + 1}}{1} = \lim_{x \rightarrow \infty} \frac{12x^2}{4x^3} = \lim_{x \rightarrow \infty} \frac{3}{x} = 0$$

In Exercises 17–54, evaluate the limit.

$$17. \lim_{x \rightarrow 1} \frac{\sqrt{8+x} - 3x^{1/3}}{x^2 - 3x + 2}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow 1} \frac{\sqrt{8+x} - 3x^{1/3}}{x^2 - 3x + 2} = \lim_{x \rightarrow 1} \frac{\frac{1}{2}(8+x)^{-1/2} - x^{-2/3}}{2x - 3} = \frac{\frac{1}{6} - 1}{-1} = \frac{5}{6}$$

$$19. \lim_{x \rightarrow -\infty} \frac{3x - 2}{1 - 5x}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow -\infty} \frac{3x - 2}{1 - 5x} = \lim_{x \rightarrow -\infty} \frac{3}{-5} = -\frac{3}{5}$$

$$21. \lim_{x \rightarrow -\infty} \frac{7x^2 + 4x}{9 - 3x^2}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow -\infty} \frac{7x^2 + 4x}{9 - 3x^2} = \lim_{x \rightarrow -\infty} \frac{14x + 4}{-6x} = \lim_{x \rightarrow -\infty} \frac{14}{-6} = -\frac{7}{3}$$

$$23. \lim_{x \rightarrow 1} \frac{(1 + 3x)^{1/2} - 2}{(1 + 7x)^{1/3} - 2}$$

SOLUTION Apply L'Hôpital's Rule once:

$$\lim_{x \rightarrow 1} \frac{(1 + 3x)^{1/2} - 2}{(1 + 7x)^{1/3} - 2} = \lim_{x \rightarrow 1} \frac{\frac{3}{2}(1 + 3x)^{-1/2}}{\frac{7}{3}(1 + 7x)^{-2/3}} = \frac{(\frac{3}{2})^{\frac{1}{2}}}{(\frac{7}{3})(\frac{1}{4})} = \frac{9}{7}$$

$$25. \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 7x}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 7x} = \lim_{x \rightarrow 0} \frac{2 \cos 2x}{7 \cos 7x} = \frac{2}{7}$$

$$27. \lim_{x \rightarrow 0} \frac{\tan x}{x}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sec^2 x}{1} = 1$$

$$29. \lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x - \sin x}$$

SOLUTION

$$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{x \sin x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{\sin x + x \cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{\cos x + \cos x - x \sin x}{\cos x} = 2$$

$$31. \lim_{x \rightarrow 0} \frac{\cos(x + \frac{\pi}{2})}{\sin x}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow 0} \frac{\cos(x + \frac{\pi}{2})}{\sin x} = \lim_{x \rightarrow 0} \frac{-\sin(x + \frac{\pi}{2})}{\cos x} = -1$$

$$33. \lim_{x \rightarrow \pi/2} \frac{\cos x}{\sin(2x)}$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow \pi/2} \frac{\cos x}{\sin(2x)} = \lim_{x \rightarrow \pi/2} \frac{-\sin x}{2 \cos(2x)} = \frac{1}{2}$$

$$35. \lim_{x \rightarrow \pi/2} (\sec x - \tan x)$$

SOLUTION

$$\lim_{x \rightarrow \frac{\pi}{2}} (\sec x - \tan x) = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1 - \sin x}{\cos x} \right) = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{-\cos x}{-\sin x} \right) = 0$$

$$37. \lim_{x \rightarrow 1} \tan\left(\frac{\pi x}{2}\right) \ln x$$

$$\text{SOLUTION} \quad \lim_{x \rightarrow 1} \tan\left(\frac{\pi x}{2}\right) \ln x = \lim_{x \rightarrow 1} \frac{\ln x}{\cot(\frac{\pi x}{2})} = \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-\frac{\pi}{2} \csc^2(\frac{\pi x}{2})} = \lim_{x \rightarrow 1} \frac{-2}{\pi x} \sin^2\left(\frac{\pi}{2} x\right) = -\frac{2}{\pi}$$

$$39. \lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$$

SOLUTION $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x} = \lim_{x \rightarrow 0} \frac{e^x}{\cos x} = 1$

41. $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - x}{x^2}$

SOLUTION $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{2e^{2x} - 1}{2x}$ which does not exist.

43. $\lim_{t \rightarrow 0^+} (\sin t)(\ln t)$

SOLUTION

$$\lim_{t \rightarrow 0^+} (\sin t)(\ln t) = \lim_{t \rightarrow 0^+} \frac{\ln t}{\csc t} = \lim_{t \rightarrow 0^+} \frac{\frac{1}{t}}{-\csc t \cot t} = \lim_{t \rightarrow 0^+} \frac{-\sin^2 t}{t \cos t} = \lim_{t \rightarrow 0^+} \frac{-2 \sin t \cos t}{\cos t - t \sin t} = 0$$

45. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} \quad (a > 0)$

SOLUTION $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{x \rightarrow 0} \frac{\ln a \cdot a^x}{1} = \ln a$

47. $\lim_{x \rightarrow 1} (1 + \ln x)^{1/(x-1)}$

SOLUTION Use the change-of-base formula to write

$$(1 + \ln x)^{1/x} = e^{\frac{1}{x-1} \ln(1 + \ln x)}$$

Now,

$$\lim_{x \rightarrow 1} \frac{1}{x-1} \ln(1 + \ln x) = \lim_{x \rightarrow 1} \frac{\ln(1 + \ln x)}{x-1} = \lim_{x \rightarrow 1} \frac{\frac{1}{1+\ln x} \cdot \frac{1}{x}}{1} = \lim_{x \rightarrow 1} \frac{1}{x(1 + \ln x)} = 1$$

Hence,

$$\lim_{x \rightarrow 1} (1 + \ln x)^{1/(x-1)} = \lim_{x \rightarrow 1} e^{\frac{1}{x-1} \ln(1 + \ln x)} = e^{\lim_{x \rightarrow 1} \frac{1 + \ln x}{x-1}} = e^1 = e$$

49. $\lim_{x \rightarrow 0} (\cos x)^{3/x^2}$

SOLUTION Use the change-of-base formula to write

$$(\cos x)^{3/x^2} = e^{\frac{3}{x^2} \ln(\cos x)}$$

Now,

$$\lim_{x \rightarrow 0} \frac{3}{x^2} \ln(\cos x) = \lim_{x \rightarrow 0} \frac{3 \ln \cos x}{x^2} = \lim_{x \rightarrow 0} -\frac{3 \tan x}{2x} = \lim_{x \rightarrow 0} -\frac{3 \sec^2 x}{2} = -\frac{3}{2}.$$

Hence,

$$\lim_{x \rightarrow 0} (\cos x)^{3/x^2} = \lim_{x \rightarrow 0} e^{\frac{3}{x^2} \ln(\cos x)} = e^{\lim_{x \rightarrow 0} \frac{3 \ln(\cos x)}{x^2}} = e^{-3/2}$$

51. $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$

SOLUTION $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{1-x^2}}}{1} = 1$

53. $\lim_{x \rightarrow 1} \frac{\tan^{-1} x - \frac{\pi}{4}}{\tan \frac{\pi}{4} x - 1}$

SOLUTION $\lim_{x \rightarrow 1} \frac{\tan^{-1} x - \frac{\pi}{4}}{\tan(\pi x/4) - 1} = \lim_{x \rightarrow 1} \frac{\frac{1}{1+x^2}}{\frac{\pi}{4} \sec^2(\pi x/4)} = \frac{1}{\frac{\pi}{2}} = \frac{1}{\pi}$

55. Evaluate $\lim_{x \rightarrow \pi/2} \frac{\cos mx}{\cos nx}$, where $m, n \neq 0$ are integers.

SOLUTION Suppose m and n are even. Then there exist integers k and l such that $m = 2k$ and $n = 2l$ and

$$\lim_{x \rightarrow \pi/2} \frac{\cos mx}{\cos nx} = \frac{\cos k\pi}{\cos l\pi} = (-1)^{k-l}$$

Now, suppose m is even and n is odd. Then

$$\lim_{x \rightarrow \pi/2} \frac{\cos mx}{\cos nx}$$

does not exist (from one side the limit tends toward $-\infty$, while from the other side the limit tends toward $+\infty$). Third, suppose m is odd and n is even. Then

$$\lim_{x \rightarrow \pi/2} \frac{\cos mx}{\cos nx} = 0$$

Finally, suppose m and n are odd. This is the only case when the limit is indeterminate. Then there exist integers k and l such that $m = 2k + 1$, $n = 2l + 1$ and, by L'Hôpital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{\cos mx}{\cos nx} = \lim_{x \rightarrow \pi/2} \frac{-m \sin mx}{-n \sin nx} = (-1)^{k-l} \frac{m}{n}$$

To summarize,

$$\lim_{x \rightarrow \pi/2} \frac{\cos mx}{\cos nx} = \begin{cases} (-1)^{(m-n)/2}, & m, n \text{ even} \\ \text{does not exist,} & m \text{ even, } n \text{ odd} \\ & m \text{ odd, } n \text{ even} \\ (-1)^{(m-n)/2} \frac{m}{n}, & m, n \text{ odd} \end{cases}$$

57. Evaluate each of the following limits.

(a) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{x^2}$

(b) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^x$

SOLUTION

(a) Use the change-of-base formula to write

$$\left(1 + \frac{1}{x}\right)^{x^2} = e^{x^2 \ln(1 + \frac{1}{x})}$$

Now,

$$\lim_{x \rightarrow \infty} x^2 \ln\left(1 + \frac{1}{x}\right) = \lim_{x \rightarrow \infty} \frac{\ln(1 + \frac{1}{x})}{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{1 + \frac{1}{x}} \cdot -\frac{1}{x^2}}{-\frac{2}{x^3}} = \lim_{x \rightarrow \infty} \frac{x}{2(1 + \frac{1}{x})} = \infty$$

Hence,

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{x^2} = \lim_{x \rightarrow \infty} e^{x^2 \ln(1 + \frac{1}{x})} = e^{\lim_{x \rightarrow \infty} x^2 \ln(1 + \frac{1}{x})} = \infty$$

(b) Use the change-of-base formula to write

$$\left(1 + \frac{1}{x^2}\right)^x = e^{x \ln(1 + \frac{1}{x^2})}$$

Now,

$$\lim_{x \rightarrow \infty} x \ln\left(1 + \frac{1}{x^2}\right) = \lim_{x \rightarrow \infty} \frac{\ln(1 + \frac{1}{x^2})}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{1 + \frac{1}{x^2}} \cdot -\frac{2}{x^3}}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{2}{x + \frac{1}{x}} = 0$$

Hence,

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^x = \lim_{x \rightarrow \infty} e^{x \ln(1 + \frac{1}{x^2})} = e^{\lim_{x \rightarrow \infty} x \ln(1 + \frac{1}{x^2})} = e^0 = 1$$

In Exercises 59–60, a projectile is launched straight up in the air and is acted on by air resistance and gravity as in Example 5. The function M gives the maximum height that the projectile attains as a function of the air resistance parameter k . In each case, determine the maximum height as we let the air resistance term go to zero; that is, determine $\lim_{k \rightarrow 0} M(k)$.

59. A ball with a mass of 1 kilogram is launched upward with an initial velocity of 60 m/s, and

$$M(k) = \frac{60k - 9.8 \ln\left(\frac{300k}{49} + 1\right)}{k^2}$$

(Compare with Exercises 37 in Section 2.5 and 29 in Section 3.4.)

SOLUTION The quotient

$$M(k) = \frac{60k - 9.8 \ln\left(\frac{300k}{49} + 1\right)}{k^2}$$

has the indeterminate form $0/0$ as $k \rightarrow 0$. To evaluate the limit, we need to use L'Hôpital's Rule twice. We find

$$\lim_{k \rightarrow 0} \frac{60k - 9.8 \ln\left(\frac{300k}{49} + 1\right)}{k^2} = \lim_{k \rightarrow 0} \frac{60 - \frac{60}{\frac{300k}{49} + 1}}{2k} = \lim_{k \rightarrow 0} \frac{\frac{18000/49}{\left(\frac{300k}{49} + 1\right)^2}}{2} = \frac{9000}{49} \approx 183.67 \text{ m}$$

Thus, the maximum height as air resistance goes to zero is approximately 183.67 m. This result is in agreement with the results from Exercises 37 in Section 2.5 and 29 in Section 3.4.

61. In each case, show that the form is indeterminate by showing that if $\lim_{x \rightarrow c} f(x)^{g(x)}$ has the form, then the limit in the exponent in $e^{\lim_{x \rightarrow c} g(x) \ln f(x)}$ has a known indeterminate form.

(a) 1^∞

(b) ∞^0

SOLUTION

(a) Suppose $\lim_{x \rightarrow c} f(x)^{g(x)}$ has the form 1^∞ . Then $\lim_{x \rightarrow c} f(x) = 1$ and $\lim_{x \rightarrow c} g(x) = \infty$. It follows that $\lim_{x \rightarrow c} \ln f(x) = 0$, so $\lim_{x \rightarrow c} g(x) \ln f(x)$ has the indeterminate form $0 \cdot \infty$.

(b) Suppose $\lim_{x \rightarrow c} f(x)^{g(x)}$ has the form ∞^0 . Then $\lim_{x \rightarrow c} f(x) = \infty$ and $\lim_{x \rightarrow c} g(x) = 0$. It follows that $\lim_{x \rightarrow c} \ln f(x) = \infty$, so $\lim_{x \rightarrow c} g(x) \ln f(x)$ has the indeterminate form $0 \cdot \infty$.

63. Let $f(x) = x^{1/x}$ for $x > 0$.

(a) Calculate $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

(b) Find the maximum value of f and determine the intervals on which f is increasing or decreasing.

SOLUTION

(a) Let $f(x) = x^{1/x}$. Note that $\lim_{x \rightarrow 0^+} x^{1/x}$ is not indeterminate. As $x \rightarrow 0^+$, the base of the function tends toward 0 and the exponent tends toward $+\infty$. Both of these factors force $x^{1/x}$ toward 0. Thus, $\lim_{x \rightarrow 0^+} f(x) = 0$. On the other hand, $\lim_{x \rightarrow \infty} f(x)$ is indeterminate. To calculate this limit, use the change-of-base formula to write

$$x^{1/x} = e^{\frac{1}{x} \ln x}$$

Now,

$$\lim_{x \rightarrow \infty} \frac{1}{x} \ln x = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$$

Hence,

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} e^{\frac{1}{x} \ln x} = e^{\lim_{x \rightarrow \infty} \frac{1}{x} \ln x} = e^0 = 1$$

(b) Again, let $f(x) = x^{1/x}$, so that $\ln f(x) = \frac{1}{x} \ln x$. To find the derivative f' , we apply the derivative to both sides:

$$\begin{aligned}\frac{d}{dx} \ln f(x) &= \frac{d}{dx} \left(\frac{1}{x} \ln x \right) \\ \frac{1}{f(x)} f'(x) &= -\frac{\ln x}{x^2} + \frac{1}{x^2} \\ f'(x) &= f(x) \left(-\frac{\ln x}{x^2} + \frac{1}{x^2} \right) = \frac{x^{1/x}}{x^2} (1 - \ln x)\end{aligned}$$

Thus, f is increasing for $0 < x < e$, is decreasing for $x > e$, and has a maximum at $x = e$. The maximum value is $f(e) = e^{1/e} \approx 1.444668$.

65. Determine whether $f \ll g$ or $g \ll f$ (or neither) for the functions $f(x) = \log_{10} x$ and $g(x) = \ln x$.

SOLUTION Because

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\log_{10} x}{\ln x} = \lim_{x \rightarrow \infty} \frac{\frac{\ln x}{\ln 10}}{\ln x} = \frac{1}{\ln 10}$$

neither $g \ll f$ nor $f \ll g$ is satisfied.

67. Just as exponential functions are distinguished by their rapid rate of increase, the logarithm functions grow particularly slowly. Show that $\ln x \ll x^a$ for all $a > 0$.

SOLUTION Using L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^a} = \lim_{x \rightarrow \infty} \frac{x^{-1}}{ax^{a-1}} = \lim_{x \rightarrow \infty} \frac{1}{a} x^{-a} = 0$$

hence, $\ln x \ll x^a$.

69. Determine whether $\sqrt{x} \ll e^{\sqrt{\ln x}}$ or $e^{\sqrt{\ln x}} \ll \sqrt{x}$. *Hint:* Use the substitution $u = \ln x$ instead of L'Hôpital's Rule.

SOLUTION Let $u = \ln x$, then $x = e^u$, and as $x \rightarrow \infty$, $u \rightarrow \infty$. So

$$\lim_{x \rightarrow \infty} \frac{e^{\sqrt{\ln x}}}{\sqrt{x}} = \lim_{u \rightarrow \infty} \frac{e^{\sqrt{u}}}{e^{u/2}} = \lim_{u \rightarrow \infty} e^{\sqrt{u} - \frac{u}{2}}$$

We need to examine $\lim_{u \rightarrow \infty} (\sqrt{u} - \frac{u}{2})$. Since

$$\lim_{u \rightarrow \infty} \frac{u/2}{\sqrt{u}} = \lim_{u \rightarrow \infty} \frac{\sqrt{u}}{2} = \infty$$

$\sqrt{u} \ll u/2$ and $\lim_{u \rightarrow \infty} \left(\sqrt{u} - \frac{u}{2} \right) = -\infty$. Thus

$$\lim_{u \rightarrow \infty} e^{\sqrt{u} - \frac{u}{2}} = e^{-\infty} = 0 \quad \text{so} \quad \lim_{x \rightarrow \infty} \frac{e^{\sqrt{\ln x}}}{\sqrt{x}} = 0$$

and $e^{\sqrt{\ln x}} \ll \sqrt{x}$.

71. Assumptions Matter Suppose $f(x) = x(2 + \sin x)$ and let $g(x) = x^2 + 1$.

(a) Show directly that $\lim_{x \rightarrow \infty} f(x)/g(x) = 0$.

(b) Show that $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$, but $\lim_{x \rightarrow \infty} f'(x)/g'(x)$ does not exist.

Do (a) and (b) contradict L'Hôpital's Rule? Explain.

SOLUTION

(a) $1 \leq 2 + \sin x \leq 3$, so

$$\frac{x}{x^2 + 1} \leq \frac{x(2 + \sin x)}{x^2 + 1} \leq \frac{3x}{x^2 + 1}$$

Since,

$$\lim_{x \rightarrow \infty} \frac{x}{x^2 + 1} = \lim_{x \rightarrow \infty} \frac{3x}{x^2 + 1} = 0$$

it follows by the Squeeze Theorem that

$$\lim_{x \rightarrow \infty} \frac{x(2 + \sin x)}{x^2 + 1} = 0$$

(b) $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x(2 + \sin x) \geq \lim_{x \rightarrow \infty} x = \infty$ and $\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} (x^2 + 1) = \infty$, but

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} \frac{x(\cos x) + (2 + \sin x)}{2x}$$

does not exist since $\cos x$ oscillates. This does not violate L'Hôpital's Rule since the theorem clearly states

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

“provided the limit on the right exists.”

73. Let $G(b) = \lim_{x \rightarrow \infty} (1 + b^x)^{1/x}$.

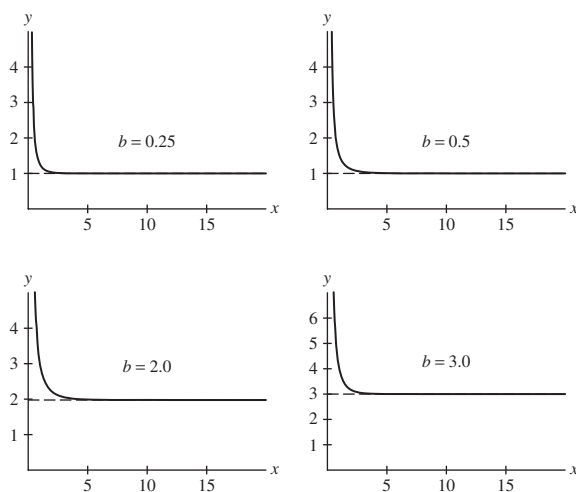
(a) Use the result of Exercise 72 to evaluate $G(b)$ for all $b > 0$.

(b)  Verify your result graphically by plotting $y = (1 + b^x)^{1/x}$ together with the horizontal line $y = G(b)$ for the values $b = 0.25, 0.5, 2, 3$.

SOLUTION

(a) Using Exercise 72, we see that $G(b) = e^{H(b)}$. Thus, $G(b) = 1$ if $0 \leq b \leq 1$ and $G(b) = b$ if $b > 1$.

(b)



In Exercises 75–77, let

$$f(x) = \begin{cases} e^{-1/x^2} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

These exercises show that f has an unusual property: All of its derivatives at $x = 0$ exist and are equal to zero.

75. Show that $\lim_{x \rightarrow 0} \frac{f(x)}{x^k} = 0$ for all k . *Hint:* Let $t = x^{-1}$ and apply the result of Exercise 74.

SOLUTION $\lim_{x \rightarrow 0} \frac{f(x)}{x^k} = \lim_{x \rightarrow 0} \frac{1}{x^k e^{1/x^2}}$. Let $t = 1/x$. As $x \rightarrow 0$, $t \rightarrow \infty$. Thus,

$$\lim_{x \rightarrow 0} \frac{1}{x^k e^{1/x^2}} = \lim_{t \rightarrow \infty} \frac{t^k}{e^{t^2}} = 0$$

by Exercise 74.

77. Show that for $k \geq 1$ and $x \neq 0$,

$$f^{(k)}(x) = \frac{P(x)e^{-1/x^2}}{x^r}$$

for some polynomial $P(x)$ and some exponent $r \geq 1$. Use the result of Exercise 75 to show that $f^{(k)}(0)$ exists and is equal to zero for all $k \geq 1$.

SOLUTION For $x \neq 0$, $f'(x) = e^{-1/x^2} \left(\frac{2}{x^3} \right)$. Here $P(x) = 2$ and $r = 3$. Assume $f^{(k)}(x) = \frac{P(x)e^{-1/x^2}}{x^r}$. Then

$$f^{(k+1)}(x) = e^{-1/x^2} \left(\frac{x^3 P'(x) + (2 - rx^2)P(x)}{x^{r+3}} \right)$$

which is of the form desired.

Moreover, from Exercise 76, $f'(0) = 0$. Suppose $f^{(k)}(0) = 0$. Then

$$f^{(k+1)}(0) = \lim_{x \rightarrow 0} \frac{f^{(k)}(x) - f^{(k)}(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{P(x)e^{-1/x^2}}{x^{r+1}} = P(0) \lim_{x \rightarrow 0} \frac{f(x)}{x^{r+1}} = 0$$

Further Insights and Challenges

79. The Second Derivative Test for critical points fails if $f''(c) = 0$. This exercise develops a **Higher Derivative Test** based on the sign of the first nonzero derivative. Suppose that

$$f'(c) = f''(c) = \cdots = f^{(n-1)}(c) = 0, \quad \text{but} \quad f^{(n)}(c) \neq 0$$

(a) Show, by applying L'Hôpital's Rule n times, that

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x - c)^n} = \frac{1}{n!} f^{(n)}(c)$$

where $n! = n(n-1)(n-2) \cdots (2)(1)$.

(b) Use (a) to show that if n is even, then $f(c)$ is a local minimum if $f^{(n)}(c) > 0$ and is a local maximum if $f^{(n)}(c) < 0$. *Hint:* If n is even, then $(x - c)^n > 0$ for $x \neq c$, so $f(x) - f(c)$ must be positive for x near c if $f^{(n)}(c) > 0$.

(c) Use (a) to show that if n is odd, then $f(c)$ is neither a local minimum nor a local maximum.

SOLUTION

(a) Repeated application of L'Hôpital's rule yields

$$\begin{aligned} \lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x - c)^n} &= \lim_{x \rightarrow c} \frac{f'(x)}{n(x - c)^{n-1}} \\ &= \lim_{x \rightarrow c} \frac{f''(x)}{n(n-1)(x - c)^{n-2}} \\ &= \lim_{x \rightarrow c} \frac{f'''(x)}{n(n-1)(n-2)(x - c)^{n-3}} \\ &= \cdots \\ &= \frac{1}{n!} f^{(n)}(c) \end{aligned}$$

(b) Suppose n is even. Then $(x - c)^n > 0$ for all $x \neq c$. If $f^{(n)}(c) > 0$, it follows that $f(x) - f(c)$ must be positive for x near c . In other words, $f(x) > f(c)$ for x near c and $f(c)$ is a local minimum. On the other hand, if $f^{(n)}(c) < 0$, it follows that $f(x) - f(c)$ must be negative for x near c . In other words, $f(x) < f(c)$ for x near c and $f(c)$ is a local maximum.

(c) If n is odd, then $(x - c)^n > 0$ for $x > c$ but $(x - c)^n < 0$ for $x < c$. If $f^{(n)}(c) > 0$, it follows that $f(x) - f(c)$ must be positive for x near c and $x > c$ but is negative for x near c and $x < c$. In other words, $f(x) > f(c)$ for x near c and $x > c$ but $f(x) < f(c)$ for x near c and $x < c$. Thus, $f(c)$ is neither a local minimum nor a local maximum. We obtain a similar result if $f^{(n)}(c) < 0$.

81.  We expended a lot of effort to evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ in Chapter 2. Show that we could have evaluated it easily using L'Hôpital's Rule. Then explain why this method would involve *circular reasoning*.

SOLUTION $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$. To use L'Hôpital's Rule to evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$, we must know that the derivative of $\sin x$ is $\cos x$, but to determine the derivative of $\sin x$, we must be able to evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

83. Patience Required Use L'Hôpital's Rule to evaluate and check your answers numerically:

$$(a) \lim_{x \rightarrow 0^+} \left(\frac{\sin x}{x} \right)^{1/x^2} \qquad (b) \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right)$$

SOLUTION

(a) Use the change-of-base formula to write

$$\left(\frac{\sin x}{x} \right)^{1/x^2} = e^{\frac{1}{x^2} \ln \left(\frac{\sin x}{x} \right)}$$

Now, repeatedly using L'Hôpital's Rule, we find

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{1}{x^2} \ln \left(\frac{\sin x}{x} \right) &= \lim_{x \rightarrow 0^+} \frac{\ln(\sin x) - \ln x}{x^2} = \lim_{x \rightarrow 0^+} \frac{\cot x - x^{-1}}{2x} = \lim_{x \rightarrow 0^+} \frac{x \cos x - \sin x}{2x^2 \sin x} = \lim_{x \rightarrow 0^+} \frac{-x \sin x}{2x^2 \cos x + 4x \sin x} \\ &= \lim_{x \rightarrow 0^+} \frac{-x \cos x - \sin x}{8x \cos x + 4 \sin x - 2x^2 \sin x} = \lim_{x \rightarrow 0^+} \frac{-2 \cos x + x \sin x}{12 \cos x - 2x^2 \cos x - 12x \sin x} \\ &= -\frac{2}{12} = -\frac{1}{6} \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow 0^+} \frac{\sin x}{x}^{1/x^2} = \lim_{x \rightarrow 0^+} e^{\frac{1}{x^2} \ln \frac{\sin x}{x}} = e^{\lim_{x \rightarrow 0^+} \frac{1}{x^2} \ln \frac{\sin x}{x}} = e^{-1/6}$$

Numerically we find:

x	1	0.1	0.01
$\frac{\sin x}{x}^{1/x^2}$	0.841471	0.846435	0.846481

Note that $e^{-1/6} \approx 0.846481724$.

(b) Repeatedly using L'Hôpital's Rule and simplifying, we find

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right) &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x}{x^2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{2x - 2 \sin x \cos x}{x^2 (2 \sin x \cos x) + 2x \sin^2 x} = \lim_{x \rightarrow 0} \frac{2x - 2 \sin 2x}{x^2 \sin 2x + 2x \sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{2 - 2 \cos 2x}{2x^2 \cos 2x + 2x \sin 2x + 4x \sin x \cos x + 2 \sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{2 - 2 \cos 2x}{2x^2 \cos 2x + 4x \sin 2x + 2 \sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{4 \sin 2x}{-4x^2 \sin 2x + 4x \cos 2x + 8x \cos 2x + 4 \sin 2x + 4 \sin x \cos x} \\ &= \lim_{x \rightarrow 0} \frac{4 \sin 2x}{(6 - 4x^2) \sin 2x + 12x \cos 2x} \\ &= \lim_{x \rightarrow 0} \frac{8 \cos 2x}{(12 - 8x^2) \cos 2x - 8x \sin 2x + 12 \cos 2x - 24x \sin 2x} = \frac{1}{3}. \end{aligned}$$

Numerically we find:

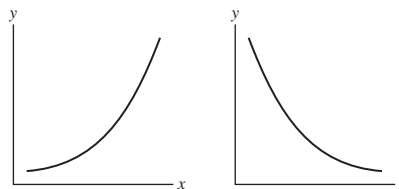
x	1	0.1	0.01
$\frac{1}{\sin^2 x} - \frac{1}{x^2}$	0.412283	0.334001	0.333340

4.6 Analyzing and Sketching Graphs of Functions

Preliminary Questions

1. Sketch an arc where f' and f'' have the sign combination $++$. Do the same for $-+$.

SOLUTION An arc with the sign combination $++$ (increasing, concave up) is shown below at the left. An arc with the sign combination $-+$ (decreasing, concave up) is shown below at the right.



2. If the sign combination of f' and f'' changes from $++$ to $-+$ at $x = c$, then (choose the correct answer)
- (a) $f(c)$ is a local min. (b) $f(c)$ is a local max.
- (c) $(c, f(c))$ is a point of inflection.

SOLUTION Because the sign of the second derivative changes at $x = c$, the correct response is (c): $(c, f(c))$ is a point of inflection.

3. The second derivative of the function $f(x) = (x - 4)^{-1}$ is $f''(x) = 2(x - 4)^{-3}$. Although $f''(x)$ changes sign at $x = 4$, f does not have a point of inflection at $x = 4$. Why not?

SOLUTION The function f does not have a point of inflection at $x = 4$ because $x = 4$ is not in the domain of f .

Exercises

1. Determine the sign combinations of f' and f'' for each interval A–G in Figure 15.

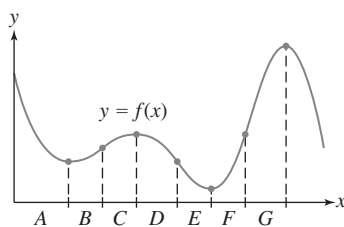


FIGURE 15

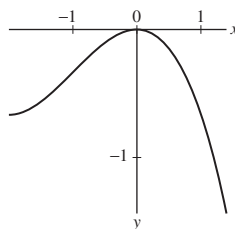
SOLUTION

- In A, f is decreasing and concave up, so $f' < 0$ and $f'' > 0$.
- In B, f is increasing and concave up, so $f' > 0$ and $f'' > 0$.
- In C, f is increasing and concave down, so $f' > 0$ and $f'' < 0$.
- In D, f is decreasing and concave down, so $f' < 0$ and $f'' < 0$.
- In E, f is decreasing and concave up, so $f' < 0$ and $f'' > 0$.
- In F, f is increasing and concave up, so $f' > 0$ and $f'' > 0$.
- In G, f is increasing and concave down, so $f' > 0$ and $f'' < 0$.

In Exercises 3–6, draw the graph of a function for which f' and f'' take on the given sign combinations in order.

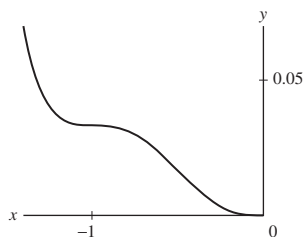
3. ++, +-, --

SOLUTION This function changes from concave up to concave down at $x = -1$ and from increasing to decreasing at $x = 0$.



5. -+, --, -+

SOLUTION The function is decreasing everywhere and changes from concave up to concave down at $x = -1$ and from concave down to concave up at $x = -\frac{1}{2}$.



7. Sketch the graph of a function that could have the graphs of f' and f'' appearing in Figure 17.

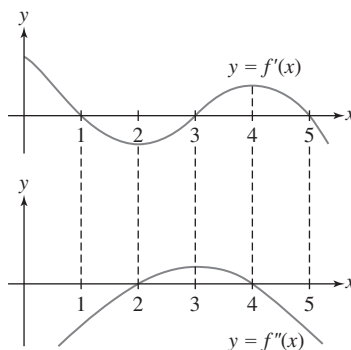
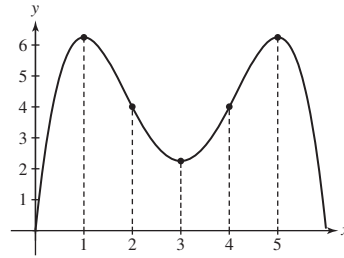


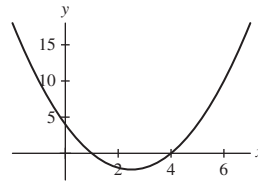
FIGURE 17

SOLUTION Based on the graphs of f' and f'' appearing in Figure 17, the graph of f needs to be increasing for $x < 1$ and $3 < x < 5$, decreasing for $1 < x < 3$ and $x > 5$, concave up for $2 < x < 4$, and concave down for $x < 2$ and $x > 4$. One possible graph for f is:



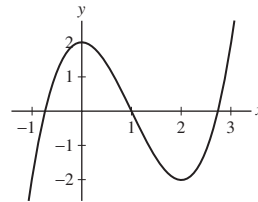
9. Investigate the behavior and sketch the graph of $y = x^2 - 5x + 4$.

SOLUTION Let $f(x) = x^2 - 5x + 4$. Then $f'(x) = 2x - 5$ and $f''(x) = 2$. Hence f is decreasing for $x < 5/2$, is increasing for $x > 5/2$, has a local minimum at $x = 5/2$ and is concave up everywhere.



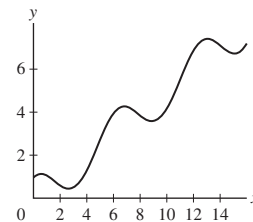
11. Investigate the behavior and sketch the graph of $f(x) = x^3 - 3x^2 + 2$. Include the zeros of f , which are $x = 1$ and $1 \pm \sqrt{3}$ (approximately $-0.73, 2.73$).

SOLUTION Let $f(x) = x^3 - 3x^2 + 2$. Then $f'(x) = 3x^2 - 6x = 3x(x - 2) = 0$ yields $x = 0, 2$ and $f''(x) = 6x - 6$. Thus f is concave down for $x < 1$, is concave up for $x > 1$, has an inflection point at $x = 1$, is increasing for $x < 0$ and for $x > 2$, is decreasing for $0 < x < 2$, has a local maximum at $x = 0$, and has a local minimum at $x = 2$.



13. Extend the sketch of the graph of $f(x) = \cos x + \frac{1}{2}x$ in Example 4 to the interval $[0, 5\pi]$.

SOLUTION Let $f(x) = \cos x + \frac{1}{2}x$. Then $f'(x) = -\sin x + \frac{1}{2} = 0$ yields critical points at $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}$, and $\frac{29\pi}{6}$. Moreover, $f''(x) = -\cos x$ so there are points of inflection at $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$, and $\frac{9\pi}{2}$.



In Exercises 15–36, find the transition points, intervals of increase/decrease, concavity, and asymptotic behavior. Then sketch the graph, with this information indicated.

15. $y = x^3 + 24x^2$

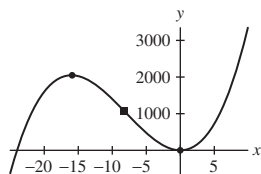
SOLUTION Let $f(x) = x^3 + 24x^2$. Then $f'(x) = 3x^2 + 48x = 3x(x + 16)$ and $f''(x) = 6x + 48$. This shows that f has critical points at $x = 0$ and $x = -16$ and a candidate for an inflection point at $x = -8$.

Interval	$(-\infty, -16)$	$(-16, -8)$	$(-8, 0)$	$(0, \infty)$
Signs of f' and f''	$+-$	$--$	$-+$	$++$

Thus, there is a local maximum at $x = -16$, a local minimum at $x = 0$, and an inflection point at $x = -8$. Moreover,

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x) = \infty$$

Here is a graph of f with these transition points highlighted as in the graphs in the textbook.

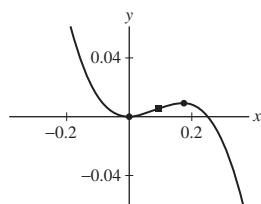


17. $y = x^2 - 4x^3$

SOLUTION Let $f(x) = x^2 - 4x^3$. Then $f'(x) = 2x - 12x^2 = 2x(1 - 6x)$ and $f''(x) = 2 - 24x$. Critical points are at $x = 0$ and $x = \frac{1}{6}$, and the sole candidate point of inflection is at $x = \frac{1}{12}$.

Interval	$(-\infty, 0)$	$(0, \frac{1}{12})$	$(\frac{1}{12}, \frac{1}{6})$	$(\frac{1}{6}, \infty)$
Signs of f' and f''	$-+$	$++$	$+-$	$--$

Thus, $f(0)$ is a local minimum, $f(\frac{1}{6})$ is a local maximum, and there is a point of inflection at $x = \frac{1}{12}$. Here is the graph of f with transition points highlighted as in the textbook:

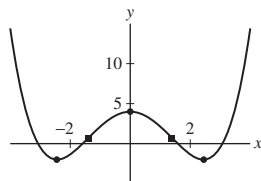


19. $y = 4 - 2x^2 + \frac{1}{6}x^4$

SOLUTION Let $f(x) = \frac{1}{6}x^4 - 2x^2 + 4$. Then $f'(x) = \frac{2}{3}x^3 - 4x = \frac{2}{3}x(x^2 - 6)$ and $f''(x) = 2x^2 - 4$. This shows that f has critical points at $x = 0$ and $x = \pm\sqrt{6}$ and has candidates for points of inflection at $x = \pm\sqrt{2}$.

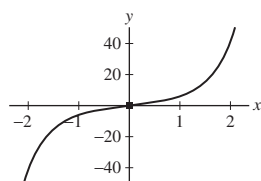
Interval	$(-\infty, -\sqrt{6})$	$(-\sqrt{6}, -\sqrt{2})$	$(-\sqrt{2}, 0)$	$(0, \sqrt{2})$	$(\sqrt{2}, \sqrt{6})$	$(\sqrt{6}, \infty)$
Signs of f' and f''	$-+$	$++$	$+-$	$--$	$-+$	$++$

Thus, f has local minima at $x = \pm\sqrt{6}$, a local maximum at $x = 0$, and inflection points at $x = \pm\sqrt{2}$. Here is a graph of f with transition points highlighted.



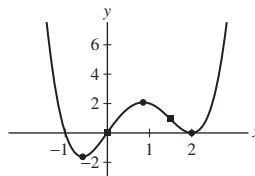
21. $y = x^5 + 5x$

SOLUTION Let $f(x) = x^5 + 5x$. Then $f'(x) = 5x^4 + 5 = 5(x^4 + 1)$ and $f''(x) = 20x^3$. $f'(x) > 0$ for all x , so the graph has no critical points and is always increasing. $f''(x) = 0$ at $x = 0$. Sign analyses reveal that $f''(x)$ changes from negative to positive at $x = 0$, so that the graph of $f(x)$ has an inflection point at $(0, 0)$. Here is a graph of f with transition points highlighted.



23. $y = x^4 - 3x^3 + 4x$

SOLUTION Let $f(x) = x^4 - 3x^3 + 4x$. Then $f'(x) = 4x^3 - 9x^2 + 4 = (4x^2 - x - 2)(x - 2)$ and $f''(x) = 12x^2 - 18x = 6x(2x - 3)$. This shows that f has critical points at $x = 2$ and $x = \frac{1 \pm \sqrt{33}}{8}$ and candidate points of inflection at $x = 0$ and $x = \frac{3}{2}$. Sign analyses reveal that $f'(x)$ changes from negative to positive at $x = \frac{1 - \sqrt{33}}{8}$, from positive to negative at $x = \frac{1 + \sqrt{33}}{8}$, and again from negative to positive at $x = 2$. Therefore, $f(\frac{1 - \sqrt{33}}{8})$ and $f(2)$ are local minima of $f(x)$, and $f(\frac{1 + \sqrt{33}}{8})$ is a local maximum. Further sign analyses reveal that $f''(x)$ changes from positive to negative at $x = 0$ and from negative to positive at $x = \frac{3}{2}$, so that there are points of inflection both at $x = 0$ and $x = \frac{3}{2}$. Here is a graph of $f(x)$ with transition points highlighted.

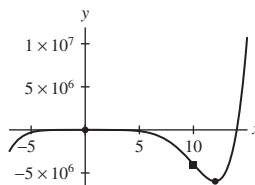


25. $y = x^7 - 14x^6$

SOLUTION Let $f(x) = x^7 - 14x^6$. Then $f'(x) = 7x^6 - 84x^5 = 7x^5(x - 12)$ and $f''(x) = 42x^5 - 420x^4 = 42x^4(x - 10)$. Critical points are at $x = 0$ and $x = 12$, and candidate inflection points are at $x = 0$ and $x = 10$. Sign analyses reveal that $f'(x)$ changes from positive to negative at $x = 0$ and from negative to positive at $x = 12$. Therefore $f(0)$ is a local maximum and $f(12)$ is a local minimum. Also, $f''(x)$ changes from negative to positive at $x = 10$. Therefore, there is a point of inflection at $x = 10$. Moreover,

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x) = \infty$$

Here is a graph of f with transition points highlighted.

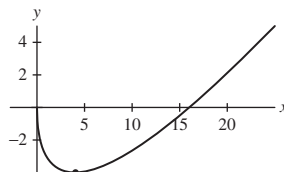


27. $y = x - 4\sqrt{x}$

SOLUTION Let $f(x) = x - 4\sqrt{x} = x - 4x^{1/2}$. Then $f'(x) = 1 - 2x^{-1/2}$. This shows that f has critical points at $x = 0$ (where the derivative does not exist) and at $x = 4$ (where the derivative is zero). Because $f'(x) < 0$ for $0 < x < 4$ and $f'(x) > 0$ for $x > 4$, $f(4)$ is a local minimum. Now $f''(x) = x^{-3/2} > 0$ for all $x > 0$, so the graph is always concave up. Moreover,

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

Here is a graph of f with transition points highlighted.



29. $y = x(8 - x)^{1/3}$

SOLUTION Let $f(x) = x(8 - x)^{1/3}$. Then

$$f'(x) = x \cdot \frac{1}{3}(8 - x)^{-2/3}(-1) + (8 - x)^{1/3} \cdot 1 = \frac{24 - 4x}{3(8 - x)^{2/3}}$$

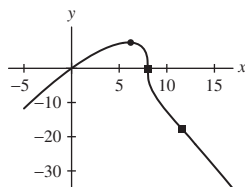
and similarly

$$f''(x) = \frac{4x - 48}{9(8 - x)^{5/3}}$$

Critical points are at $x = 8$ and $x = 6$, and candidate inflection points are $x = 8$ and $x = 12$. Sign analyses reveal that $f'(x)$ changes from positive to negative at $x = 6$ and $f'(x)$ remains negative on either side of $x = 8$. Moreover, $f''(x)$ changes from negative to positive at $x = 8$ and from positive to negative at $x = 12$. Therefore, f has a local maximum at $x = 6$ and inflection points at $x = 8$ and $x = 12$. Moreover,

$$\lim_{x \rightarrow \pm\infty} f(x) = -\infty$$

Here is a graph of f with the transition points highlighted.



31. $y = xe^{-x^2}$

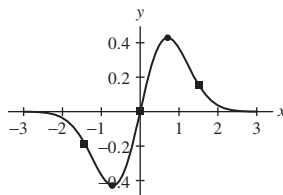
SOLUTION Let $f(x) = xe^{-x^2}$. Then

$$f'(x) = -2x^2e^{-x^2} + e^{-x^2} = (1 - 2x^2)e^{-x^2}$$

and

$$f''(x) = (4x^3 - 2x)e^{-x^2} - 4xe^{-x^2} = 2x(2x^2 - 3)e^{-x^2}$$

There are critical points at $x = \pm\frac{\sqrt{2}}{2}$, and $x = 0$ and $x = \pm\frac{\sqrt{3}}{2}$ are candidates for inflection points. Sign analysis shows that $f'(x)$ changes from negative to positive at $x = -\frac{\sqrt{2}}{2}$ and from positive to negative at $x = \frac{\sqrt{2}}{2}$. Moreover, $f''(x)$ changes from negative to positive at both $x = \pm\frac{\sqrt{3}}{2}$ and from positive to negative at $x = 0$. Therefore, f has a local minimum at $x = -\frac{\sqrt{2}}{2}$, a local maximum at $x = \frac{\sqrt{2}}{2}$ and inflection points at $x = 0$ and at $x = \pm\frac{\sqrt{3}}{2}$. Here is a graph of f with the transition points highlighted.



33. $y = x - 2 \ln x$

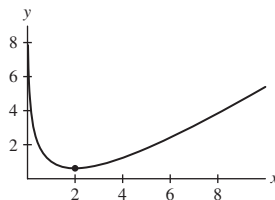
SOLUTION Let $f(x) = x - 2 \ln x$. Note that the domain of f is $x > 0$. Now,

$$f'(x) = 1 - \frac{2}{x} \quad \text{and} \quad f''(x) = \frac{2}{x^2}$$

The only critical point is $x = 2$. Sign analysis shows that $f'(x)$ changes from negative to positive at $x = 2$, so $f(2)$ is a local minimum. Further, $f''(x) > 0$ for $x > 0$, so the graph is always concave up. Moreover,

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

Here is a graph of f with the transition points highlighted.



35. $y = x^2 - 2 \ln x$

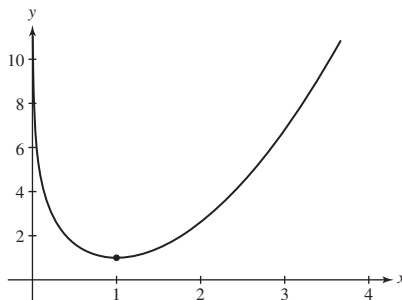
SOLUTION Let $f(x) = x^2 - 2 \ln x$. Note that the domain of f is $x > 0$. Now,

$$f'(x) = 2x - \frac{2}{x} \quad \text{and} \quad f''(x) = 2 + \frac{2}{x^2}$$

The only critical point is $x = 1$. Sign analysis shows that $f'(x)$ changes from negative to positive at $x = 1$, so $f(1)$ is a local minimum. Further, $f''(x) > 0$ for $x > 0$, so the graph is always concave up. Moreover,

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

Here is a graph of f with the transition points highlighted.



37. Investigate the behavior and sketch the graph of $f(x) = 18(x-3)(x-1)^{2/3}$ using the formulas

$$f'(x) = \frac{30(x - \frac{9}{5})}{(x-1)^{1/3}}, \quad f''(x) = \frac{20(x - \frac{3}{5})}{(x-1)^{4/3}}$$

SOLUTION

$$f'(x) = \frac{30(x - \frac{9}{5})}{(x-1)^{1/3}}$$

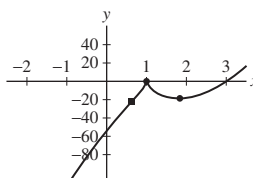
yields critical points at $x = \frac{9}{5}$, $x = 1$.

$$f''(x) = \frac{20(x - \frac{3}{5})}{(x-1)^{4/3}}$$

yields potential inflection points at $x = \frac{3}{5}$, $x = 1$.

Interval	signs of f' and f''
$(-\infty, \frac{3}{5})$	$+-$
$(\frac{3}{5}, 1)$	$++$
$(1, \frac{9}{5})$	$-+$
$(\frac{9}{5}, \infty)$	$++$

The graph has an inflection point at $x = \frac{3}{5}$, a local maximum at $x = 1$ (at which the graph has a cusp), and a local minimum at $x = \frac{9}{5}$. The sketch looks something like this.



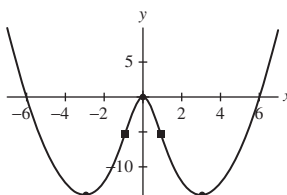
 In Exercises 39–42, sketch the graph of the function, indicating all transition points. If necessary, use a graphing utility or computer algebra system to locate the transition points numerically.

39. $y = x^2 - 10 \ln(x^2 + 1)$

SOLUTION Let $f(x) = x^2 - 10 \ln(x^2 + 1)$. Then $f'(x) = 2x - \frac{20x}{x^2 + 1}$, and

$$f''(x) = 2 - \frac{(x^2 + 1)(20) - (20x)(2x)}{(x^2 + 1)^2} = \frac{x^4 + 12x^2 - 9}{(x^2 + 1)^2}$$

There are critical points at $x = 0$ and $x = \pm 3$, and $x = \pm \sqrt{-6 + 3\sqrt{5}}$ are candidates for inflection points. Sign analysis shows that $f'(x)$ changes from negative to positive at $x = \pm 3$ and from positive to negative at $x = 0$. Moreover, $f''(x)$ changes from positive to negative at $x = -\sqrt{-6 + 3\sqrt{5}}$ and from negative to positive at $x = \sqrt{-6 + 3\sqrt{5}}$. Therefore, f has a local maximum at $x = 0$, local minima at $x = \pm 3$ and points of inflection at $x = \pm \sqrt{-6 + 3\sqrt{5}}$. Here is a graph of f with the transition points highlighted.

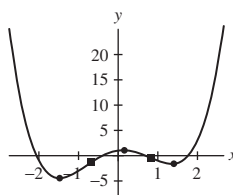


41. $y = x^4 - 4x^2 + x + 1$

SOLUTION Let $f(x) = x^4 - 4x^2 + x + 1$. Then $f'(x) = 4x^3 - 8x + 1$ and $f''(x) = 12x^2 - 8$. The critical points are $x = -1.473$, $x = 0.126$, and $x = 1.347$, while the candidates for points of inflection are $x = \pm \sqrt{\frac{2}{3}}$. Sign analysis reveals that $f'(x)$ changes from negative to positive at $x = -1.473$, from positive to negative at $x = 0.126$ and from negative to positive at $x = 1.347$. For the second derivative, $f''(x)$ changes from positive to negative at $x = -\sqrt{\frac{2}{3}}$ and from negative to positive at $x = \sqrt{\frac{2}{3}}$. Therefore, f has local minima at $x = -1.473$ and $x = 1.347$, a local maximum at $x = 0.126$ and points of inflection at $x = \pm \sqrt{\frac{2}{3}}$. Moreover,

$$\lim_{x \rightarrow \pm\infty} f(x) = \infty$$

Here is a graph of f with the transition points highlighted.



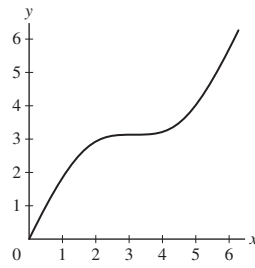
In Exercises 43–48, sketch the graph over the given interval, with all transition points indicated.

43. $y = x + \sin x$, $[0, 2\pi]$

SOLUTION Let $f(x) = x + \sin x$. Setting $f'(x) = 1 + \cos x = 0$ yields $\cos x = -1$, so that $x = \pi$ is the lone critical point on the interval $[0, 2\pi]$. Setting $f''(x) = -\sin x = 0$ yields potential points of inflection at $x = 0, \pi, 2\pi$ on the interval $[0, 2\pi]$.

Interval	signs of f' and f''
$(0, \pi)$	$+-$
$(\pi, 2\pi)$	$++$

The graph has an inflection point at $x = \pi$, and no local maxima or minima. Here is a graph of f without transition points highlighted.

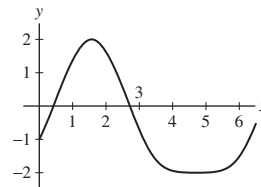


45. $y = 2 \sin x - \cos^2 x$, $[0, 2\pi]$

SOLUTION Let $f(x) = 2 \sin x - \cos^2 x$. Then $f'(x) = 2 \cos x - 2 \cos x(-\sin x) = \sin 2x + 2 \cos x$ and $f''(x) = 2 \cos 2x - 2 \sin x$. Setting $f'(x) = 0$ yields $\sin 2x = -2 \cos x$, so that $2 \sin x \cos x = -2 \cos x$. This implies $\cos x = 0$ or $\sin x = -1$, so that $x = \frac{\pi}{2}$ or $\frac{3\pi}{2}$. Setting $f''(x) = 0$ yields $2 \cos 2x = 2 \sin x$, so that $2 \sin(\frac{\pi}{2} - 2x) = 2 \sin x$, or $\frac{\pi}{2} - 2x = x \pm 2n\pi$. This yields $3x = \frac{\pi}{2} + 2n\pi$, or $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6} = \frac{3\pi}{2}$.

Interval	signs of f' and f''
$(0, \frac{\pi}{6})$	++
$(\frac{\pi}{6}, \frac{\pi}{2})$	+-
$(\frac{\pi}{2}, \frac{5\pi}{6})$	--
$(\frac{5\pi}{6}, \frac{3\pi}{2})$	-+
$(\frac{3\pi}{2}, 2\pi)$	++

The graph has a local maximum at $x = \frac{\pi}{2}$, a local minimum at $x = \frac{3\pi}{2}$, and inflection points at $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$. Here is a graph of f without transition points highlighted.

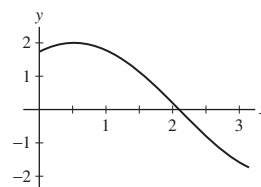


47. $y = \sin x + \sqrt{3} \cos x$, $[0, \pi]$

SOLUTION Let $f(x) = \sin x + \sqrt{3} \cos x$. Setting $f'(x) = \cos x - \sqrt{3} \sin x = 0$ yields $\tan x = \frac{1}{\sqrt{3}}$. In the interval $[0, \pi]$, the solution is $x = \frac{\pi}{6}$. Setting $f''(x) = -\sin x - \sqrt{3} \cos x = 0$ yields $\tan x = -\sqrt{3}$. In the interval $[0, \pi]$, the lone solution is $x = \frac{2\pi}{3}$.

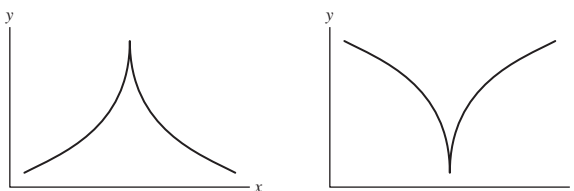
Interval	signs of f' and f''
$(0, \pi/6)$	+-
$(\pi/6, 2\pi/3)$	--
$(2\pi/3, \pi)$	-+

The graph has a local maximum at $x = \frac{\pi}{6}$ and a point of inflection at $x = \frac{2\pi}{3}$. Here is a graph of f without transition points highlighted.



49.  Are all sign transitions possible? Explain with a sketch why the transitions $++ \rightarrow --$ and $-- \rightarrow ++$ do not occur if the function is differentiable. (See Exercise 80 for a proof.)

SOLUTION In both cases, there is a point where f is not differentiable at the transition from increasing to decreasing or decreasing to increasing.



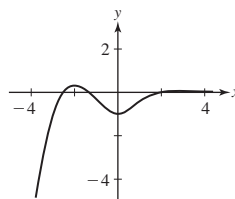
In Exercises 51–52, draw the graph of a function f having the given limits at $\pm\infty$ and for which f' and f'' take on the given sign combinations in order.

51. $\lim_{x \rightarrow -\infty} f(x) = -\infty$, $\lim_{x \rightarrow \infty} f(x) = 0$; $+-$, $--$, $-+$, $++$, $+-$

SOLUTION The graph of one possible function f satisfying $\lim_{x \rightarrow -\infty} f(x) = -\infty$ and $\lim_{x \rightarrow \infty} f(x) = 0$ and for which f' and f'' take on the sign combinations

$+-$, $--$, $-+$, $++$, $+-$

is shown below.



53. Match the graphs in Figure 19 with the two functions $y = \frac{3x}{x^2 - 1}$ and $y = \frac{3x^2}{x^2 - 1}$. Explain.

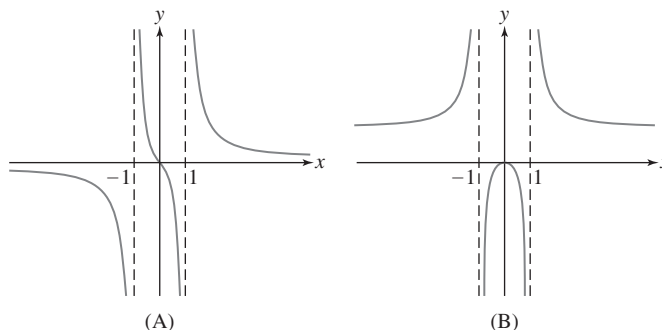


FIGURE 19

SOLUTION Since $\lim_{x \rightarrow \pm\infty} \frac{3x^2}{x^2 - 1} = \frac{3}{1} \cdot \lim_{x \rightarrow \pm\infty} 1 = 3$, the graph of $y = \frac{3x^2}{x^2 - 1}$ has a horizontal asymptote of $y = 3$; hence, the right curve is the graph of $f(x) = \frac{3x^2}{x^2 - 1}$. Since

$$\lim_{x \rightarrow \pm\infty} \frac{3x}{x^2 - 1} = \frac{3}{1} \cdot \lim_{x \rightarrow \pm\infty} x^{-1} = 0$$

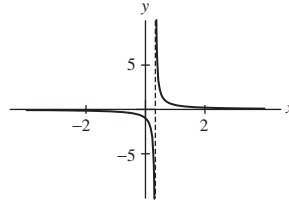
the graph of $y = \frac{3x}{x^2 - 1}$ has a horizontal asymptote of $y = 0$; hence, the left curve is the graph of $f(x) = \frac{3x}{x^2 - 1}$.

In Exercises 55–72, sketch the graph of the function. Indicate the transition points and asymptotes.

55. $y = \frac{1}{3x - 1}$

SOLUTION Let $f(x) = \frac{1}{3x-1}$. Then $f'(x) = \frac{-3}{(3x-1)^2}$, so that f is decreasing for all $x \neq \frac{1}{3}$. Moreover, $f''(x) = \frac{18}{(3x-1)^3}$, so that f is concave up for $x > \frac{1}{3}$ and concave down for $x < \frac{1}{3}$. Because $\lim_{x \rightarrow \pm\infty} \frac{1}{3x-1} = 0$, f has a horizontal asymptote at $y = 0$. Finally, f has a vertical asymptote at $x = \frac{1}{3}$ with

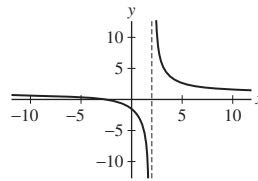
$$\lim_{x \rightarrow \frac{1}{3}^-} \frac{1}{3x-1} = -\infty \quad \text{and} \quad \lim_{x \rightarrow \frac{1}{3}^+} \frac{1}{3x-1} = \infty$$



57. $y = \frac{x+3}{x-2}$

SOLUTION Let $f(x) = \frac{x+3}{x-2}$. Then $f'(x) = \frac{-5}{(x-2)^2}$, so that f is decreasing for all $x \neq 2$. Moreover, $f''(x) = \frac{10}{(x-2)^3}$, so that f is concave up for $x > 2$ and concave down for $x < 2$. Because $\lim_{x \rightarrow \pm\infty} \frac{x+3}{x-2} = 1$, f has a horizontal asymptote at $y = 1$. Finally, f has a vertical asymptote at $x = 2$ with

$$\lim_{x \rightarrow 2^-} \frac{x+3}{x-2} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 2^+} \frac{x+3}{x-2} = \infty$$



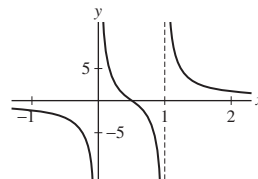
59. $y = \frac{1}{x} + \frac{1}{x-1}$

SOLUTION Let $f(x) = \frac{1}{x} + \frac{1}{x-1}$. Then $f'(x) = -\frac{2x^2-2x+1}{x^2(x-1)^2}$, so that f is decreasing for all $x \neq 0, 1$. Moreover, $f''(x) = \frac{2(2x^3-3x^2+3x-1)}{x^3(x-1)^3}$, so that f is concave up for $0 < x < \frac{1}{2}$ and $x > 1$ and concave down for $x < 0$ and $\frac{1}{2} < x < 1$. Because $\lim_{x \rightarrow \pm\infty} \left(\frac{1}{x} + \frac{1}{x-1}\right) = 0$, f has a horizontal asymptote at $y = 0$. Finally, f has vertical asymptotes at $x = 0$ and $x = 1$ with

$$\lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x-1}\right) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \left(\frac{1}{x} + \frac{1}{x-1}\right) = \infty$$

and

$$\lim_{x \rightarrow 1^-} \left(\frac{1}{x} + \frac{1}{x-1}\right) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 1^+} \left(\frac{1}{x} + \frac{1}{x-1}\right) = \infty$$



61. $y = \frac{1}{x(x-2)}$

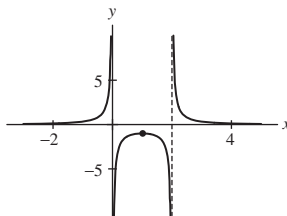
SOLUTION Let $f(x) = \frac{1}{x(x-2)}$. Then $f'(x) = \frac{2(1-x)}{x^2(x-2)^2}$, so that f is increasing for $x < 0$ and $0 < x < 1$ and decreasing for $1 < x < 2$ and $x > 2$. Moreover, $f''(x) = \frac{2(3x^2-6x+4)}{x^3(x-2)^3}$, so that f is concave up for $x < 0$ and $x > 2$ and

concave down for $0 < x < 2$. Because $\lim_{x \rightarrow \pm\infty} \left(\frac{1}{x(x-2)} \right) = 0$, f has a horizontal asymptote at $y = 0$. Finally, f has vertical asymptotes at $x = 0$ and $x = 2$ with

$$\lim_{x \rightarrow 0^-} \left(\frac{1}{x(x-2)} \right) = +\infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \left(\frac{1}{x(x-2)} \right) = -\infty$$

and

$$\lim_{x \rightarrow 2^-} \left(\frac{1}{x(x-2)} \right) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 2^+} \left(\frac{1}{x(x-2)} \right) = \infty$$



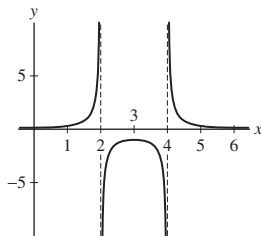
63. $y = \frac{1}{x^2 - 6x + 8}$

SOLUTION Let $f(x) = \frac{1}{x^2 - 6x + 8} = \frac{1}{(x-2)(x-4)}$. Then $f'(x) = \frac{6-2x}{(x^2 - 6x + 8)^2}$, so that f is increasing for $x < 2$ and for $2 < x < 3$, is decreasing for $3 < x < 4$ and for $x > 4$, and has a local maximum at $x = 3$. Moreover, $f''(x) = \frac{2(3x^2 - 18x + 28)}{(x^2 - 6x + 8)^3}$, so that f is concave up for $x < 2$ and for $x > 4$ and is concave down for $2 < x < 4$. Because $\lim_{x \rightarrow \pm\infty} \frac{1}{x^2 - 6x + 8} = 0$, f has a horizontal asymptote at $y = 0$. Finally, f has vertical asymptotes at $x = 2$ and $x = 4$, with

$$\lim_{x \rightarrow 2^-} \left(\frac{1}{x^2 - 6x + 8} \right) = \infty \quad \text{and} \quad \lim_{x \rightarrow 2^+} \left(\frac{1}{x^2 - 6x + 8} \right) = -\infty$$

and

$$\lim_{x \rightarrow 4^-} \left(\frac{1}{x^2 - 6x + 8} \right) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 4^+} \left(\frac{1}{x^2 - 6x + 8} \right) = \infty$$



65. $y = 1 - \frac{3}{x} + \frac{4}{x^3}$

SOLUTION Let $f(x) = 1 - \frac{3}{x} + \frac{4}{x^3}$. Then

$$f'(x) = \frac{3}{x^2} - \frac{12}{x^4} = \frac{3(x-2)(x+2)}{x^4}$$

so that f is increasing for $|x| > 2$ and decreasing for $-2 < x < 0$ and for $0 < x < 2$. Moreover,

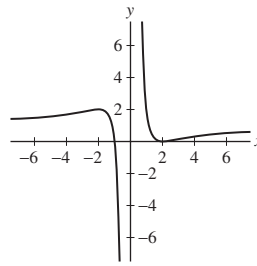
$$f''(x) = -\frac{6}{x^3} + \frac{48}{x^5} = \frac{6(8-x^2)}{x^5}$$

so that f is concave down for $-2\sqrt{2} < x < 0$ and for $x > 2\sqrt{2}$, while f is concave up for $x < -2\sqrt{2}$ and for $0 < x < 2\sqrt{2}$. Because

$$\lim_{x \rightarrow \pm\infty} \left(1 - \frac{3}{x} + \frac{4}{x^3} \right) = 1$$

f has a horizontal asymptote at $y = 1$. Finally, f has a vertical asymptote at $x = 0$ with

$$\lim_{x \rightarrow 0^-} \left(1 - \frac{3}{x} + \frac{4}{x^3} \right) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \left(1 - \frac{3}{x} + \frac{4}{x^3} \right) = \infty$$



67. $y = \frac{1}{x^2} - \frac{1}{(x-2)^2}$

SOLUTION Let $f(x) = \frac{1}{x^2} - \frac{1}{(x-2)^2}$. Then $f'(x) = -2x^{-3} + 2(x-2)^{-3}$, so that f is increasing for $x < 0$ and for $x > 2$ and is decreasing for $0 < x < 2$. Moreover,

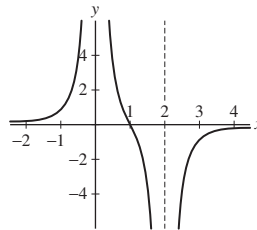
$$f''(x) = 6x^{-4} - 6(x-2)^{-4} = -48(x-1)(x^2 - 2x + 2)$$

so that f is concave up for $x < 0$ and for $0 < x < 1$, is concave down for $1 < x < 2$ and for $x > 2$, and has a point of inflection at $x = 1$. Because $\lim_{x \rightarrow \pm\infty} \left(\frac{1}{x^2} - \frac{1}{(x-2)^2} \right) = 0$, f has a horizontal asymptote at $y = 0$. Finally, f has vertical asymptotes at $x = 0$ and $x = 2$ with

$$\lim_{x \rightarrow 0^-} \left(\frac{1}{x^2} - \frac{1}{(x-2)^2} \right) = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \left(\frac{1}{x^2} - \frac{1}{(x-2)^2} \right) = \infty$$

and

$$\lim_{x \rightarrow 2^-} \left(\frac{1}{x^2} - \frac{1}{(x-2)^2} \right) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 2^+} \left(\frac{1}{x^2} - \frac{1}{(x-2)^2} \right) = -\infty$$

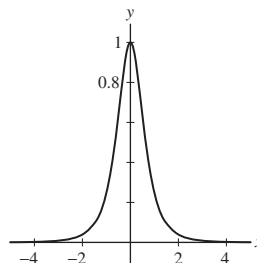


69. $y = \frac{1}{(x^2 + 1)^2}$

SOLUTION Let $f(x) = \frac{1}{(x^2 + 1)^2}$. Then $f'(x) = \frac{-4x}{(x^2 + 1)^3}$, so that f is increasing for $x < 0$, is decreasing for $x > 0$ and has a local maximum at $x = 0$. Moreover,

$$f''(x) = \frac{-4(x^2 + 1)^3 + 4x \cdot 3(x^2 + 1)^2 \cdot 2x}{(x^2 + 1)^6} = \frac{20x^2 - 4}{(x^2 + 1)^4}$$

so that f is concave up for $|x| > 1/\sqrt{5}$, is concave down for $|x| < 1/\sqrt{5}$, and has points of inflection at $x = \pm 1/\sqrt{5}$. Because $\lim_{x \rightarrow \pm\infty} \frac{1}{(x^2 + 1)^2} = 0$, f has a horizontal asymptote at $y = 0$. Finally, f has no vertical asymptotes.



71. $y = \frac{1}{\sqrt{x^2 + 1}}$

SOLUTION Let $f(x) = \frac{1}{\sqrt{x^2+1}}$. Then

$$f'(x) = -\frac{x}{\sqrt{(x^2+1)^3}} = -x(x^2+1)^{-3/2}$$

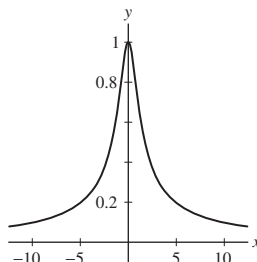
so that f is increasing for $x < 0$ and decreasing for $x > 0$. Moreover,

$$f''(x) = -\frac{3}{2}x(x^2+1)^{-5/2}(-2x) - (x^2+1)^{-3/2} = (2x^2-1)(x^2+1)^{-5/2}$$

so that f is concave down for $|x| < \frac{\sqrt{2}}{2}$ and concave up for $|x| > \frac{\sqrt{2}}{2}$. Because

$$\lim_{x \rightarrow \pm\infty} \frac{1}{\sqrt{x^2+1}} = 0$$

f has a horizontal asymptote at $y = 0$. Finally, f has no vertical asymptotes.



73. Consider the general logistic function, $P(x) = \frac{M}{1 + Ae^{-kx}}$, with A , M , and k all positive. Show that

- (a) $P'(x) = \frac{MAke^{-kx}}{(1 + Ae^{-kx})^2}$ and $P''(x) = \frac{MAk^2e^{-kx}(Ae^{-kx} - 1)}{(1 + Ae^{-kx})^3}$
- (b) $\lim_{x \rightarrow -\infty} P(x) = 0$ and $\lim_{x \rightarrow \infty} P(x) = M$, and therefore $P = 0$ and $P = M$ are horizontal asymptotes of P .
- (c) P is increasing for all x .
- (d) The only inflection point of P is at $(\frac{\ln A}{k}, \frac{M}{2})$. To the left of it P is concave up, and to the right of it P is concave down.

SOLUTION Let $P(x) = \frac{M}{1 + Ae^{-kx}}$, with A , M , and k all positive.

(a) Then

$$P'(x) = -\frac{M}{(1 + Ae^{-kx})^2}(-Ake^{-kx}) = \frac{MAke^{-kx}}{(1 + Ae^{-kx})^2}$$

and

$$P''(x) = \frac{(1 + Ae^{-kx})^2(-MAk^2e^{-kx}) - MAke^{-kx}(2)(1 + Ae^{-kx})(-Ake^{-kx})}{(1 + Ae^{-kx})^4} = \frac{MAk^2e^{-kx}(Ae^{-kx} - 1)}{(1 + Ae^{-kx})^3}$$

(b) Because k is positive,

$$\lim_{x \rightarrow -\infty} e^{-kx} = \infty \quad \text{and} \quad \lim_{x \rightarrow \infty} e^{-kx} = 0$$

It then follows that

$$\lim_{x \rightarrow -\infty} P(x) = \frac{M}{1 + A \lim_{x \rightarrow -\infty} e^{-kx}} = 0$$

and

$$\lim_{x \rightarrow \infty} P(x) = \frac{M}{1 + A \lim_{x \rightarrow \infty} e^{-kx}} = \frac{M}{1 + A(0)} = M$$

Therefore, $P = 0$ and $P = M$ are horizontal asymptotes of P .

(c) Because M , A , and k are all positive,

$$P'(x) = \frac{MAke^{-kx}}{(1 + Ae^{-kx})^2} > 0$$

for all x . Therefore, P is increasing for all x .

(d) The second derivative of P is defined for all x and is equal to zero when

$$Ae^{-kx} - 1 = 0 \quad \text{or} \quad x = \frac{\ln A}{k}$$

Note

$$P\left(\frac{\ln A}{k}\right) = \frac{M}{1 + Ae^{-\ln A}} = \frac{M}{1 + A \cdot A^{-1}} = \frac{M}{2}$$

Now, for $x < \frac{\ln A}{k}$, $P''(x) > 0$, while for $x > \frac{\ln A}{k}$, $P''(x) < 0$. Thus, the only inflection point of P is at $\left(\frac{\ln A}{k}, \frac{M}{2}\right)$. To the left of this point P is concave up, and to the right P is concave down.

Further Insights and Challenges

In Exercises 75–79, we explore functions whose graphs approach a nonhorizontal line as $x \rightarrow \infty$. A line $y = ax + b$ is called a **slant asymptote** if

$$\lim_{x \rightarrow \infty} (f(x) - (ax + b)) = 0$$

or

$$\lim_{x \rightarrow -\infty} (f(x) - (ax + b)) = 0$$

75. Let $f(x) = \frac{x^2}{x-1}$ (Figure 21). Verify the following:

- (a) $f(0)$ is a local max and $f(2)$ a local min.
- (b) f is concave down on $(-\infty, 1)$ and concave up on $(1, \infty)$.
- (c) $\lim_{x \rightarrow 1^-} f(x) = -\infty$ and $\lim_{x \rightarrow 1^+} f(x) = \infty$.
- (d) $y = x + 1$ is a slant asymptote of f as $x \rightarrow \pm\infty$.
- (e) The slant asymptote lies above the graph of f for $x < 1$ and below the graph for $x > 1$.

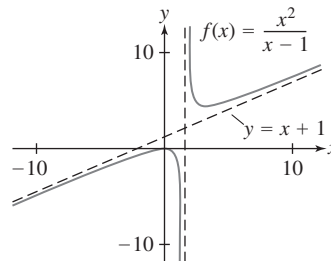


FIGURE 21

SOLUTION Let $f(x) = \frac{x^2}{x-1}$. Then $f'(x) = \frac{x(x-2)}{(x-1)^2}$ and $f''(x) = \frac{2}{(x-1)^3}$.

- (a) Critical points of $f'(x)$ occur at $x = 0$ and $x = 2$. $x = 1$ is not a critical point because it is not in the domain of f . Sign analysis reveals that f' changes from positive to negative at $x = 0$ and from negative to positive at $x = 2$. Thus, $f(0)$ is a local maximum, and $f(2)$ is a local minimum.
- (b) Sign analysis of $f''(x)$ reveals that $f''(x) < 0$ and f is concave down on $(-\infty, 1)$ whereas $f''(x) > 0$ and f is concave up on $(1, \infty)$.
- (c)

$$\lim_{x \rightarrow 1^-} f(x) = -1 \lim_{x \rightarrow 1^-} \frac{1}{1-x} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 1^+} f(x) = 1 \lim_{x \rightarrow 1^+} \frac{1}{x-1} = \infty$$

(d) Note that using polynomial division, $f(x) = \frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}$. Then

$$\lim_{x \rightarrow \pm\infty} (f(x) - (x + 1)) = \lim_{x \rightarrow \pm\infty} x + 1 + \frac{1}{x-1} - (x + 1) = \lim_{x \rightarrow \pm\infty} \frac{1}{x-1} = 0.$$

- (e) For $x > 1$, $f(x) - (x + 1) = \frac{1}{x-1} > 0$, so $f(x)$ approaches $x + 1$ from above. Similarly, for $x < 1$, $f(x) - (x + 1) = \frac{1}{x-1} < 0$, so $f(x)$ approaches $x + 1$ from below.

77. Sketch the graph of

$$f(x) = \frac{x^2}{x+1}$$

Proceed as in the previous exercise to find the slant asymptote.

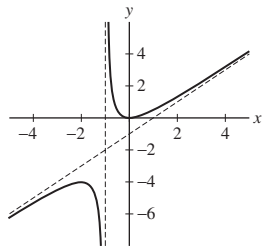
SOLUTION Let $f(x) = \frac{x^2}{x+1}$. Then $f'(x) = \frac{x(x+2)}{(x+1)^2}$ and $f''(x) = \frac{2}{(x+1)^3}$. Thus, f is increasing for $x < -2$ and for $x > 0$, is decreasing for $-2 < x < -1$ and for $-1 < x < 0$, has a local minimum at $x = 0$, has a local maximum at $x = -2$, is concave down on $(-\infty, -1)$ and concave up on $(-1, \infty)$. Limit analyses give a vertical asymptote at $x = -1$, with

$$\lim_{x \rightarrow -1^-} \frac{x^2}{x+1} = -\infty \quad \text{and} \quad \lim_{x \rightarrow -1^+} \frac{x^2}{x+1} = \infty$$

By polynomial division, $f(x) = x - 1 + \frac{1}{x+1}$ and

$$\lim_{x \rightarrow \pm\infty} \left(x - 1 + \frac{1}{x+1} - (x-1) \right) = 0$$

which implies that the slant asymptote is $y = x - 1$. Notice that f approaches the slant asymptote as in Exercise 78.

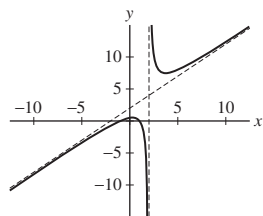


79. Sketch the graph of $f(x) = \frac{1-x^2}{2-x}$.

SOLUTION Let $f(x) = \frac{1-x^2}{2-x}$. Using polynomial division, $f(x) = x + 2 + \frac{3}{x-2}$. Then

$$\lim_{x \rightarrow \pm\infty} (f(x) - (x+2)) = \lim_{x \rightarrow \pm\infty} (x+2) + \frac{3}{x-2} - (x+2) = \lim_{x \rightarrow \pm\infty} \frac{3}{x-2} = \frac{3}{1} \cdot \lim_{x \rightarrow \pm\infty} x^{-1} = 0$$

which implies that $y = x + 2$ is the slant asymptote of $f(x)$. Since $f(x) - (x+2) = \frac{3}{x-2} > 0$ for $x > 2$, $f(x)$ approaches the slant asymptote from above for $x > 2$; similarly, $\frac{3}{x-2} < 0$ for $x < 2$ so $f(x)$ approaches the slant asymptote from below for $x < 2$. Moreover, $f'(x) = \frac{x^2 - 4x + 1}{(2-x)^2}$ and $f''(x) = \frac{-6}{(2-x)^3}$. Sign analyses reveal a local minimum at $x = 2 + \sqrt{3}$, a local maximum at $x = 2 - \sqrt{3}$ and that f is concave down on $(-\infty, 2)$ and concave up on $(2, \infty)$. Limit analyses give a vertical asymptote at $x = 2$.



81.  Assume that f'' exists and $f''(x) > 0$ for all x . Show that $f(x)$ cannot be negative for all x . *Hint:* Show that $f'(b) \neq 0$ for some b and use the result of Exercise 74 in Section 4.4.

SOLUTION Let $f(x)$ be a function such that $f''(x)$ exists and $f''(x) > 0$ for all x . Since $f''(x) > 0$, there is at least one point $x = b$ such that $f'(b) \neq 0$. If not, $f'(x) = 0$ for all x , so $f''(x) = 0$. By the result of Exercise 74 in Section 4.4, $f(x) \geq f(b) + f'(b)(x-b)$. Now, if $f'(b) > 0$, we find that $f(b) + f'(b)(x-b) > 0$ whenever

$$x > \frac{bf'(b) - f(b)}{f'(b)}$$

a condition that must be met for some x sufficiently large. For such x , $f(x) > f(b) + f'(b)(x-b) > 0$. On the other hand, if $f'(b) < 0$, we find that $f(b) + f'(b)(x-b) > 0$ whenever

$$x < \frac{bf'(b) - f(b)}{f'(b)}$$

For such an x , $f(x) > f(b) + f'(b)(x-b) > 0$.

4.7 Applied Optimization

Preliminary Questions

1. The problem is to find the right triangle of perimeter 10 whose area is as large as possible. What is the constraint equation relating the base b and height h of the triangle?

SOLUTION The perimeter of a right triangle is the sum of the lengths of the base, the height, and the hypotenuse. If the base has length b and the height is h , then the length of the hypotenuse is $\sqrt{b^2 + h^2}$ and the perimeter of the triangle is $P = b + h + \sqrt{b^2 + h^2}$. The requirement that the perimeter be 10 translates to the constraint equation

$$b + h + \sqrt{b^2 + h^2} = 10$$

2. Describe a way of showing that a continuous function on an open interval (a, b) has a minimum value.

SOLUTION If the function tends to infinity at the endpoints of the interval, then the function must take on a minimum value at a critical point.

3. Is there a rectangle of area 100 of largest perimeter? Explain.

SOLUTION No. Even by fixing the area at 100, we can take one of the dimensions as large as we like, thereby allowing the perimeter to become as large as we like.

Exercises

1. Find the dimensions x and y of the rectangle of maximum area that can be formed using 3 m of wire.

- (a) What is the constraint equation relating x and y ?
- (b) Find a formula for the area in terms of x alone.
- (c) What is the interval of optimization? Is it open or closed?
- (d) Solve the optimization problem.

SOLUTION

- (a) The perimeter of the rectangle is 3 meters, so $3 = 2x + 2y$, which is equivalent to $y = \frac{3}{2} - x$.
- (b) Using part (a), $A = xy = x(\frac{3}{2} - x) = \frac{3}{2}x - x^2$.
- (c) This problem requires optimization over the closed interval $[0, \frac{3}{2}]$, since both x and y must be nonnegative.
- (d) $A'(x) = \frac{3}{2} - 2x = 0$, which yields $x = \frac{3}{4}$ and consequently, $y = \frac{3}{4}$. Because $A(0) = A(3/2) = 0$ and $A(\frac{3}{4}) = 0.5625$, the maximum area 0.5625 m^2 is achieved with $x = y = \frac{3}{4} \text{ m}$.

3. A rectangular bird sanctuary is being created with one side along a straight riverbank. The remaining three sides are to be enclosed with a protective fence. If there are 12 km of fence available, find the dimension of the rectangle to maximize the area of the sanctuary.

SOLUTION Let x denote the length of the rectangular sanctuary running perpendicular to the riverbank, and y denote the length of the sanctuary parallel to the riverbank. The area enclosed by the sanctuary is $A = xy$, and the amount of protective fencing used to construct the sanctuary is $2x + y$ km. There are 12 km of protective fencing available, so $2x + y = 12$, or $y = 12 - 2x$. Substituting for y in the formula for the area yields

$$A = x(12 - 2x) = 12x - 2x^2$$

Because both x and y must be nonnegative, the domain for the area function is the closed interval $[0, 6]$. Now,

$$A' = 12 - 4x = 0$$

when $x = 3$. Evaluating A at the endpoints of the domain and at the critical point $x = 3$ yields

$$A(0) = 0, \quad A(3) = 18, \quad \text{and} \quad A(6) = 0$$

Thus, the area of the sanctuary is maximized when 3 km of fencing is used perpendicular to the riverbank and 6 km of fencing is used parallel to the riverbank.

5. Find two positive real numbers such that the sum of the first number squared and the second number is 48 and their product is a maximum.

SOLUTION Let x denote the first number, and y denote the second number. The objective is to maximize $P = xy$ subject to the condition that $x^2 + y = 48$, or $y = 48 - x^2$. Substituting for y in the formula for P yields

$$P = x(48 - x^2) = 48x - x^3$$

Because x and y must both be positive, the domain for the product function is the open interval $(0, 4\sqrt{3})$. Now,

$$P' = 48 - 3x^2 = 0$$

when $x = 4$. Because $P(4) = 128$, while $P \rightarrow 0$ as $x \rightarrow 0^+$ and as $x \rightarrow 4\sqrt{3}^-$, we conclude that P achieves its absolute maximum when $x = 4$. Thus, the two positive real numbers such that the sum of the first number squared and the second number is 48 and their product is a maximum are 4 and 32.

7. A wire of length 12 m is divided into two pieces and the pieces are bent into a square and a circle. How should this be done in order to minimize the sum of their areas?

SOLUTION Suppose the wire is divided into one piece of length x m that is bent into a circle and a piece of length $12 - x$ m that is bent into a square. Because the circle has circumference x , it follows that the radius of the circle is $x/2\pi$; therefore, the area of the circle is

$$\pi \left(\frac{x}{2\pi} \right)^2 = \frac{x^2}{4\pi}$$

As for the square, because the perimeter is $12 - x$, the length of each side is $3 - x/4$ and the area is $(3 - x/4)^2$. Then

$$A(x) = \frac{x^2}{4\pi} + \left(3 - \frac{1}{4}x \right)^2$$

Now

$$A'(x) = \frac{x}{2\pi} - \frac{1}{2} \left(3 - \frac{1}{4}x \right) = 0$$

when

$$x = \frac{12\pi}{4 + \pi} \text{ m} \approx 5.28 \text{ m}$$

Because $A(0) = 9 \text{ m}^2$, $A(12) = 36/\pi \approx 11.46 \text{ m}^2$, and

$$A \left(\frac{12\pi}{4 + \pi} \right) \approx 5.04 \text{ m}^2$$

we see that the sum of the areas is minimized when approximately 5.28 m of the wire is allotted to the circle.

9. Find two positive real numbers such that they add to 40 and their product is as large as possible.

SOLUTION Let x and y denote the two numbers. The objective is to maximize $P = xy$ subject to the condition that $x + y = 40$, or $y = 40 - x$. Substituting for y in the formula for P yields

$$P = x(40 - x) = 40x - x^2$$

Because x and y must both be positive, the domain for the product function is the open interval $(0, 40)$. Now,

$$P' = 40 - 2x = 0$$

when $x = 20$. Because $P(20) = 400$, while $P \rightarrow 0$ as $x \rightarrow 0^+$ and as $x \rightarrow 40^-$, we conclude that P achieves its absolute maximum when $x = 20$. Thus, the two positive real numbers such that they add to 40 and their product is as large as possible are 20 and 20.

11. Find two positive real numbers x and y such that their product is 800 and $x + 2y$ is as small as possible.

SOLUTION The objective is to minimize $S = x + 2y$ subject to the condition that $xy = 800$, or $y = \frac{800}{x}$. Substituting for y in the formula for S yields

$$S = x + \frac{1600}{x}$$

The domain for this function is $x > 0$. Now,

$$S' = 1 - \frac{1600}{x^2} = 0$$

when $x = 40$. As $x \rightarrow 0^+$ and as $x \rightarrow \infty$, $S \rightarrow \infty$, so S achieves its absolute minimum when $x = 40$. Thus, the two positive real numbers x and y such that their product is 800 and $x + 2y$ is as small as possible are $x = 40$ and $y = 20$.

13. Find the dimensions of the box with square base with

(a) Volume 12 and the minimal surface area.

(b) Surface area 20 and maximal volume.

SOLUTION A box has a square base of side x and height y where $x, y > 0$. Its volume is $V = x^2y$ and its surface area is $S = 2x^2 + 4xy$.

(a) If $V = x^2y = 12$, then $y = 12/x^2$ and $S(x) = 2x^2 + 4x(12/x^2) = 2x^2 + 48x^{-1}$. Solve $S'(x) = 4x - 48x^{-2} = 0$ to obtain $x = 12^{1/3}$. Since $S(x) \rightarrow \infty$ as $x \rightarrow 0^+$ and as $x \rightarrow \infty$, the minimum surface area is $S(12^{1/3}) = 6(12)^{2/3} \approx 31.45$, when $x = 12^{1/3}$ and $y = 12^{1/3}$.

(b) If $S = 2x^2 + 4xy = 20$, then $y = 5x^{-1} - \frac{1}{2}x$ and $V(x) = x^2y = 5x - \frac{1}{2}x^3$. Note that x must lie on the closed interval $[0, \sqrt{10}]$. Solve $V'(x) = 5 - \frac{3}{2}x^2$ for $x > 0$ to obtain $x = \frac{\sqrt{30}}{3}$. Since $V(0) = V(\sqrt{10}) = 0$ and $V\left(\frac{\sqrt{30}}{3}\right) = \frac{10\sqrt{30}}{9}$, the maximum volume is $V\left(\frac{\sqrt{30}}{3}\right) = \frac{10}{9}\sqrt{30} \approx 6.086$, when $x = \frac{\sqrt{30}}{3}$ and $y = \frac{\sqrt{30}}{3}$.

15. A rancher will use 600 m of fencing to build a corral in the shape of a semicircle on top of a rectangle (Figure 11). Find the dimensions that maximize the area of the corral.

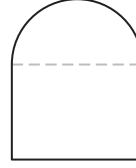


FIGURE 11

SOLUTION Let x be the width of the corral and therefore the diameter of the semicircle, and let y be the height of the rectangular section. Then the perimeter of the corral can be expressed by the equation $2y + x + \frac{\pi}{2}x = 2y + (1 + \frac{\pi}{2})x = 600$ m or equivalently, $y = \frac{1}{2}(600 - (1 + \frac{\pi}{2})x)$. Since x and y must both be nonnegative, it follows that x must be restricted to the interval $[0, \frac{600}{1 + \pi/2}]$. The area of the corral is the sum of the area of the rectangle and semicircle, $A = xy + \frac{\pi}{8}x^2$. Making the substitution for y from the constraint equation,

$$A(x) = \frac{1}{2}x \left(600 - (1 + \frac{\pi}{2})x \right) + \frac{\pi}{8}x^2 = 300x - \frac{1}{2} \left(1 + \frac{\pi}{2} \right) x^2 + \frac{\pi}{8}x^2$$

Now, $A'(x) = 300 - (1 + \frac{\pi}{2})x + \frac{\pi}{4}x = 0$ implies $x = \frac{300}{(1 + \pi/4)} \approx 168.029746$ m. With $A(0) = 0$ m²,

$$A\left(\frac{300}{1 + \pi/4}\right) \approx 25204.5 \text{ m}^2 \quad \text{and} \quad A\left(\frac{600}{1 + \pi/2}\right) \approx 21390.8 \text{ m}^2$$

it follows that the corral of maximum area has dimensions

$$x = \frac{300}{1 + \pi/4} \text{ m} \quad \text{and} \quad y = \frac{150}{1 + \pi/4} \text{ m}$$

17. Find the dimensions of the rectangle of maximum area that can be inscribed in a circle of radius $r = 4$ (Figure 13).

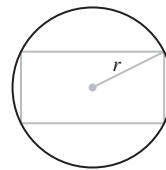


FIGURE 13

SOLUTION Place the center of the circle at the origin with the sides of the rectangle (of lengths $2x > 0$ and $2y > 0$) parallel to the coordinate axes. By the Pythagorean Theorem, $x^2 + y^2 = r^2 = 16$, so that $y = \sqrt{16 - x^2}$. Thus the area of the rectangle is $A(x) = 2x \cdot 2y = 4x\sqrt{16 - x^2}$. To guarantee both x and y are real and nonnegative, we must restrict x to the interval $[0, 4]$. Solve

$$A'(x) = 4\sqrt{16 - x^2} - \frac{4x^2}{\sqrt{16 - x^2}} = 0$$

for $x > 0$ to obtain $x = \frac{4}{\sqrt{2}} = 2\sqrt{2}$. Since $A(0) = A(4) = 0$ and $A(2\sqrt{2}) = 32$, the rectangle of maximum area has dimensions $2x = 2y = 4\sqrt{2}$.

19. In the setting of Examples 2 and 3, let r denote the speed along the road, and h denote the speed along the highway.

(a) Show that the travel-time function $T(x)$ has a critical point at $x = \frac{30}{\sqrt{(h/r)^2 - 1}}$ and explain why this indicates that if $r \geq h$ there is no critical point.

(b) Explain why there cannot be a critical point at $x = 0$, but depending on the speeds, the critical point can be arbitrarily close to 0.

SOLUTION

(a) In the setting of Examples 2 and 3, if r is the speed along the road and h is the speed along the highway, then the travel–time function is

$$T(x) = \frac{\sqrt{900 + x^2}}{r} + \frac{50 - x}{h}$$

Therefore,

$$T'(x) = \frac{x}{r\sqrt{900 + x^2}} - \frac{1}{h}$$

Setting $T'(x) = 0$ and solving for x yields

$$\begin{aligned}\frac{x}{r\sqrt{900 + x^2}} &= \frac{1}{h} \\ \frac{xh}{r} &= \sqrt{900 + x^2} \\ \frac{x^2h^2}{r^2} &= 900 + x^2 \\ x^2 \frac{h^2}{r^2} - 1 &= 900 \\ x &= \frac{30}{(h/r)^2 - 1} \quad \triangleright\end{aligned}$$

If $r \geq h$, then $(h/r)^2 - 1 \leq 0$, implying there is no solution to $T'(x) = 0$ and therefore no critical point.

(b) Because the numerator in the expression for the critical point is nonzero, the critical point must be nonzero. Note that for fixed r ,

$$\lim_{h \rightarrow \infty} \frac{30}{(h/r)^2 - 1} = 0$$

so by making h large enough, the critical point can be made arbitrarily close to 0. Also, for fixed h ,

$$\lim_{r \rightarrow 0} \frac{30}{(h/r)^2 - 1} = 0$$

so by making r small enough, the critical point can be made arbitrarily close to 0.

21. In the article “Do Dogs Know Calculus?” the author Timothy Pennings explained how he noticed that when he threw a ball diagonally into Lake Michigan along a straight shoreline, his dog Elvis seemed to pick the optimal point in which to enter the water so as to minimize his time to reach the ball, as in Figure 14. He timed the dog and found Elvis could run at 6.4 m/s on the sand and swim at 0.91 m/s. If Tim stood at point A and threw the ball to a point B in the water, which was a perpendicular distance 10 m from point C on the shore, where C is a distance 15 m from where he stood, at what distance x from point C did Elvis enter the water if the dog effectively minimized his time to reach the ball?

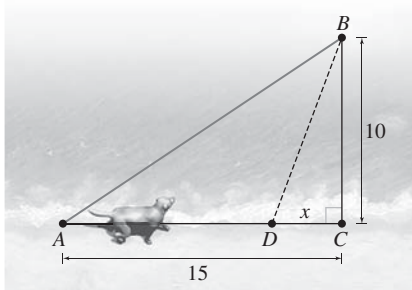


FIGURE 14

SOLUTION Based on the diagram in Figure 14, it follows that Elvis runs a distance of $15 - x$ m along the shoreline and swims a distance of $\sqrt{x^2 + 100}$ m. The time it takes for Elvis to reach point B is then

$$T = \frac{15 - x}{6.4} + \frac{\sqrt{x^2 + 100}}{0.91}$$

The domain of this function is the closed interval $[0, 15]$.

$$T' = -\frac{1}{6.4} + \frac{x}{0.91\sqrt{x^2 + 100}} = 0$$

when

$$\frac{0.91}{6.4} = \frac{x}{\sqrt{x^2 + 100}} \quad \text{or} \quad x = \frac{9.1}{\sqrt{6.4^2 + 0.91^2}} \approx 1.44$$

Evaluating T at the endpoints of the domain and at the critical point $x \approx 1.44$ yields

$$T(0) \approx 13.33, \quad T(1.44) \approx 13.22, \quad \text{and} \quad T(15) \approx 19.81$$

Therefore, Elvis entered the water a distance of 1.44 m from point C .

23. Find the point on the line $y = x$ closest to the point $(1, 0)$. *Hint:* It is equivalent and easier to minimize the *square* of the distance.

SOLUTION With $y = x$, let's equivalently minimize the square of the distance, $f(x) = (x - 1)^2 + y^2 = 2x^2 - 2x + 1$, which is computationally easier (when working by hand). Solve $f'(x) = 4x - 2 = 0$ to obtain $x = \frac{1}{2}$. Since $f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$, $(\frac{1}{2}, \frac{1}{2})$ is the point on $y = x$ closest to $(1, 0)$.

25. CAS Find a good numerical approximation to the coordinates of the point on the graph of $y = \ln x - x$ closest to the origin (Figure 16).

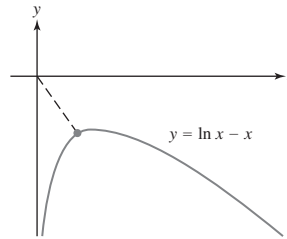


FIGURE 16

SOLUTION The distance from the origin to the point $(x, \ln x - x)$ on the graph of $y = \ln x - x$ is $d = \sqrt{x^2 + (\ln x - x)^2}$. As usual, we will minimize d^2 . Let $d^2 = f(x) = x^2 + (\ln x - x)^2$. Then

$$f'(x) = 2x + 2(\ln x - x) \cdot \frac{1}{x} - 1$$

To determine x , we need to solve

$$4x + \frac{2 \ln x}{x} - 2 \ln x - 2 = 0$$

This yields $x \approx 0.632784$. Thus, the point on the graph of $y = \ln x - x$ that is closest to the origin is approximately $(0.632784, -1.090410)$.

27. Find the angle θ that maximizes the area of the isosceles triangle whose legs have length ℓ (Figure 17), using the fact the area is given by $A = \frac{1}{2}\ell^2 \sin \theta$.

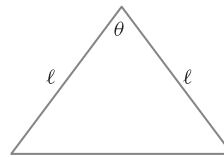


FIGURE 17

SOLUTION The area of the triangle is

$$A(\theta) = \frac{1}{2}\ell^2 \sin \theta$$

where $0 \leq \theta \leq \pi$. Setting

$$A'(\theta) = \frac{1}{2}\ell^2 \cos \theta = 0$$

yields $\theta = \frac{\pi}{2}$. Since $A(0) = A(\pi) = 0$ and $A(\frac{\pi}{2}) = \frac{1}{2}\ell^2$, the angle that maximizes the area of the isosceles triangle is $\theta = \frac{\pi}{2}$.

29. Find the area of the largest isosceles triangle that can be inscribed in a circle of radius 1 (Figure 19).

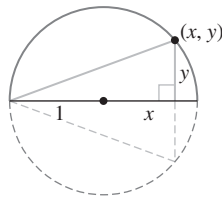


FIGURE 19

SOLUTION The area of the isosceles triangle is given by $A = (x + 1)y$ where $x^2 + y^2 = 1$. Hence, we wish to maximize $A = x\sqrt{1 - x^2}$ for $0 \leq x \leq 1$. Then $A'(x) = \sqrt{1 - x^2} + (1 + x)\frac{1}{2}(1 - x^2)^{-1/2}(-2x) = \frac{1 - 2x^2 - x}{\sqrt{1 - x^2}} = 0$. Thus, $(1 - 2x)(1 + x) = 0$, and $x = 1/2$. $A(1/2) = \frac{3\sqrt{3}}{4}$, whereas $A(0) = 1$ and $A(1) = 0$, so the maximum area is $\frac{3\sqrt{3}}{4}$.

31. A poster of area 6000 cm^2 has blank margins of width 10 cm on the top and bottom and 6 cm on the sides. Find the dimensions that maximize the printed area.

SOLUTION Let x be the width of the printed region, and let y be the height. The total printed area is $A = xy$. Because the total area of the poster is 6000 cm^2 , we have the constraint $(x + 12)(y + 20) = 6000$, so that $xy + 12y + 20x + 240 = 6000$, or $y = \frac{5760 - 20x}{x + 12}$. Therefore, $A(x) = 20\frac{288x - x^2}{x + 12}$, where $0 \leq x \leq 288$.

$A(0) = A(288) = 0$, so we are looking for a critical point on the interval $[0, 288]$. Setting $A'(x) = 0$ yields

$$\begin{aligned} 20 \frac{(x + 12)(288 - 2x) - (288x - x^2)}{(x + 12)^2} &= 0 \\ \frac{-x^2 - 24x + 3456}{(x + 12)^2} &= 0 \\ x^2 + 24x - 3456 &= 0 \\ (x - 48)(x + 72) &= 0 \end{aligned}$$

Therefore $x = 48$ or $x = -72$. $x = 48$ is the only critical point of $A(x)$ in the interval $[0, 288]$, so $A(48) = 3840$ is the maximum value of $A(x)$ in the interval $[0, 288]$. Now, $y = 20\frac{288 - 48}{48 + 12} = 80$ cm, so the poster with maximum printed area is $48 + 12 = 60$ cm. wide by $80 + 20 = 100$ cm. tall.

33. Kepler's Wine Barrel Problem In his work *Nova stereometria doliorum vinariorum* (New Solid Geometry of a Wine Barrel), published in 1615, astronomer Johannes Kepler stated and solved the following problem: Find the dimensions of the cylinder of largest volume that can be inscribed in a sphere of radius R . *Hint:* Show that an inscribed cylinder has volume $2\pi x(R^2 - x^2)$, where x is one-half the height of the cylinder.

SOLUTION Place the center of the sphere at the origin in three-dimensional space. Let the cylinder be of radius y and half-height x . The Pythagorean Theorem states, $x^2 + y^2 = R^2$, so that $y^2 = R^2 - x^2$. The volume of the cylinder is $V(x) = \pi y^2 (2x) = 2\pi(R^2 - x^2)x = 2\pi R^2 x - 2\pi x^3$. Allowing for degenerate cylinders, we have $0 \leq x \leq R$. Solve $V'(x) = 2\pi R^2 - 6\pi x^2 = 0$ for $x \geq 0$ to obtain $x = \frac{R}{\sqrt{3}}$. Since $V(0) = V(R) = 0$, the largest volume is $V(\frac{R}{\sqrt{3}}) = \frac{4}{3}\pi \sqrt{3}R^3$ when $x = \frac{R}{\sqrt{3}}$ and $y = \sqrt{\frac{2}{3}}R$.

35. A landscape architect wishes to enclose a rectangular garden of area 1000 m^2 on one side by a brick wall costing \$90/m and on the other three sides by a metal fence costing \$30/m. Which dimensions minimize the total cost?

SOLUTION Let x be the length of the brick wall and y the length of an adjacent side with $x, y > 0$. With $xy = 1000$ or $y = \frac{1000}{x}$, the total cost is

$$C(x) = 90x + 30(x + 2y) = 120x + 60000x^{-1}$$

Solve $C'(x) = 120 - 60000x^{-2} = 0$ for $x > 0$ to obtain $x = 10\sqrt{5}$. Since $C(x) \rightarrow \infty$ as $x \rightarrow 0+$ and as $x \rightarrow \infty$, the minimum cost is $C(10\sqrt{5}) = 2400\sqrt{5} \approx \5366.56 when $x = 10\sqrt{5} \approx 22.36$ m and $y = 20\sqrt{5} \approx 44.72$ m.

37. Find the maximum area of a rectangle inscribed in the region bounded by the graph of $y = \frac{4 - x}{2 + x}$ and the axes (Figure 21).

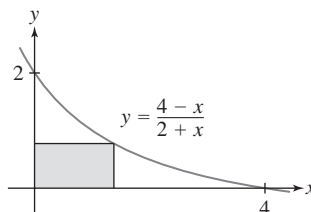


FIGURE 21

SOLUTION Let s be the width of the rectangle. The height of the rectangle is $h = \frac{4-s}{2+s}$, so that the area is

$$A(s) = s \frac{4-s}{2+s} = \frac{4s-s^2}{2+s}$$

We are maximizing on the closed interval $[0, 4]$. It is obvious from the pictures that $A(0) = A(4) = 0$, so we look for critical points of A .

$$A'(s) = \frac{(2+s)(4-2s) - (4s-s^2)}{(2+s)^2} = -\frac{s^2+4s-8}{(s+2)^2}$$

The only point where $A'(s)$ doesn't exist is $s = -2$ which isn't under consideration.

Setting $A'(s) = 0$ gives, by the quadratic formula,

$$s = \frac{-4 \pm \sqrt{48}}{2} = -2 \pm 2\sqrt{3}$$

Of these, only $-2 + 2\sqrt{3}$ is positive, so this is our lone critical point. $A(-2 + 2\sqrt{3}) \approx 1.0718 > 0$. Since we are finding the maximum over a closed interval and $-2 + 2\sqrt{3}$ is the only critical point, the maximum area is $A(-2 + 2\sqrt{3}) \approx 1.0718$.

39. Find the maximum area of a rectangle circumscribed around a rectangle of sides L and H . *Hint:* Express the area in terms of the angle θ (Figure 22).

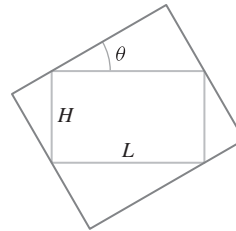


FIGURE 22

SOLUTION Position the $L \times H$ rectangle in the first quadrant of the xy -plane with its “northwest” corner at the origin. Let θ be the angle the base of the circumscribed rectangle makes with the positive x -axis, where $0 \leq \theta \leq \frac{\pi}{2}$. Then the area of the circumscribed rectangle is $A = LH + 2 \cdot \frac{1}{2}(H \sin \theta)(H \cos \theta) + 2 \cdot \frac{1}{2}(L \sin \theta)(L \cos \theta) = LH + \frac{1}{2}(L^2 + H^2) \sin 2\theta$, which has a maximum value of $LH + \frac{1}{2}(L^2 + H^2)$ when $\theta = \frac{\pi}{4}$ because $\sin 2\theta$ achieves its maximum when $\theta = \frac{\pi}{4}$.

41. Find the equation of the line through $P = (4, 12)$ such that the triangle bounded by this line and the axes in the first quadrant has minimal area.

SOLUTION Let $P = (4, 12)$ be a point in the first quadrant and $y - 12 = m(x - 4)$, $-\infty < m < 0$, be a line through P that cuts the positive x - and y -axes. Then $y = L(x) = m(x - 4) + 12$. The line $L(x)$ intersects the y -axis at $H(0, 12 - 4m)$ and the x -axis at $W(4 - \frac{12}{m}, 0)$. Hence the area of the triangle is

$$A(m) = \frac{1}{2}(12 - 4m)\left(4 - \frac{12}{m}\right) = 48 - 8m - 72m^{-1}$$

Solve $A'(m) = 72m^{-2} - 8 = 0$ for $m < 0$ to obtain $m = -3$. Since $A \rightarrow \infty$ as $m \rightarrow -\infty$ or $m \rightarrow 0^-$, we conclude that the minimal triangular area is obtained when $m = -3$. The equation of the line through $P = (4, 12)$ is $y = -3(x - 4) + 12 = -3x + 24$.

43. Archimedes's Problem A spherical cap (Figure 24) of radius r and height h has volume $V = \pi h^2(r - \frac{1}{3}h)$ and surface area $S = 2\pi rh$. Prove that the hemisphere encloses the largest volume among all spherical caps of fixed surface area S .

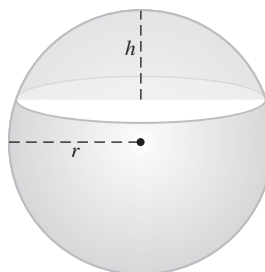


FIGURE 24

SOLUTION Consider all spherical caps of fixed surface area S . Because $S = 2\pi rh$, it follows that

$$r = \frac{S}{2\pi h}$$

and

$$V(h) = \pi h^2 \left(\frac{S}{2\pi h} - \frac{1}{3}h \right) = \frac{S}{2}h - \frac{\pi}{3}h^3$$

Now

$$V'(h) = \frac{S}{2} - \pi h^2 = 0$$

when

$$h^2 = \frac{S}{2\pi} \quad \text{or} \quad h = \frac{S}{2\pi h} = r$$

Hence, the hemisphere encloses the largest volume among all spherical caps of fixed surface area S .

45. A box of volume 72 m^3 with a square bottom and no top is constructed out of two different materials. The cost of the bottom is $\$40/\text{m}^2$ and the cost of the sides is $\$30/\text{m}^2$. Find the dimensions of the box that minimize total cost.

SOLUTION Let s denote the length of the side of the square bottom of the box and h denote the height of the box. Then

$$V = s^2h = 72 \quad \text{or} \quad h = \frac{72}{s^2}$$

The cost of the box is

$$C = 40s^2 + 120sh = 40s^2 + \frac{8640}{s}$$

so

$$C'(s) = 80s - \frac{8640}{s^2} = 0$$

when $s = 3\sqrt[3]{4} \text{ m}$ and $h = 2\sqrt[3]{4} \text{ m}$. Because $C \rightarrow \infty$ as $s \rightarrow 0^-$ and as $s \rightarrow \infty$, we conclude that the critical point gives the minimum cost.

47. Your task is to design a rectangular industrial warehouse consisting of three separate spaces of equal size as in Figure 26. The wall materials cost $\$500$ per linear meter and your company allocates $\$2,400,000$ for that part of the project involving the walls.

- (a) Which dimensions maximize the area of the warehouse?
- (b) What is the area of each compartment in this case?

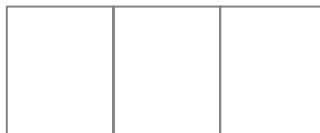


FIGURE 26

SOLUTION Let one compartment have length x and width y . Then total length of the wall of the warehouse is $P = 4x + 6y$ and the constraint equation is cost $= 2,400,000 = 500(4x + 6y)$, which gives $y = 800 - \frac{2}{3}x$.

(a) Area is given by $A = 3xy = 3x(800 - \frac{2}{3}x) = 2400x - 2x^2$, where $0 \leq x \leq 1200$. Then $A'(x) = 2400 - 4x = 0$ yields $x = 600$ and consequently $y = 400$. Since $A(0) = A(1200) = 0$ and $A(600) = 720,000$, the area of the warehouse is maximized when each compartment has length of 600 m and width of 400 m .

(b) The area of one compartment is $600 \cdot 400 = 240,000$ square meters.

49. According to a model developed by economists E. Heady and J. Pesek, if fertilizer made from N pounds of nitrogen and P lb of phosphate is used on an acre of farmland, then the yield of corn (in bushels per acre) is

$$Y = 7.5 + 0.6N + 0.7P - 0.001N^2 - 0.002P^2 + 0.001NP$$

A farmer intends to spend $\$30/\text{acre}$ on fertilizer. If nitrogen costs 25 cents/lb and phosphate costs 20 cents/lb, which combination of N and P produces the highest yield of corn?

SOLUTION The farmer's budget for fertilizer is \$30 per acre, so we have the constraint equation

$$0.25N + 0.2P = 30 \quad \text{or} \quad P = 150 - 1.25N$$

Substituting for P in the equation for Y , we find

$$\begin{aligned} Y(N) &= 7.5 + 0.6N + 0.7(150 - 1.25N) - 0.001N^2 - 0.002(150 - 1.25N)^2 + 0.001N(150 - 1.25N) \\ &= 67.5 + 0.625N - 0.005375N^2 \end{aligned}$$

Both N and P must be nonnegative. Since $P = 150 - 1.25N \geq 0$, we require that $0 \leq N \leq 120$. Next,

$$\frac{dY}{dN} = 0.625 - 0.01075N = 0 \quad \Rightarrow \quad N = \frac{0.625}{0.01075} \approx 58.14 \text{ pounds}$$

Now, $Y(0) = 67.5$, $Y(120) = 65.1$ and $Y(58.14) \approx 85.67$, so the maximum yield of corn occurs for $N \approx 58.14$ pounds and $P \approx 77.33$ pounds.

51. All units in a 100-unit apartment building are rented out when the monthly rent is set at $r = \$900/\text{month}$. Suppose that one unit becomes vacant with each \$10 increase in rent and that each occupied unit costs \$80/month in maintenance. Which rent r maximizes monthly profit?

SOLUTION Let n denote the number of \$10 increases in rent. Then the monthly profit is given by

$$P(n) = (100 - n)(900 + 10n - 80) = 82000 + 180n - 10n^2$$

and

$$P'(n) = 180 - 20n = 0$$

when $n = 9$. We know this results in maximum profit because this gives the location of vertex of a downward opening parabola. Thus, monthly profit is maximized with a rent of \$990.

53. The monthly output of a Spanish light bulb factory is $P = 2LK^2$ (in millions), where L is the cost of labor and K is the cost of equipment (in millions of euros). The company needs to produce 1.7 million units per month. Which values of L and K would minimize the total cost $L + K$?

SOLUTION Since $P = 1.7$ and $P = 2LK^2$, we have $L = \frac{0.85}{K^2}$. Accordingly, the cost of production is

$$C(K) = L + K = K + \frac{0.85}{K^2}$$

Solve $C'(K) = 1 - \frac{1.7}{K^3}$ for $K \geq 0$ to obtain $K = \sqrt[3]{1.7}$. Since $C(K) \rightarrow \infty$ as $K \rightarrow 0+$ and as $K \rightarrow \infty$, the minimum cost of production is achieved for $K = \sqrt[3]{1.7} \approx 1.2$ and $L = 0.6$. The company should invest 1.2 million euros in equipment and 600,000 euros in labor.

55. Brandon is on one side of a river that is 50 m wide and wants to reach a point 200 m downstream on the opposite side as quickly as possible by swimming diagonally across the river and then running the rest of the way. Find the best route if Brandon can swim at 1.5 m/s and run at 4 m/s.

SOLUTION Let lengths be in meters, times in seconds, and speeds in m/s. Suppose that Brandon swims diagonally to a point located x meters downstream on the opposite side. Then Brandon then swims a distance $\sqrt{x^2 + 50^2}$ and runs a distance $200 - x$. The total time of the trip is

$$f(x) = \frac{\sqrt{x^2 + 2500}}{1.5} + \frac{200 - x}{4}, \quad 0 \leq x \leq 200$$

Solve

$$f'(x) = \frac{2x}{3\sqrt{x^2 + 2500}} - \frac{1}{4} = 0$$

to obtain $x = 30\frac{5}{11} \approx 20.2$ and $f(20.2) \approx 80.9$. Since $f(0) \approx 83.3$ and $f(200) \approx 137.4$, we conclude that the minimal time is 80.9 s. This occurs when Brandon swims diagonally to a point located 20.2 m downstream and then runs the rest of the way.

57. Vascular Branching A small blood vessel of radius r branches off at an angle θ from a larger vessel of radius R to supply blood along a path from A to B . According to Poiseuille's Law, the total resistance to blood flow is proportional to

$$T = \left(\frac{a - b \cot \theta}{R^4} + \frac{b \csc \theta}{r^4} \right)$$

where a and b are as in Figure 29. Show that the total resistance is minimized when $\cos \theta = (r/R)^4$.

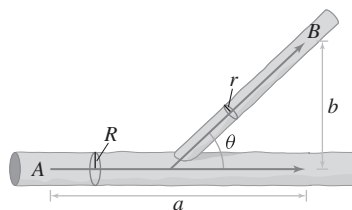


FIGURE 29

SOLUTION With $a, b, r, R > 0$ and $R > r$, let $T(\theta) = \frac{a - b \cot \theta}{R^4} + \frac{b \csc \theta}{r^4}$. Set

$$T'(\theta) = \frac{b \csc^2 \theta}{R^4} - \frac{b \csc \theta \cot \theta}{r^4} = 0$$

Then

$$\frac{b(r^4 - R^4 \cos \theta)}{R^4 r^4 \sin^2 \theta} = 0$$

so that $\cos \theta = \frac{r^4}{R^4}$. Since $\lim_{\theta \rightarrow 0^+} T(\theta) = \infty$ and $\lim_{\theta \rightarrow \pi^-} T(\theta) = \infty$, the minimum value of $T(\theta)$ occurs when $\cos \theta = \frac{r^4}{R^4}$.

In Exercises 58–59, a box (with no top) is to be constructed from a piece of cardboard with sides of length A and B by cutting out squares of length h from the corners and folding up the sides (Figure 30).

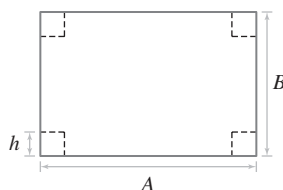


FIGURE 30

59. Which values of A and B maximize the volume of the box if $h = 10$ cm and $AB = 900$ cm²?

SOLUTION With $h = 10$ and $AB = 900$ (which means that $B = 900/A$), the volume of the box is

$$V(A) = 10(A - 20) \left(\frac{900}{A} - 20 \right) = 13000 - 200A - \frac{180000}{A}$$

where $20 \leq A \leq 45$. Now, solving

$$V'(A) = -200 + \frac{180000}{A^2} = 0$$

yields $A = 30$. Because $V(20) = V(45) = 0$ and $V(30) = 1000$ cm³, maximum volume is achieved with $A = B = 30$ cm.

61. Given n numbers $x_1, \dots, x_n$, find the value of x minimizing the sum of the squares:

$$(x - x_1)^2 + (x - x_2)^2 + \cdots + (x - x_n)^2$$

First, solve for $n = 2, 3$ and then try it for arbitrary n .

SOLUTION Note that the sum of squares approaches ∞ as $x \rightarrow \pm\infty$, so the minimum must occur at a critical point.

- For $n = 2$: Let $f(x) = (x - x_1)^2 + (x - x_2)^2$. Then setting $f'(x) = 2(x - x_1) + 2(x - x_2) = 0$ yields $x = \frac{1}{2}(x_1 + x_2)$.
- For $n = 3$: Let $f(x) = (x - x_1)^2 + (x - x_2)^2 + (x - x_3)^2$, so that setting $f'(x) = 2(x - x_1) + 2(x - x_2) + 2(x - x_3) = 0$ yields $x = \frac{1}{3}(x_1 + x_2 + x_3)$.
- Let $f(x) = \sum_{k=1}^n (x - x_k)^2$. Solve $f'(x) = 2 \sum_{k=1}^n (x - x_k) = 0$ to obtain $x = \bar{x} = \frac{1}{n} \sum_{k=1}^n x_k$.

Note that the optimum value for x is the average of $x_1, \dots, x_n$.

63. Solve Exercise 62 again using geometry rather than calculus. There is a unique circle passing through points B and C that is tangent to the street. Let R be the point of tangency. Note that the two angles labeled ψ in Figure 31 are equal because they subtend equal arcs on the circle.

- Show that the maximum value of θ is $\theta = \psi$. *Hint:* Show that $\psi = \theta + \angle PBA$, where A is the intersection of the circle with PC .
- Prove that this agrees with the answer to Exercise 62.
- Show that $\angle QRB = \angle RCQ$ for the maximal angle ψ .

SOLUTION

(a) We note that $\angle PAB$ is supplementary to both ψ and $\theta + \angle PBA$; hence, $\psi = \theta + \angle PBA$. From here, it is clear that θ is at a maximum when $\angle PBA = 0$; that is, when A coincides with P . This occurs when $P = R$.

(b) To show that the two answers agree, let O be the center of the circle. One observes that if d is the distance from R to the wall, then O has coordinates $(-d, \frac{b}{2} + h)$. This is because the height of the center is equidistant from points B and C and because the center must lie directly above R if the circle is tangent to the floor.

Now we can solve for d . The radius of the circle is clearly $\frac{b}{2} + h$, by the distance formula:

$$\overline{OB}^2 = d^2 + \left(\frac{b}{2} + h - h\right)^2 = \left(\frac{b}{2} + h\right)^2$$

This gives

$$d^2 = \left(\frac{b}{2} + h\right)^2 - \left(\frac{b}{2}\right)^2 = bh + h^2$$

or $d = \sqrt{bh + h^2}$ as claimed.

(c) Observe that the arc RB on the dashed circle is subtended by $\angle QRB$ and also by $\angle RCQ$. Thus, both are equal to one-half the angular measure of the arc.

65. Victor Klee's Endpoint Maximum Problem Given 40 m of straight fence, your goal is to build a rectangular enclosure using 80 additional meters of fence that encompasses the greatest area. Let $A(x)$ be the area of the enclosure, with x as in Figure 32.

(a) Find the maximum value of $A(x)$.

(b) Which interval of x values is relevant to our problem? Find the maximum value of $A(x)$ on this interval.

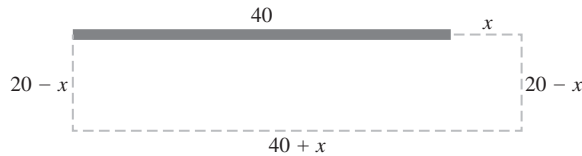


FIGURE 32

SOLUTION

(a) From the diagram, $A(x) = (40 + x)(20 - x) = 800 - 20x - x^2 = 900 - (x + 10)^2$. Thus, the maximum value of $A(x)$ is 900 square meters, occurring when $x = -10$.

(b) For our problem, $x \in [0, 20]$. On this interval, $A(x)$ has no critical points and $A(0) = 800$, while $A(20) = 0$. Thus, on the relevant interval, the maximum enclosed area is 800 square meters.

67. The force F (in Newtons) required to move a box of mass m kg in motion by pulling on an attached rope (Figure 33) is

$$F(\theta) = \frac{fmg}{\cos \theta + f \sin \theta}$$

where θ is the angle between the rope and the horizontal, f is the coefficient of static friction, and $g = 9.8 \text{ m/s}^2$. Find the angle θ that minimizes the required force F , assuming $f = 0.4$. *Hint:* Find the maximum value of $\cos \theta + f \sin \theta$.

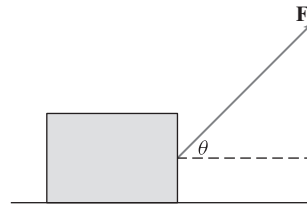


FIGURE 33

SOLUTION Let $F(\theta) = \frac{k}{\sin \theta + \frac{2}{5} \cos \theta}$, where $k > 0$ is a proportionality constant and $0 \leq \theta \leq \frac{\pi}{2}$. Solve

$$F'(\theta) = \frac{k\left(\frac{2}{5} \sin \theta - \cos \theta\right)}{\left(\sin \theta + \frac{2}{5} \cos \theta\right)^2} = 0$$

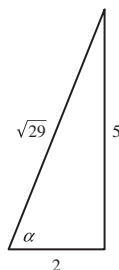
for $0 \leq \theta \leq \frac{\pi}{2}$ to obtain $\tan \theta = \frac{5}{2}$. From the diagram below, we note that when $\tan \theta = \frac{5}{2}$,

$$\sin \theta = \frac{5}{\sqrt{29}} \quad \text{and} \quad \cos \theta = \frac{2}{\sqrt{29}}$$

Therefore, at the critical point the force is

$$\frac{k}{\frac{5}{\sqrt{29}} + \frac{2}{5} \cdot \frac{2}{\sqrt{29}}} = \frac{5\sqrt{29}}{29}k \approx 0.93k$$

Since $F(0) = \frac{5}{2}k = 2.5k$ and $F\left(\frac{\pi}{2}\right) = 1k$, we conclude that the minimum force occurs when $\tan \theta = \frac{5}{2}$.



69. Bird Migration Ornithologists have found that the power (in joules per second) consumed by a certain pigeon flying at velocity v m/s is described well by the function $P(v) = 17v^{-1} + 10^{-3}v^3$ joules/s. Assume that the pigeon can store 5×10^4 joules of usable energy as body fat.

- Show that at velocity v , a pigeon can fly a total distance of $D(v) = (5 \times 10^4)v/P(v)$ if it uses all of its stored energy.
- Find the velocity v_p that *minimizes* P .
- Migrating birds are smart enough to fly at the velocity that maximizes distance traveled rather than minimizes power consumption. Show that the velocity v_d which maximizes $D(v)$ satisfies $P'(v_d) = P(v_d)/v_d$. Show that v_d is obtained graphically as the velocity coordinate of the point where a line through the origin is tangent to the graph of P (Figure 34).
- Find v_d and the maximum distance $D(v_d)$.

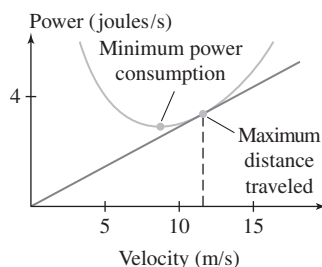


FIGURE 34

SOLUTION

(a) Flying at a velocity v , the birds will exhaust their energy store after $T = \frac{5 \cdot 10^4 \text{ joules}}{P(v) \text{ joules/sec}} = \frac{5 \cdot 10^4 \text{ sec}}{P(v)}$. The total distance traveled is then $D(v) = vT = \frac{5 \cdot 10^4 v}{P(v)}$.

(b) Let $P(v) = 17v^{-1} + 10^{-3}v^3$. Then $P'(v) = -17v^{-2} + 0.003v^2 = 0$ implies $v_p = \left(\frac{17}{0.003}\right)^{1/4} \approx 8.676247$. This critical point is a minimum, because it is the only critical point and $P(v) \rightarrow \infty$ both as $v \rightarrow 0+$ and as $v \rightarrow \infty$.

(c) $D'(v) = \frac{P(v) \cdot 5 \cdot 10^4 - 5 \cdot 10^4 v \cdot P'(v)}{(P(v))^2} = 5 \cdot 10^4 \frac{P(v) - vP'(v)}{(P(v))^2} = 0$ implies $P(v) - vP'(v) = 0$, or $P'(v) = \frac{P(v)}{v}$. Since $D(v) \rightarrow 0$ as $v \rightarrow 0$ and as $v \rightarrow \infty$, the critical point determined by $P'(v) = P(v)/v$ corresponds to a maximum. Graphically, the expression

$$\frac{P(v)}{v} = \frac{P(v) - 0}{v - 0}$$

is the slope of the line passing through the origin and $(v, P(v))$. The condition $P'(v) = P(v)/v$ which defines v_d therefore indicates that v_d is the velocity component of the point where a line through the origin is tangent to the graph of $P(v)$.

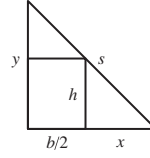
(d) Using $P'(v) = \frac{P(v)}{v}$ gives

$$-17v^{-2} + 0.003v^2 = \frac{17v^{-1} + 0.001v^3}{v} = 17v^{-2} + 0.001v^2$$

which simplifies to $0.002v^4 = 34$ and thus $v_d \approx 11.418583$. The maximum total distance is given by $D(v_d) = \frac{5 \cdot 10^4 \cdot v_d}{P(v_d)} = 191.741$ kilometers.

71. Redo Exercise 70 for arbitrary b and h .

SOLUTION Consider the right triangle formed by the right half of the rectangle and its “roof.” This triangle has hypotenuse s .



As shown, let y be the height of the roof, and let x be the distance from the right base of the rectangle to the base of the roof. By similar triangles applied to the smaller right triangles at the top and right of the larger triangle, we get:

$$\frac{y-h}{b/2} = \frac{h}{x} \quad \text{or} \quad y = \frac{bh}{2x} + h$$

s , y , and x are related by the Pythagorean Theorem:

$$s^2 = \left(\frac{b}{2} + x\right)^2 + y^2 = \left(\frac{b}{2} + x\right)^2 + \left(\frac{bh}{2x} + h\right)^2$$

Since $s > 0$, s^2 is least whenever s is least, so we can maximize s^2 instead of s . Setting the derivative equal to zero yields

$$\begin{aligned} 2\left(\frac{b}{2} + x\right) + 2\left(\frac{bh}{2x} + h\right)\left(-\frac{bh}{2x^2}\right) &= 0 \\ 2\left(\frac{b}{2} + x\right) + 2\frac{h}{x}\left(\frac{b}{2} + x\right)\left(-\frac{bh}{2x^2}\right) &= 0 \\ 2\left(\frac{b}{2} + x\right)\left(1 - \frac{bh^2}{2x^3}\right) &= 0 \end{aligned}$$

The zeros are $x = -\frac{b}{2}$ (irrelevant) and

$$x = \frac{b^{1/3}h^{2/3}}{2^{1/3}}$$

Since this is the only critical point of s with $x > 0$, and since $s \rightarrow \infty$ as $x \rightarrow 0$ and $s \rightarrow \infty$ as $x \rightarrow \infty$, this is the point where s attains its minimum. For this value of x ,

$$\begin{aligned} s^2 &= \left(\frac{b}{2} + \frac{b^{1/3}h^{2/3}}{2^{1/3}}\right)^2 + \left(\frac{b^{2/3}h^{1/3}}{2^{2/3}} + h\right)^2 \\ &= \frac{b^{2/3}}{2^{2/3}}\left(\frac{b^{2/3}}{2^{2/3}} + h^{2/3}\right)^2 + h^{2/3}\left(\frac{b^{2/3}}{2^{2/3}} + h^{2/3}\right)^2 = \left(\frac{b^{2/3}}{2^{2/3}} + h^{2/3}\right)^3 \end{aligned}$$

so the smallest roof length is

$$s = \left(\frac{b^{2/3}}{2^{2/3}} + h^{2/3}\right)^{3/2}$$

73. Redo Exercise 72 for arbitrary widths a and b .

SOLUTION If the corridors have widths a and b , and if θ is the angle between the beam and the line perpendicular to the corridor of width a , then we have to *minimize*

$$f(\theta) = \frac{a}{\cos \theta} + \frac{b}{\sin \theta}$$

Setting the derivative equal to zero,

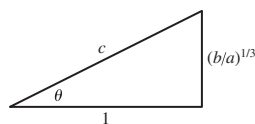
$$a \sec \theta \tan \theta - b \cot \theta \csc \theta = 0$$

we obtain the critical value θ_0 defined by

$$\tan \theta_0 = \left(\frac{b}{a}\right)^{1/3}$$

and from this we conclude (witness the diagram below) that

$$\cos \theta_0 = \frac{1}{\sqrt{1 + (b/a)^{2/3}}} \quad \text{and} \quad \sin \theta_0 = \frac{(b/a)^{1/3}}{\sqrt{1 + (b/a)^{2/3}}}$$



This gives the minimum value as

$$\begin{aligned} f(\theta_0) &= a \sqrt{1 + (b/a)^{2/3}} + b(b/a)^{-1/3} \sqrt{1 + (b/a)^{2/3}} \\ &= a^{2/3} \sqrt{a^{2/3} + b^{2/3}} + b^{2/3} \sqrt{a^{2/3} + b^{2/3}} \\ &= (a^{2/3} + b^{2/3})^{3/2} \end{aligned}$$

75. A basketball player stands d feet from the basket. Let h and α be as in Figure 38. Using physics, one can show that if the player releases the ball at an angle θ , then the initial velocity required to make the ball go through the basket satisfies

$$v^2 = \frac{16d}{\cos^2 \theta (\tan \theta - \tan \alpha)}$$

- (a) Explain why this formula is meaningful only for $\alpha < \theta < \frac{\pi}{2}$. Why does v approach infinity at the endpoints of this interval?
- (b) Take $\alpha = \frac{\pi}{6}$ and plot v^2 as a function of θ for $\frac{\pi}{6} < \theta < \frac{\pi}{2}$. Verify that the minimum occurs at $\theta = \frac{\pi}{3}$.
- (c) Set $F(\theta) = \cos^2 \theta (\tan \theta - \tan \alpha)$. Explain why v is minimized for θ such that $F(\theta)$ is maximized.
- (d) Verify that $F'(\theta) = \cos(\alpha - 2\theta) \sec \alpha$ (you will need to use the addition formula for cosine) and show that the maximum value of F on $[\alpha, \frac{\pi}{2}]$ occurs at $\theta_0 = \frac{\alpha}{2} + \frac{\pi}{4}$.
- (e) For a given α , the optimal angle for shooting the basket is θ_0 because it minimizes v^2 and therefore minimizes the energy required to make the shot (energy is proportional to v^2). Show that the velocity v_{opt} at the optimal angle θ_0 satisfies

$$v_{\text{opt}}^2 = \frac{32d \cos \alpha}{1 - \sin \alpha} = \frac{32d^2}{-h + \sqrt{d^2 + h^2}}$$

- (f) Show with a graph that for fixed d (say, $d = 15$ ft, the distance of a free throw), v_{opt}^2 is an increasing function of h . Use this to explain why taller players have an advantage and why it can help to jump while shooting.

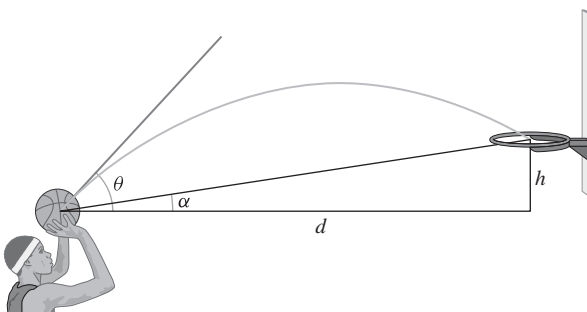


FIGURE 38

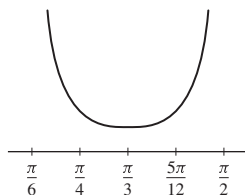
SOLUTION

(a) $\alpha = 0$ corresponds to shooting the ball directly at the basket while $\alpha = \pi/2$ corresponds to shooting the ball directly upward. In neither case is it possible for the ball to go into the basket.

If the angle α is extremely close to 0, the ball is shot almost directly at the basket, so that it must be launched with great speed, as it can only fall an extremely short distance on the way to the basket.

On the other hand, if the angle α is extremely close to $\pi/2$, the ball is launched almost vertically. This requires the ball to travel a great distance upward in order to travel the horizontal distance. In either one of these cases, the ball has to travel at an enormous speed.

(b)



The minimum clearly occurs where $\theta = \pi/3$.

(c) If $F(\theta) = \cos^2 \theta (\tan \theta - \tan \alpha)$,

$$v^2 = \frac{16d}{\cos^2 \theta (\tan \theta - \tan \alpha)} = \frac{16d}{F(\theta)}$$

Since $\alpha \leq \theta$, $F(\theta) > 0$, hence v^2 is smallest whenever $F(\theta)$ is greatest.

(d) $F'(\theta) = -2 \sin \theta \cos \theta (\tan \theta - \tan \alpha) + \cos^2 \theta (\sec^2 \theta) = -2 \sin \theta \cos \theta \tan \theta + 2 \sin \theta \cos \theta \tan \alpha + 1$. We will apply all the double angle formulas:

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta; \sin 2\theta = 2 \sin \theta \cos \theta$$

getting:

$$\begin{aligned} F'(\theta) &= 2 \sin \theta \cos \theta \tan \alpha - 2 \sin \theta \cos \theta \tan \theta + 1 \\ &= 2 \sin \theta \cos \theta \frac{\sin \alpha}{\cos \alpha} - 2 \sin \theta \cos \theta \frac{\sin \theta}{\cos \theta} + 1 \\ &= \sec \alpha (-2 \sin^2 \theta \cos \alpha + 2 \sin \theta \cos \theta \sin \alpha + \cos \alpha) \\ &= \sec \alpha (\cos \alpha (1 - 2 \sin^2 \theta) + \sin \alpha (2 \sin \theta \cos \theta)) \\ &= \sec \alpha (\cos \alpha (\cos 2\theta) + \sin \alpha (\sin 2\theta)) \\ &= \sec \alpha \cos(\alpha - 2\theta) \end{aligned}$$

A critical point of $F(\theta)$ occurs where $\cos(\alpha - 2\theta) = 0$, so that $\alpha - 2\theta = -\frac{\pi}{2}$ (negative because $2\theta > \theta > \alpha$), and this gives us $\theta = \alpha/2 + \pi/4$. The minimum value $F(\theta_0)$ takes place at $\theta_0 = \alpha/2 + \pi/4$.

(e) Plug in $\theta_0 = \alpha/2 + \pi/4$. To find v_{opt}^2 we must simplify

$$\cos^2 \theta_0 (\tan \theta_0 - \tan \alpha) = \frac{\cos \theta_0 (\sin \theta_0 \cos \alpha - \cos \theta_0 \sin \alpha)}{\cos \alpha}$$

By the addition law for sine:

$$\sin \theta_0 \cos \alpha - \cos \theta_0 \sin \alpha = \sin(\theta_0 - \alpha) = \sin(-\alpha/2 + \pi/4)$$

and so

$$\cos \theta_0 (\sin \theta_0 \cos \alpha - \cos \theta_0 \sin \alpha) = \cos(\alpha/2 + \pi/4) \sin(-\alpha/2 + \pi/4)$$

Now use the identity (that follows from the addition law):

$$\sin x \cos y = \frac{1}{2}(\sin(x+y) + \sin(x-y))$$

to get

$$\cos(\alpha/2 + \pi/4) \sin(-\alpha/2 + \pi/4) = (1/2)(1 - \sin \alpha)$$

So we finally get

$$\cos^2 \theta_0 (\tan \theta_0 - \tan \alpha) = \frac{(1/2)(1 - \sin \alpha)}{\cos \alpha}$$

and therefore

$$v_{\text{opt}}^2 = \frac{32d \cos \alpha}{1 - \sin \alpha}$$

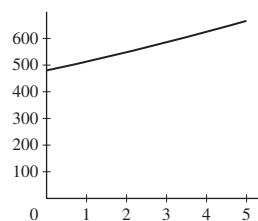
as claimed. From Figure 38 we see that

$$\cos \alpha = \frac{d}{\sqrt{d^2 + h^2}} \quad \text{and} \quad \sin \alpha = \frac{h}{\sqrt{d^2 + h^2}}$$

Substituting these values into the expression for v_{opt}^2 yields

$$v_{\text{opt}}^2 = \frac{32d^2}{-h + \sqrt{d^2 + h^2}}$$

(f) A sketch of the graph of v_{opt}^2 versus h for $d = 15$ feet is given below: v_{opt}^2 increases with respect to basket height relative to the shooter. This shows that the minimum velocity required to launch the ball to the basket drops as shooter height increases. This shows one of the ways height is an advantage in free throws; a taller shooter need not shoot the ball as hard to reach the basket.



Further Insights and Challenges

77. Tom and Ali drive along a highway represented by the graph of f in Figure 40. During the trip, Ali views a billboard represented by the segment $\overline{BC}$ along the y -axis. Let Q be the y -intercept of the tangent line to $y = f(x)$. Show that θ is maximized at the value of x for which the angles $\angle QPB$ and $\angle QCP$ are equal. This generalizes Exercise 63 (c) [which corresponds to the case $f(x) = 0$]. *Hints:*

(a) Show that $d\theta/dx$ is equal to

$$(b - c) \cdot \frac{(x^2 + (xf'(x))^2) - (b - (f(x) - xf'(x)))(c - (f(x) - xf'(x)))}{(x^2 + (b - f(x))^2)(x^2 + (c - f(x))^2)}$$

(b) Show that the y -coordinate of Q is $f(x) - xf'(x)$.

(c) Show that the condition $d\theta/dx = 0$ is equivalent to

$$PQ^2 = BQ \cdot CQ$$

(d) Conclude that $\triangle QPB$ and $\triangle QCP$ are similar triangles.

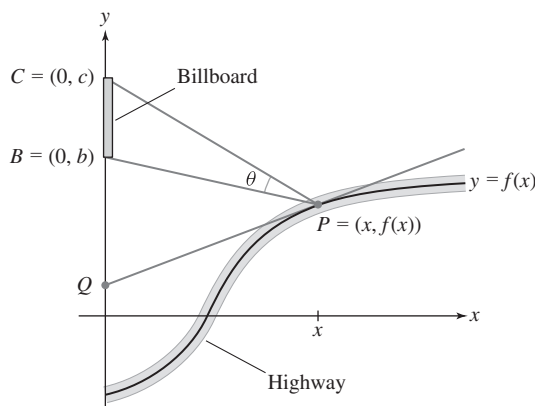


FIGURE 40

SOLUTION

(a) From the figure, we see that

$$\theta(x) = \tan^{-1} \frac{c - f(x)}{x} - \tan^{-1} \frac{b - f(x)}{x}$$

Then

$$\begin{aligned} \theta'(x) &= \frac{b - (f(x) - xf'(x))}{x^2 + (b - f(x))^2} - \frac{c - (f(x) - xf'(x))}{x^2 + (c - f(x))^2} \\ &= (b - c) \frac{x^2 - bc + (b + c)(f(x) - xf'(x)) - (f(x))^2 + 2xf(x)f'(x)}{(x^2 + (b - f(x))^2)(x^2 + (c - f(x))^2)} \\ &= (b - c) \frac{(x^2 + (xf'(x))^2) - (bc - (b + c)(f(x) - xf'(x)) + (f(x) - xf'(x))^2)}{(x^2 + (b - f(x))^2)(x^2 + (c - f(x))^2)} \\ &= (b - c) \frac{(x^2 + (xf'(x))^2) - (b - (f(x) - xf'(x)))(c - (f(x) - xf'(x)))}{(x^2 + (b - f(x))^2)(x^2 + (c - f(x))^2)} \end{aligned}$$

(b) The point Q is the y -intercept of the line tangent to the graph of $f(x)$ at point P . The equation of this tangent line is

$$Y - f(x) = f'(x)(X - x)$$

The y -coordinate of Q is then $f(x) - xf'(x)$.

(c) From the figure, we see that

$$BQ = b - (f(x) - xf'(x)),$$

$$CQ = c - (f(x) - xf'(x))$$

and

$$PQ = \sqrt{x^2 + (f(x) - (f(x) - xf'(x)))^2} = \sqrt{x^2 + (xf'(x))^2}$$

Comparing these expressions with the numerator of $d\theta/dx$, it follows that $\frac{d\theta}{dx} = 0$ is equivalent to

$$PQ^2 = BQ \cdot CQ$$

(d) The equation $PQ^2 = BQ \cdot CQ$ is equivalent to

$$\frac{PQ}{BQ} = \frac{CQ}{PQ}$$

In other words, the sides CQ and PQ from the triangle $\triangle QCP$ are proportional in length to the sides PQ and BQ from the triangle $\triangle QPB$. As $\angle PQB = \angle CQP$, it follows that triangles $\triangle QCP$ and $\triangle QPB$ are similar.

Seismic Prospecting Exercises 78–80 are concerned with determining the thickness d of a layer of soil that lies on top of a rock formation. Geologists send two sound pulses from point A to point D separated by a distance s . The first pulse travels directly from A to D along the surface of the earth. The second pulse travels down to the rock formation, then along its surface, and then back up to D (path $ABCD$), as in Figure 41. The pulse travels with velocity v_1 in the soil and v_2 in the rock.

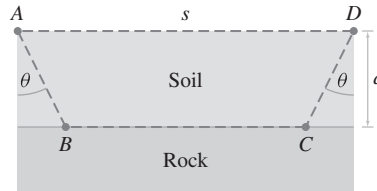


FIGURE 41

79. In this exercise, assume that $v_2/v_1 \geq \sqrt{1 + 4(d/s)^2}$.

(a) Show that inequality (1) holds if $\sin \theta = v_1/v_2$.

(b) Show that the minimal time for the second pulse is

$$t_2 = \frac{2d}{v_1}(1 - k^2)^{1/2} + \frac{s}{v_2}$$

where $k = v_1/v_2$.

(c) Conclude that $\frac{t_2}{t_1} = \frac{2d(1 - k^2)^{1/2}}{s} + k$.

SOLUTION

(a) If $\sin \theta = \frac{v_1}{v_2}$, then

$$\tan \theta = \frac{v_1}{\sqrt{v_2^2 - v_1^2}} = \frac{1}{\sqrt{\left(\frac{v_2}{v_1}\right)^2 - 1}}$$

Because $\frac{v_2}{v_1} \geq \sqrt{1 + 4\left(\frac{d}{s}\right)^2}$, it follows that

$$\sqrt{\left(\frac{v_2}{v_1}\right)^2 - 1} \geq \sqrt{1 + 4\left(\frac{d}{s}\right)^2 - 1} = \frac{2d}{s}$$

Hence, $\tan \theta \leq \frac{s}{2d}$ as required.

(b) For the time-minimizing choice of θ , we have $\sin \theta = \frac{v_1}{v_2}$ from which $\sec \theta = \frac{v_2}{\sqrt{v_2^2 - v_1^2}}$ and $\tan \theta = \frac{v_1}{\sqrt{v_2^2 - v_1^2}}$. Thus

$$\begin{aligned} t_2 &= \frac{2d}{v_1} \sec \theta + \frac{s - 2d \tan \theta}{v_2} = \frac{2d}{v_1} \frac{v_2}{\sqrt{v_2^2 - v_1^2}} + \frac{s - 2d \frac{v_1}{\sqrt{v_2^2 - v_1^2}}}{v_2} \\ &= \frac{2d}{v_1} \left(\frac{v_2}{\sqrt{v_2^2 - v_1^2}} - \frac{v_1^2}{v_2 \sqrt{v_2^2 - v_1^2}} \right) + \frac{s}{v_2} \\ &= \frac{2d}{v_1} \left(\frac{v_2^2 - v_1^2}{v_2 \sqrt{v_2^2 - v_1^2}} \right) + \frac{s}{v_2} = \frac{2d}{v_1} \left(\frac{\sqrt{v_2^2 - v_1^2}}{\sqrt{v_2^2}} \right) + \frac{s}{v_2} \\ &= \frac{2d}{v_1} \sqrt{1 - \left(\frac{v_1}{v_2} \right)^2} + \frac{s}{v_2} = \frac{2d(1 - k^2)^{1/2}}{v_1} + \frac{s}{v_2} \end{aligned}$$

(c) Recall that $t_1 = \frac{s}{v_1}$. We therefore have

$$\begin{aligned} \frac{t_2}{t_1} &= \frac{\frac{2d(1-k^2)^{1/2}}{v_1} + \frac{s}{v_2}}{\frac{s}{v_1}} \\ &= \frac{2d(1-k^2)^{1/2}}{s} + \frac{v_1}{v_2} = \frac{2d(1-k^2)^{1/2}}{s} + k \end{aligned}$$

81.  In this exercise, we use Figure 42 to prove Heron's principle of Example 7 without calculus. By definition, C is the reflection of B across the line $\overline{MN}$ (so that $\overline{BC}$ is perpendicular to $\overline{MN}$ and $BN = CN$). Let P be the intersection of $\overline{AC}$ and $\overline{MN}$. Use geometry to justify the following:

- (a) $\triangle PNB$ and $\triangle PNC$ are congruent and $\theta_1 = \theta_2$.
- (b) The paths APB and APC have equal length.
- (c) Similarly, AQB and AQC have equal length.
- (d) The path APC is shorter than AQC for all $Q \neq P$.

Conclude that the shortest path AQB occurs for $Q = P$.

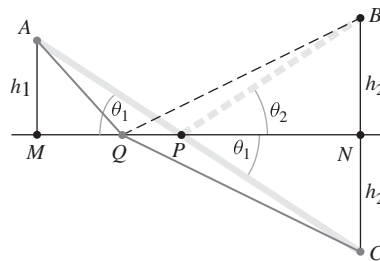


FIGURE 42

SOLUTION

(a) By definition, $\overline{BC}$ is orthogonal to $\overline{MN}$, so triangles $\triangle PNB$ and $\triangle PNC$ are congruent by side-angle-side. Therefore $\theta_1 = \theta_2$.

(b) Because $\triangle PNB$ and $\triangle PNC$ are congruent, it follows that $\overline{PB}$ and $\overline{PC}$ are of equal length. Thus, paths APB and APC have equal length.

(c) The same reasoning used in parts (a) and (b) lead us to conclude that $\triangle QNB$ and $\triangle QNC$ are congruent and that $\overline{QB}$ and $\overline{QC}$ are of equal length. Thus, paths AQB and AQC are of equal length.

(d) Consider triangle $\triangle AQC$. By the triangle inequality, the length of side $\overline{AC}$ is less than or equal to the sum of the lengths of the sides $\overline{AQ}$ and $\overline{QC}$. Thus, the path APC is shorter than AQC for all $Q \neq P$.

Finally, the shortest path AQB occurs for $Q = P$.

4.8 Newton's Method

Preliminary Questions

1. How many iterations of Newton's Method are required to compute a root if f is a linear function?

SOLUTION Remember that Newton's Method uses the linear approximation of a function to estimate the location of a root. If the original function is linear, then only one iteration of Newton's Method will be required to compute the root.

2. What happens in Newton's Method if your initial guess happens to be a zero of f ?

SOLUTION If x_0 happens to be a zero of f , then

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = x_0 - 0 = x_0$$

in other words, every term in the Newton's Method sequence will remain x_0 .

3. What happens in Newton's Method if your initial guess happens to be a local min or max of f ?

SOLUTION Assuming that the function is differentiable, then the derivative is zero at a local maximum or a local minimum. If Newton's Method is started with an initial guess such that $f'(x_0) = 0$, then Newton's Method will fail in the sense that x_1 will not be defined. That is, the tangent line will be parallel to the x -axis and will never intersect it.

4. Is the following a reasonable description of Newton's Method: "A root of the equation of the tangent line to the graph of f is used as an approximation to a root of f itself"? Explain.

SOLUTION Yes, that is a reasonable description. The iteration formula for Newton's Method was derived by solving the equation of the tangent line to $y = f(x)$ at x_0 for its x -intercept.

Exercises

In this exercise set, all approximations should be carried out using Newton's Method.

In Exercises 1–6, apply Newton's Method to f and initial guess x_0 to calculate x_1, x_2, x_3 .

1. $f(x) = x^2 - 6, \quad x_0 = 2$

SOLUTION Let $f(x) = x^2 - 6$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 6}{2x_n}$$

With $x_0 = 2$, we compute

n	1	2	3
x_n	2.5	2.45	2.44948980

3. $f(x) = x^3 - 10, \quad x_0 = 2$

SOLUTION Let $f(x) = x^3 - 10$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 10}{3x_n^2}$$

With $x_0 = 2$ we compute

n	1	2	3
x_n	2.16666667	2.15450362	2.15443469

5. $f(x) = \cos x - 4x, \quad x_0 = 1$

SOLUTION Let $f(x) = \cos x - 4x$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\cos x_n - 4x_n}{-\sin x_n - 4}$$

With $x_0 = 1$ we compute

n	1	2	3
x_n	0.28540361	0.24288009	0.24267469

7. Use Figure 6 to choose an initial guess x_0 to the unique real root of $x^3 + 2x + 5 = 0$ and compute the first three Newton iterates.

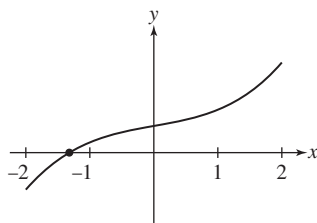


FIGURE 6 Graph of $y = x^3 + 2x + 5$.

SOLUTION Let $f(x) = x^3 + 2x + 5$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 + 2x_n + 5}{3x_n^2 + 2}$$

We take $x_0 = -1.4$, based on the figure, and then calculate

n	1	2	3
x_n	-1.330964467	-1.328272820	-1.328268856

9. Approximate both solutions of $e^x = 5x$ to three decimal places (Figure 7).

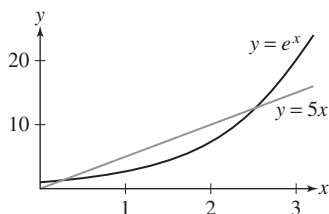


FIGURE 7 Graphs of $y = e^x$ and $y = 5x$.

SOLUTION We need to solve $e^x - 5x = 0$, so let $f(x) = e^x - 5x$. Then $f'(x) = e^x - 5$. With an initial guess of $x_0 = 0.2$, we calculate

Newton's Method (First root) $x_0 = .2$ (guess)	
$x_1 = 0.2 - \frac{f(0.2)}{f'(0.2)}$	$x_1 \approx .25859$
$x_2 = .25859 - \frac{f(.25859)}{f'(.25859)}$	$x_2 \approx .25917$
$x_3 = .25917 - \frac{f(.25917)}{f'(.25917)}$	$x_3 \approx .25917$

For the second root, we use an initial guess of $x_0 = 2.5$.

Newton's Method (Second root) $x_0 = 2.5$ (guess)	
$x_1 = 2.5 - \frac{f(2.5)}{f'(2.5)}$	$x_1 \approx 2.54421$
$x_2 = 2.54421 - \frac{f(2.54421)}{f'(2.54421)}$	$x_2 \approx 2.54264$
$x_3 = 2.54264 - \frac{f(2.54264)}{f'(2.54264)}$	$x_3 \approx 2.54264$

Thus the two solutions of $e^x = 5x$ are approximately $r_1 \approx .25917$ and $r_2 \approx 2.54264$.

In Exercises 11–14, approximate to three decimal places using Newton's Method and compare with the value from a calculator.

11. $\sqrt{11}$

SOLUTION Let $f(x) = x^2 - 11$, and let $x_0 = 3$. Newton's Method yields:

n	1	2	3
x_n	3.33333333	3.31666667	3.31662479

A calculator yields 3.31662479.

13. $2^{7/3}$

SOLUTION Note that $2^{7/3} = 4 \cdot 2^{1/3}$. Let $f(x) = x^3 - 2$, and let $x_0 = 1$. Newton's Method yields:

n	1	2	3
x_n	1.33333333	1.26388889	1.25993349

Thus, $2^{7/3} \approx 4 \cdot 1.25993349 = 5.03973397$. A calculator yields 5.0396842.

15. Approximate the largest positive root of $f(x) = x^4 - 6x^2 + x + 5$ to within an error of at most 10^{-4} . Refer to Figure 5.

SOLUTION Figure 5 from the text suggests the largest positive root of $f(x) = x^4 - 6x^2 + x + 5$ is near 2. So let $f(x) = x^4 - 6x^2 + x + 5$ and take $x_0 = 2$.

n	1	2	3	4
x_n	2.11111111	2.093568458	2.093064768	2.093064358

The largest positive root of $x^4 - 6x^2 + x + 5$ is approximately 2.093064358.

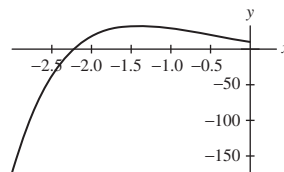
 In Exercises 16–21, approximate the value specified to three decimal places using Newton's Method. Use a plot to choose an initial guess.

17. Negative root of $f(x) = x^5 - 20x + 10$

SOLUTION Let $f(x) = x^5 - 20x + 10$. The graph of $f(x)$ shown below suggests taking $x_0 = -2.2$. Starting from $x_0 = -2.2$, the first three iterates of Newton's Method are:

n	1	2	3
x_n	-2.22536529	-2.22468998	-2.22468949

Thus, to three decimal places, the negative root of $f(x) = x^5 - 20x + 10$ is -2.225.



19. Positive solution of $2 \tan^{-1} x = x$

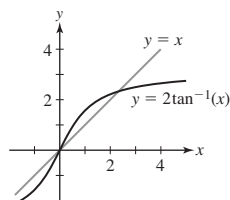
SOLUTION From the graph below, we see that the positive solution to the equation $2 \tan^{-1} x = x$ is approximately $x = 2.3$. Now, let $f(x) = 2 \tan^{-1} x - x$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{2 \tan^{-1} x_n - x_n}{\frac{2}{1+x_n^2} - 1}$$

With $x_0 = 2.3$, we find

n	1	2	3
x_n	2.33128574	2.33112237	2.33112237

Thus, to three decimal places, the positive solution to the equation $2 \tan^{-1} x = x$ is 2.331.

**21.** Solution of $\ln(x+4) = x$

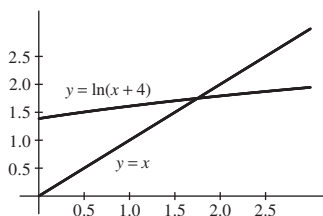
SOLUTION From the graph below, we see that the positive solution to the equation $\ln(x+4) = x$ is approximately $x = 2$. Now, let $f(x) = \ln(x+4) - x$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\ln(x_n+4) - x_n}{\frac{1}{x_n+4} - 1}$$

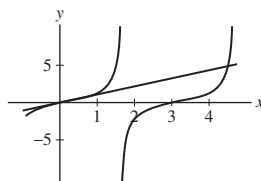
With $x_0 = 2$ we find

n	1	2	3
x_n	1.750111363	1.749031407	1.749031386

Thus, to three decimal places, the positive solution to the equation $\ln(x+4) = x$ is 1.749.

**23.** Find the smallest positive value of x at which $y = x$ and $y = \tan x$ intersect. *Hint:* Draw a plot.

SOLUTION Here is a plot of $\tan x$ and x on the same axes:



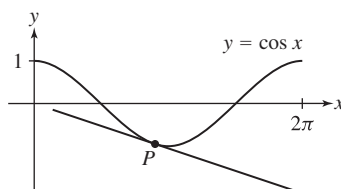
The first intersection with $x > 0$ lies on the second “branch” of $y = \tan x$, between $x = \frac{5\pi}{4}$ and $x = \frac{3\pi}{2}$. Let $f(x) = \tan x - x$. The graph suggests an initial guess $x_0 = \frac{5\pi}{4}$, from which we get the following table:

n	1	2	3	4
x_n	6.85398	21.921	4480.8	7456.27

This is clearly leading nowhere, so we need to try a better initial guess. *Note: This happens with Newton’s Method—it is sometimes difficult to choose an initial guess.* We try the point directly between $\frac{5\pi}{4}$ and $\frac{3\pi}{2}$, $x_0 = \frac{11\pi}{8}$:

n	1	2	3	4	5	6	7
x_n	4.64662	4.60091	4.54662	4.50658	4.49422	4.49341	4.49341

The first point where $y = x$ and $y = \tan x$ cross is at approximately $x = 4.49341$, which is approximately 1.4303π .

25. Find (to two decimal places) the coordinates of the point P in Figure 9 where the tangent line to $y = \cos x$ passes through the origin.**FIGURE 9**

SOLUTION Let $(x_r, \cos(x_r))$ be the coordinates of the point P . The slope of the tangent line is $-\sin(x_r)$, so we are looking for a tangent line:

$$y = -\sin(x_r)(x - x_r) + \cos(x_r)$$

such that $y = 0$ when $x = 0$. This gives us the equation:

$$-\sin(x_r)(-x_r) + \cos(x_r) = 0$$

Let $f(x) = \cos x + x \sin x$. We are looking for the first point $x = r$ where $f(r) = 0$. The sketch given indicates that $x_0 = 3\pi/4$ would be a good initial guess. The following table gives successive Newton's Method approximations:

n	1	2	3	4
x_n	2.931781309	2.803636974	2.798395826	2.798386046

The point P has approximate coordinates $(2.7984, -0.941684)$.

Newton's Method is often used to determine interest rates in financial calculations. In Exercises 26–28, r denotes a yearly interest rate expressed as a decimal (rather than as a percent).

27. If you borrow L dollars for N years at a yearly interest rate r , your monthly payment of P dollars is calculated using the equation

$$L = P \left(\frac{1 - b^{-12N}}{b - 1} \right), \quad \text{where } b = 1 + \frac{r}{12}$$

(a) Find P if $L = \$5000$, $N = 3$, and $r = 0.08$ (8%).

(b) You are offered a loan of $L = \$5000$ to be paid back over 3 years with monthly payments of $P = \$200$. Use Newton's Method to compute b and find the implied interest rate r of this loan. *Hint:* Show that

$$(L/P)b^{12N+1} - (1 + L/P)b^{12N} + 1 = 0$$

SOLUTION

(a) $b = (1 + 0.08/12) = 1.00667$

$$P = L \left(\frac{b - 1}{1 - b^{-12N}} \right) = 5000 \left(\frac{\sqrt[12]{1.00667} - 1}{1 - 1.00667^{-36}} \right) \approx \$156.69$$

(b) Starting from

$$L = P \left(\frac{1 - b^{-12N}}{b - 1} \right)$$

divide by P , multiply by $b - 1$, multiply by b^{12N} and collect like terms to arrive at

$$(L/P)b^{12N+1} - (1 + L/P)b^{12N} + 1 = 0$$

Since $L/P = 5000/200 = 25$, we must solve

$$25b^{37} - 26b^{36} + 1 = 0$$

Newton's Method gives $b \approx 1.02121$ and

$$r = 12(b - 1) = 12(0.02121) \approx 0.25452$$

So the interest rate is around 25.45%.

29. There is no simple formula for the position at time t of a planet P in its orbit (an ellipse) around the sun. Introduce the auxiliary circle and angle θ in Figure 10 (note that P determines θ because it is the central angle of point B on the circle). Let $a = OA$ and $e = OS/OA$ (the eccentricity of the orbit).

(a) Show that sector BSA has area $(a^2/2)(\theta - e \sin \theta)$.

(b) By Kepler's Second Law, the area of sector BSA is proportional to the time t elapsed since the planet passed point A , and because the circle has area πa^2 , BSA has area $(\pi a^2)(t/T)$, where T is the period of the orbit. Deduce **Kepler's Equation**:

$$\frac{2\pi t}{T} = \theta - e \sin \theta$$

(c) The eccentricity of Mercury's orbit is approximately $e = 0.2$. Use Newton's Method to find θ after a quarter of Mercury's year has elapsed ($t = T/4$). Convert θ to degrees. Has Mercury covered more than a quarter of its orbit at $t = T/4$?

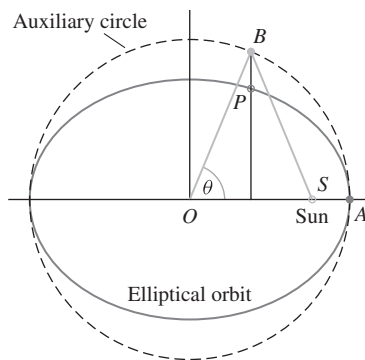


FIGURE 10

SOLUTION

(a) The sector SAB is the slice OAB with the triangle OPS removed. OAB is a central sector with arc θ and radius $\overline{OA} = a$, and therefore has area $\frac{a^2\theta}{2}$. OPS is a triangle with height $a \sin \theta$ and base length $\overline{OS} = ea$. Hence, the area of the sector is

$$\frac{a^2}{2}\theta - \frac{1}{2}ea^2 \sin \theta = \frac{a^2}{2}(\theta - e \sin \theta)$$

(b) Since Kepler's second law indicates that the area of the sector is proportional to the time t since the planet passed point A , we get

$$\pi a^2 (t/T) = a^2/2 (\theta - e \sin \theta)$$

$$2\pi \frac{t}{T} = \theta - e \sin \theta$$

(c) If $t = T/4$, the last equation in (b) gives:

$$\frac{\pi}{2} = \theta - e \sin \theta = \theta - .2 \sin \theta$$

Let $f(\theta) = \theta - .2 \sin \theta - \frac{\pi}{2}$. We will use Newton's Method to find the point where $f(\theta) = 0$. Since a quarter of the year on Mercury has passed, a good first estimate θ_0 would be $\frac{\pi}{2}$.

n	1	2	3	4
x_n	1.7708	1.76696	1.76696	1.76696

From the point of view of the Sun, Mercury has traversed an angle of approximately 1.76696 radians $\approx 101.24^\circ$. Mercury has therefore traveled more than one-fourth of the way around (from the point of view of central angle) during this time.

31. Let $f(x) = xe^{-x}$.

(a) Show that a sequence obtained by applying Newton's Method to f satisfies $x_{n+1} = \frac{x_n^2}{x_n - 1}$.

(b) $\square \nabla \subseteq$ Compute $x_1, x_2, \dots, x_{10}$ separately with $x_0 = 0.8$ and $x_0 = 5$. Discuss what appears to be happening in each case. (Note: the only root of f is at $x = 0$.)

SOLUTION

(a) Let $f(x) = xe^{-x}$. Then $f'(x) = e^{-x} - xe^{-x}$, and the sequence obtained by applying Newton's Method to f is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n e^{-x_n}}{e^{-x_n} - x_n e^{-x_n}} = \frac{x_n e^{-x_n} - x_n^2 e^{-x_n} - x_n e^{-x_n}}{e^{-x_n} - x_n e^{-x_n}} = \frac{x_n^2 e^{-x_n}}{x_n e^{-x_n} - e^{-x_n}} = \frac{x_n^2}{x_n - 1}$$

(b) Taking $x_0 = 0.8$, we have

n	1	2	3	4	5	6
x_n	-3.200000	-2.438095	-1.728954	-1.095395	-0.572632	-0.208509

n	7	8	9	10
x_n	-0.035975	-0.001249	-1.5587×10^{-6}	-2.4294×10^{-12}

This sequence appears to be converging toward the root at zero.

On the other hand, taking $x_0 = 5$, we have

n	1	2	3	4	5	6
x_n	6.250000	7.440476	8.595744	9.727397	10.841979	11.943584

n	7	8	9	10
x_n	13.034962	14.118053	15.194284	16.264735

This sequence appears to be diverging.

33. What happens when you apply Newton's Method to find a zero of $f(x) = x^{1/3}$? Note that $x = 0$ is the only zero.

SOLUTION Let $f(x) = x^{1/3}$. Define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^{1/3}}{\frac{1}{3}x_n^{-2/3}} = x_n - 3x_n = -2x_n$$

Take $x_0 = 0.5$. Then the sequence of iterates is $-1, 2, -4, 8, -16, 32, -64, \dots$. That is, for any nonzero starting value, the sequence of iterates diverges spectacularly, since $x_n = (-2)^n x_0$. Thus $\lim_{n \rightarrow \infty} |x_n| = \lim_{n \rightarrow \infty} 2^n |x_0| = \infty$.

Further Insights and Challenges

35. Newton's Method can be used to compute reciprocals without performing division. Let $c > 0$ and set $f(x) = x^{-1} - c$.

(a) Show that $x - (f(x)/f'(x)) = 2x - cx^2$.

(b) Calculate the first three iterates of Newton's Method with $c = 10.3$ and the two initial guesses $x_0 = 0.1$ and $x_0 = 0.5$.

(c) Explain graphically why $x_0 = 0.5$ does not yield a sequence converging to $1/10.3$.

SOLUTION

(a) Let $f(x) = \frac{1}{x} - c$. Then

$$x - \frac{f(x)}{f'(x)} = x - \frac{\frac{1}{x} - c}{-x^{-2}} = 2x - cx^2$$

(b) For $c = 10.324$, we have $f(x) = \frac{1}{x} - 10.324$ and thus $x_{n+1} = 2x_n - 10.324x_n^2$.

• Take $x_0 = 0.1$.

n	1	2	3
x_n	0.09676	0.096861574	0.096861681

• Take $x_0 = 0.5$.

n	1	2	3
x_n	-1.581	-28.96746776	-8720.949819

(c) The graph is disconnected. If $x_0 = .5$, $(x_1, f(x_1))$ is on the other portion of the graph, which will never converge to any point under Newton's Method.

In Exercises 36 and 37, consider a metal rod of length L fastened at both ends. If you cut the rod and weld on an additional segment of length m , leaving the ends fixed, the rod will bow up into a circular arc of radius R (unknown), as indicated in Figure 12.

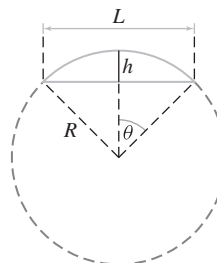


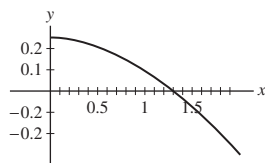
FIGURE 12 The bold circular arc has length $L + m$.

37. Let $L = 3$ and $m = 1$. Apply Newton's Method to Eq. (1) to estimate θ , and use this to estimate h .

SOLUTION We let $L = 3$ and $m = 1$. We want the solution of:

$$\begin{aligned}\frac{\sin \theta}{\theta} &= \frac{L}{L+m} \\ \frac{\sin \theta}{\theta} - \frac{L}{L+m} &= 0 \\ \frac{\sin \theta}{\theta} - \frac{3}{4} &= 0\end{aligned}$$

Let $f(\theta) = \frac{\sin \theta}{\theta} - \frac{3}{4}$.



The figure above suggests that $\theta_0 = 1.5$ would be a good initial guess. The Newton's Method approximations for the solution follow:

n	1	2	3	4
θ_n	1.2854388	1.2757223	1.2756981	1.2756981

The angle where $\frac{\sin \theta}{\theta} = \frac{L}{L+m}$ is approximately 1.2757. Hence

$$h = L \frac{1 - \cos \theta}{2 \sin \theta} \approx 1.11181$$

In Exercises 39–41, a flexible chain of length L is suspended between two poles of equal height separated by a distance $2M$ (Figure 13). By Newton's laws, the chain describes a **catenary**

$$y = a \cosh\left(\frac{x}{a}\right)$$

where a is the number such that

$$L = 2a \sinh\left(\frac{M}{a}\right)$$

The sag s is the vertical distance from the highest to the lowest point on the chain.

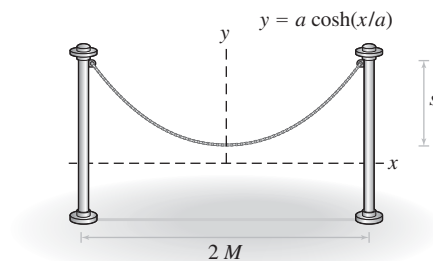


FIGURE 13 Chain hanging between two poles.

39. Suppose that $L = 120$ and $M = 50$.

(a) Use Newton's Method to find a value of a (to two decimal places) satisfying $L = 2a \sinh(M/a)$.

(b) Compute the sag s .

SOLUTION

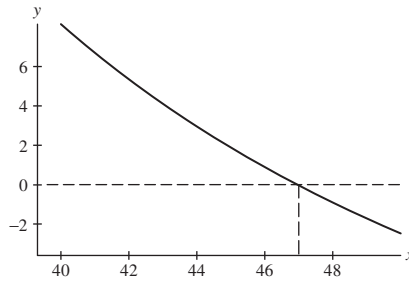
(a) Let

$$f(a) = 2a \sinh\left(\frac{50}{a}\right) - 120$$

The graph of f shown below suggests $a \approx 47$ is a root of f . Starting with $a_0 = 47$, we find the following approximations using Newton's method:

$$a_1 = 46.95408 \quad \text{and} \quad a_2 = 46.95415$$

Thus, to two decimal places, $a = 46.95$.



(b) The sag is given by

$$s = y(M) - y(0) = \left(a \cosh \frac{M}{a} + C \right) - \left(a \cosh \frac{0}{a} + C \right) = a \cosh \frac{M}{a} - a$$

Using $M = 50$ and $a = 46.95$, we find $s = 29.24$.

41. Suppose that $L = 160$ and $M = 50$.

(a) Use Newton's Method to find a value of a (to two decimal places) satisfying $L = 2a \sinh(M/a)$.

(b) Use Eq. (2) and the Linear Approximation to estimate the increase in sag Δs for changes in length $\Delta L = 1$ and $\Delta L = 5$.

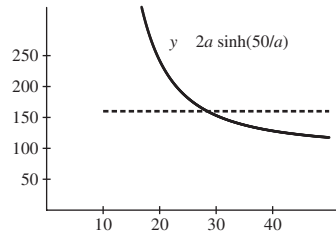
(c) $\square \nabla \square$ Compute $s(161) - s(160)$ and $s(165) - s(160)$ directly and compare with your estimates in (b).

SOLUTION

(a) Let $f(x) = 2x \sinh(50/x) - 160$. Using the graph below, we select an initial guess of $x_0 = 30$. Newton's Method then yields:

n	1	2	3
x_n	28.30622107	28.45653356	28.45797517

Thus, to two decimal places, $a \approx 28.46$



(b) With $M = 50$ and $a \approx 28.46$, we find using equation (2) that

$$\frac{ds}{dL} = 0.61$$

By the Linear Approximation,

$$\Delta s \approx \frac{ds}{dL} \cdot \Delta L$$

If L increases from 160 to 161, then $\Delta L = 1$ foot and $\Delta s \approx 0.61$; if L increases from 160 to 165, then $\Delta L = 5$ and $\Delta s \approx 3.05$.

(c) When $L = 160$, $a \approx 28.46$ and

$$s(160) = 28.46 \cosh\left(\frac{50}{28.46}\right) - 28.46 \approx 56.45$$

whereas, when $L = 161$, $a \approx 28.25$ and

$$s(161) = 28.25 \cosh\left(\frac{50}{28.25}\right) - 28.25 \approx 57.07$$

Therefore, $s(161) - s(160) = 0.62$, very close to the approximation obtained from the Linear Approximation. Moreover, when $L = 165$, $a \approx 27.49$ and

$$s(165) = 27.49 \cosh\left(\frac{50}{27.49}\right) - 27.49 \approx 59.47$$

thus, $s(165) - s(160) = 3.02$, again very close to the approximation obtained from the Linear Approximation.

CHAPTER REVIEW EXERCISES

In Exercises 1–6, estimate using the Linear Approximation or linearization, and use a calculator to estimate the error.

1. $8.1^{1/3} - 2$

SOLUTION Let $f(x) = x^{1/3}$, $a = 8$ and $\Delta x = 0.1$. Then $f'(x) = \frac{1}{3}x^{-2/3}$, $f'(a) = \frac{1}{12}$ and, by the Linear Approximation,

$$\Delta f = 8.1^{1/3} - 2 \approx f'(a)\Delta x = \frac{1}{12}(0.1) = 0.00833333$$

Using a calculator, $8.1^{1/3} - 2 = 0.00829885$. The error in the Linear Approximation is therefore

$$|0.00829885 - 0.00833333| = 3.445 \times 10^{-5}$$

3. $625^{1/4} - 624^{1/4}$

SOLUTION Let $f(x) = x^{1/4}$, $a = 625$ and $\Delta x = -1$. Then $f'(x) = \frac{1}{4}x^{-3/4}$, $f'(a) = \frac{1}{500}$ and, by the Linear Approximation,

$$\Delta f = 624^{1/4} - 625^{1/4} \approx f'(a)\Delta x = \frac{1}{500}(-1) = -0.002$$

Thus $625^{1/4} - 624^{1/4} \approx 0.002$. Using a calculator,

$$625^{1/4} - 624^{1/4} = 0.00200120$$

The error in the Linear Approximation is therefore

$$|0.00200120 - (0.002)| = 1.201 \times 10^{-6}$$

5. $\frac{1}{1.02}$

SOLUTION Let $f(x) = x^{-1}$ and $a = 1$. Then $f(a) = 1$, $f'(x) = -x^{-2}$ and $f'(a) = -1$. The linearization of $f(x)$ at $a = 1$ is therefore

$$L(x) = f(a) + f'(a)(x - a) = 1 - (x - 1) = 2 - x$$

and $\frac{1}{1.02} \approx L(1.02) = 0.98$. Using a calculator, $\frac{1}{1.02} = 0.980392$, so the error in the Linear Approximation is

$$|0.980392 - 0.98| = 3.922 \times 10^{-4}$$

In Exercises 7–12, find the linearization at the point indicated.

7. $y = \sqrt{x}$, $a = 25$

SOLUTION Let $y = \sqrt{x}$ and $a = 25$. Then $y(a) = 5$, $y' = \frac{1}{2}x^{-1/2}$ and $y'(a) = \frac{1}{10}$. The linearization of y at $a = 25$ is therefore

$$L(x) = y(a) + y'(a)(x - 25) = 5 + \frac{1}{10}(x - 25)$$

9. $A(r) = \frac{4}{3}\pi r^3$, $a = 3$

SOLUTION Let $A(r) = \frac{4}{3}\pi r^3$ and $a = 3$. Then $A(a) = 36\pi$, $A'(r) = 4\pi r^2$ and $A'(a) = 36\pi$. The linearization of $A(r)$ at $a = 3$ is therefore

$$L(r) = A(a) + A'(a)(r - a) = 36\pi + 36\pi(r - 3) = 36\pi(r - 2)$$

11. $P(x) = e^{-x^2/2}$, $a = 1$

SOLUTION Let $P(x) = e^{-x^2/2}$ and $a = 1$. Then $P(a) = e^{-1/2}$, $P'(x) = -xe^{-x^2/2}$, and $P'(a) = -e^{-1/2}$. The linearization of $P(x)$ at $a = 1$ is therefore

$$L(x) = P(a) + P'(a)(x - a) = e^{-1/2} - e^{-1/2}(x - 1) = \frac{1}{\sqrt{e}}(2 - x)$$

In Exercises 13–16, use the Linear Approximation.

13. The position of an object in linear motion at time t is $s(t) = 0.4t^2 + (t + 1)^{-1}$. Estimate the distance traveled over the time interval $[4, 4.2]$.

SOLUTION Let $s(t) = 0.4t^2 + (t + 1)^{-1}$, $a = 4$ and $\Delta t = 0.2$. Then $s'(t) = 0.8t - (t + 1)^{-2}$ and $s'(a) = 3.16$. Using the Linear Approximation, the distance traveled over the time interval $[4, 4.2]$ is approximately

$$\Delta s = s(4.2) - s(4) \approx s'(a)\Delta t = 3.16(0.2) = 0.632$$

15. When a bus pass from Albuquerque to Los Alamos is priced at p dollars, a bus company takes in a monthly revenue of $R(p) = 1.5p - 0.01p^2$ (in thousands of dollars).

(a) Estimate ΔR if the price rises from \$50 to \$53.

(b) If $p = 80$, how will revenue be affected by a small increase in price? Explain using the Linear Approximation.

SOLUTION

(a) If the price is raised from \$50 to \$53, then $\Delta p = 3$ and

$$\Delta R \approx R'(50)\Delta p = (1.5 - 0.02(50))(3) = 1.5$$

We therefore estimate an increase of \$1500 in revenue.

(b) Because $R'(80) = 1.5 - 0.02(80) = -0.1$, the Linear Approximation gives $\Delta R \approx -0.1\Delta p$. A small increase in price would thus result in a decrease in revenue.

17. Use the Intermediate Value Theorem to prove that $\sin x - \cos x = 3x$ has a solution, and use Rolle's Theorem to show that this solution is unique.

SOLUTION Let $f(x) = \sin x - \cos x - 3x$, and observe that each root of this function corresponds to a solution of the equation $\sin x - \cos x = 3x$. Now,

$$f\left(-\frac{\pi}{2}\right) = -1 + \frac{3\pi}{2} > 0 \quad \text{and} \quad f(0) = -1 < 0$$

Because f is continuous on $(-\frac{\pi}{2}, 0)$ and $f(-\frac{\pi}{2})$ and $f(0)$ are of opposite sign, the Intermediate Value Theorem guarantees there exists a $c \in (-\frac{\pi}{2}, 0)$ such that $f(c) = 0$. Thus, the equation $\sin x - \cos x = 3x$ has at least one solution.

Next, suppose that the equation $\sin x - \cos x = 3x$ has two solutions, and therefore f has two roots, say a and b . Because f is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b) = 0$, Rolle's Theorem guarantees there exists $c \in (a, b)$ such that $f'(c) = 0$. However,

$$f'(x) = \cos x + \sin x - 3 \leq -1$$

for all x . We have reached a contradiction. Consequently, f has a unique root, and the equation $\sin x - \cos x = 3x$ has a unique solution.

19. Verify the MVT for $f(x) = \ln x$ on $[1, 4]$.

SOLUTION Let $f(x) = \ln x$. On the interval $[1, 4]$, this function is continuous and differentiable, so the MVT applies. Now, $f'(x) = \frac{1}{x}$, so

$$\frac{1}{c} = f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{\ln 4 - \ln 1}{4 - 1} = \frac{1}{3} \ln 4$$

or

$$c = \frac{3}{\ln 4} \approx 2.164 \in (1, 4)$$

21. Use the MVT to prove that if $f'(x) \leq 2$ for $x > 0$ and $f(0) = 4$, then $f(x) \leq 2x + 4$ for all $x \geq 0$.

SOLUTION Let $x > 0$. Because f is continuous on $[0, x]$ and differentiable on $(0, x)$, the Mean Value Theorem guarantees there exists a $c \in (0, x)$ such that

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \quad \text{or} \quad f(x) = f(0) + xf'(c)$$

Now, we are given that $f(0) = 4$ and that $f'(x) \leq 2$ for $x > 0$. Therefore, for all $x \geq 0$,

$$f(x) \leq 4 + x(2) = 2x + 4$$

In Exercises 23–28, find the critical points and determine whether they are minima, maxima, or neither.

23. $f(x) = x^3 - 4x^2 + 4x$

SOLUTION Let $f(x) = x^3 - 4x^2 + 4x$. Then $f'(x) = 3x^2 - 8x + 4 = (3x - 2)(x - 2)$, so that $x = \frac{2}{3}$ and $x = 2$ are critical points. Next, $f''(x) = 6x - 8$, so $f''(\frac{2}{3}) = -4 < 0$ and $f''(2) = 4 > 0$. Therefore, by the Second Derivative Test, $f(\frac{2}{3})$ is a local maximum while $f(2)$ is a local minimum.

25. $f(x) = x^2(x + 2)^3$

SOLUTION Let $f(x) = x^2(x + 2)^3$. Then

$$f'(x) = 3x^2(x + 2)^2 + 2x(x + 2)^3 = x(x + 2)^2(3x + 2x + 4) = x(x + 2)^2(5x + 4)$$

so that $x = 0$, $x = -2$ and $x = -\frac{4}{5}$ are critical points. The sign of the first derivative on the intervals surrounding the critical points is indicated in the table below. Based on this information, $f(-2)$ is neither a local maximum nor a local minimum, $f(-\frac{4}{5})$ is a local maximum and $f(0)$ is a local minimum.

Interval	$(-\infty, -2)$	$(-2, -\frac{4}{5})$	$(-\frac{4}{5}, 0)$	$(0, \infty)$
Sign of f'	+	+	−	+

27. $g(\theta) = \sin^2 \theta + \theta$

SOLUTION Let $g(\theta) = \sin^2 \theta + \theta$. Then

$$g'(\theta) = 2 \sin \theta \cos \theta + 1 = 2 \sin 2\theta + 1$$

so the critical points are

$$\theta = \frac{3\pi}{4} + n\pi$$

for all integers n . Because $g'(\theta) \geq 0$ for all θ , it follows that $g(\frac{3\pi}{4} + n\pi)$ is neither a local maximum nor a local minimum for all integers n .

In Exercises 29–36, find the extreme values on the interval.

29. $f(x) = x(10 - x)$, $[-1, 3]$

SOLUTION Let $f(x) = x(10 - x) = 10x - x^2$. Then $f'(x) = 10 - 2x$, so that $x = 5$ is the only critical point. As this critical point is not in the interval $[-1, 3]$, we only need to check the value of f at the endpoints to determine the extreme values. Because $f(-1) = -11$ and $f(3) = 21$, the maximum value of $f(x) = x(10 - x)$ on the interval $[-1, 3]$ is 21 while the minimum value is -11 .

31. $g(\theta) = \sin^2 \theta - \cos \theta$, $[0, 2\pi]$

SOLUTION Let $g(\theta) = \sin^2 \theta - \cos \theta$. Then

$$g'(\theta) = 2 \sin \theta \cos \theta + \sin \theta = \sin \theta(2 \cos \theta + 1) = 0$$

when $\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi$. The table below lists the value of g at each of the critical points and the endpoints of the interval $[0, 2\pi]$. Based on this information, the minimum value of g on the interval $[0, 2\pi]$ is -1 and the maximum value is $\frac{5}{4}$.

θ	0	$2\pi/3$	π	$4\pi/3$	2π
$g(\theta)$	-1	$5/4$	1	$5/4$	-1

33. $f(x) = x^{2/3} - 2x^{1/3}$, $[-1, 3]$

SOLUTION Let $f(x) = x^{2/3} - 2x^{1/3}$. Then $f'(x) = \frac{2}{3}x^{-1/3} - \frac{2}{3}x^{-2/3} = \frac{2}{3}x^{-2/3}(x^{1/3} - 1)$, so that the critical points are $x = 0$ and $x = 1$. With $f(-1) = 3$, $f(0) = 0$, $f(1) = -1$ and $f(3) = \sqrt[3]{9} - 2\sqrt[3]{3} \approx -0.804$, it follows that the minimum value of f on the interval $[-1, 3]$ is -1 and the maximum value is 3.

35. $f(x) = x - 12 \ln x$, $[5, 40]$

SOLUTION Let $f(x) = x - 12 \ln x$. Then $f'(x) = 1 - \frac{12}{x}$, whence $x = 12$ is the only critical point. The minimum value of f is then $12 - 12 \ln 12 \approx -17.818880$, and the maximum value is $40 - 12 \ln 40 \approx -4.266553$. Note that $f(5) = 5 - 12 \ln 5 \approx -14.313255$.

37. Find the critical points and extreme values of $f(x) = |x - 1| + |2x - 6|$ in $[0, 8]$.

SOLUTION Let

$$f(x) = |x - 1| + |2x - 6| = \begin{cases} 7 - 3x, & x < 1 \\ 5 - x, & 1 \leq x < 3 \\ 3x - 7, & x \geq 3 \end{cases}$$

The derivative of f is never zero but does not exist at the transition points $x = 1$ and $x = 3$. Thus, the critical points of f are $x = 1$ and $x = 3$. With $f(0) = 7$, $f(1) = 4$, $f(3) = 2$, and $f(8) = 17$, it follows that the minimum value of f on the interval $[0, 8]$ is 2 and the maximum value is 17.

In Exercises 39–44, find the points of inflection.

39. $y = x^3 - 4x^2 + 4x$

SOLUTION Let $y = x^3 - 4x^2 + 4x$. Then $y' = 3x^2 - 8x + 4$ and $y'' = 6x - 8$. Thus, $y'' > 0$ and y is concave up for $x > \frac{4}{3}$, while $y'' < 0$ and y is concave down for $x < \frac{4}{3}$. Hence, there is a point of inflection at $x = \frac{4}{3}$.

41. $y = \frac{x^2}{x^2 + 4}$

SOLUTION Let $y = \frac{x^2}{x^2 + 4} = 1 - \frac{4}{x^2 + 4}$. Then $y' = \frac{8x}{(x^2 + 4)^2}$ and

$$y'' = \frac{(x^2 + 4)^2(8) - 8x(2)(2x)(x^2 + 4)}{(x^2 + 4)^4} = \frac{8(4 - 3x^2)}{(x^2 + 4)^3}$$

Thus, $y'' > 0$ and y is concave up for

$$-\frac{2}{\sqrt{3}} < x < \frac{2}{\sqrt{3}}$$

while $y'' < 0$ and y is concave down for

$$|x| \geq \frac{2}{\sqrt{3}}$$

Hence, there are points of inflection at

$$x = \pm \frac{2}{\sqrt{3}}$$

43. $f(x) = (x^2 - x)e^{-x}$

SOLUTION Let $f(x) = (x^2 - x)e^{-x}$. Then

$$f'(x) = -(x^2 - x)e^{-x} + (2x - 1)e^{-x} = -(x^2 - 3x + 1)e^{-x}$$

and

$$f''(x) = (x^2 - 3x + 1)e^{-x} - (2x - 3)e^{-x} = e^{-x}(x^2 - 5x + 4) = e^{-x}(x - 1)(x - 4)$$

Thus, $f''(x) > 0$ and f is concave up for $x < 1$ and for $x > 4$, while $f''(x) < 0$ and f is concave down for $1 < x < 4$. Hence, there are points of inflection at $x = 1$ and $x = 4$.

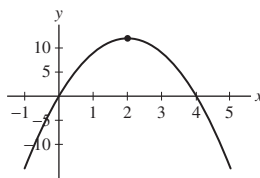
In Exercises 45–54, sketch the graph, noting the transition points and asymptotic behavior.

45. $y = 12x - 3x^2$

SOLUTION Let $y = 12x - 3x^2$. Then $y' = 12 - 6x$ and $y'' = -6$. It follows that the graph of $y = 12x - 3x^2$ is increasing for $x < 2$, decreasing for $x > 2$, has a local maximum at $x = 2$ and is concave down for all x . Because

$$\lim_{x \rightarrow \pm\infty} (12x - 3x^2) = -\infty$$

the graph has no horizontal asymptotes. There are also no vertical asymptotes. The graph is shown below.

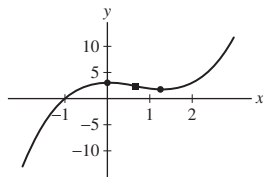


47. $y = x^3 - 2x^2 + 3$

SOLUTION Let $y = x^3 - 2x^2 + 3$. Then $y' = 3x^2 - 4x$ and $y'' = 6x - 4$. It follows that the graph of $y = x^3 - 2x^2 + 3$ is increasing for $x < 0$ and $x > \frac{4}{3}$, is decreasing for $0 < x < \frac{4}{3}$, has a local maximum at $x = 0$, has a local minimum at $x = \frac{4}{3}$, is concave up for $x > \frac{2}{3}$, is concave down for $x < \frac{2}{3}$ and has a point of inflection at $x = \frac{2}{3}$. Because

$$\lim_{x \rightarrow -\infty} (x^3 - 2x^2 + 3) = -\infty \quad \text{and} \quad \lim_{x \rightarrow \infty} (x^3 - 2x^2 + 3) = \infty$$

the graph has no horizontal asymptotes. There are also no vertical asymptotes. The graph is shown below.



49. $y = \frac{x}{x^3 + 1}$

SOLUTION Let $y = \frac{x}{x^3 + 1}$. Then

$$y' = \frac{x^3 + 1 - x(3x^2)}{(x^3 + 1)^2} = \frac{1 - 2x^3}{(x^3 + 1)^2}$$

and

$$y'' = \frac{(x^3 + 1)^2(-6x^2) - (1 - 2x^3)(2)(x^3 + 1)(3x^2)}{(x^3 + 1)^4} = -\frac{6x^2(2 - x^3)}{(x^3 + 1)^3}$$

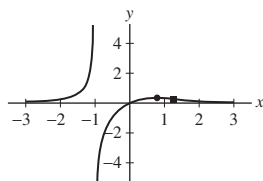
It follows that the graph of $y = \frac{x}{x^3 + 1}$ is increasing for $x < -1$ and $-1 < x < \sqrt[3]{\frac{1}{2}}$, is decreasing for $x > \sqrt[3]{\frac{1}{2}}$, has a local maximum at $x = \sqrt[3]{\frac{1}{2}}$, is concave up for $x < -1$ and $x > \sqrt[3]{2}$, is concave down for $-1 < x < 0$ and $0 < x < \sqrt[3]{2}$ and has a point of inflection at $x = \sqrt[3]{2}$. Note that $x = -1$ is not an inflection point because $x = -1$ is not in the domain of the function. Now,

$$\lim_{x \rightarrow \pm\infty} \frac{x}{x^3 + 1} = 0$$

so $y = 0$ is a horizontal asymptote. Moreover,

$$\lim_{x \rightarrow -1^-} \frac{x}{x^3 + 1} = \infty \quad \text{and} \quad \lim_{x \rightarrow -1^+} \frac{x}{x^3 + 1} = -\infty$$

so $x = -1$ is a vertical asymptote. The graph is shown below.

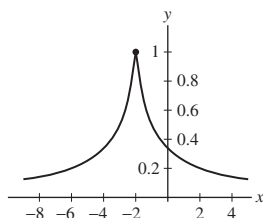


51. $y = \frac{1}{|x + 2| + 1}$

SOLUTION Let $y = \frac{1}{|x + 2| + 1}$. Because

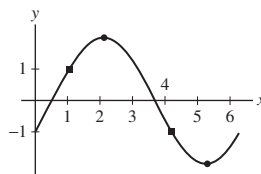
$$\lim_{x \rightarrow \pm\infty} \frac{1}{|x + 2| + 1} = 0$$

the graph of this function has a horizontal asymptote of $y = 0$. The graph has no vertical asymptotes as $|x + 2| + 1 \geq 1$ for all x . The graph is shown below. From this graph we see there is a local maximum at $x = -2$.



53. $y = \sqrt{3} \sin x - \cos x$ on $[0, 2\pi]$

SOLUTION Let $y = \sqrt{3} \sin x - \cos x$. Then $y' = \sqrt{3} \cos x + \sin x$ and $y'' = -\sqrt{3} \sin x + \cos x$. It follows that the graph of $y = \sqrt{3} \sin x - \cos x$ is increasing for $0 < x < 5\pi/6$ and $11\pi/6 < x < 2\pi$, is decreasing for $5\pi/6 < x < 11\pi/6$, has a local maximum at $x = 5\pi/6$, has a local minimum at $x = 11\pi/6$, is concave up for $0 < x < \pi/3$ and $4\pi/3 < x < 2\pi$, is concave down for $\pi/3 < x < 4\pi/3$ and has points of inflection at $x = \pi/3$ and $x = 4\pi/3$. The graph is shown below.



55. Draw a curve $y = f(x)$ for which f' and f'' have signs as indicated in Figure 2.

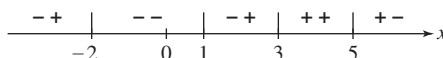
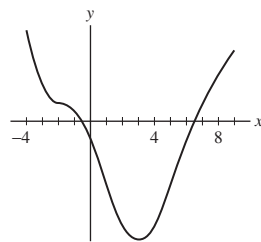


FIGURE 2

SOLUTION The figure below depicts a curve for which f' and f'' have the required signs.



57. A rectangular open-topped box of height h with a square base of side b has volume $V = 4 \text{ m}^3$. Two of the side faces are made of material costing $\$40/\text{m}^2$. The remaining sides cost $\$20/\text{m}^2$. Which values of b and h minimize the cost of the box?

SOLUTION Because the volume of the box is

$$V = b^2 h = 4 \quad \text{it follows that} \quad h = \frac{4}{b^2}$$

Now, the cost of the box is

$$C = 40(2bh) + 20(2bh) + 20b^2 = 120bh + 20b^2 = \frac{480}{b} + 20b^2$$

Thus,

$$C'(b) = -\frac{480}{b^2} + 40b = 0$$

when $b = \sqrt[3]{12}$ meters. Because $C(b) \rightarrow \infty$ as $b \rightarrow 0+$ and as $b \rightarrow \infty$, it follows that cost is minimized when $b = \sqrt[3]{12}$ meters and $h = \frac{1}{3} \sqrt[3]{12}$ meters.

59. Let $N(t)$ be the size of a tumor (in units of 10^6 cells) at time t (in days). According to the **Gompertz Model**, $dN/dt = N(a - b \ln N)$, where a, b are positive constants. Show that the maximum value of N is $e^{a/b}$ and that the tumor increases most rapidly when $N = e^{a/b-1}$.

SOLUTION Given $dN/dt = N(a - b \ln N)$, the critical points of N occur when $N = 0$ and when $N = e^{a/b}$. The sign of $N'(t)$ changes from positive to negative at $N = e^{a/b}$ so the maximum value of N is $e^{a/b}$. To determine when N changes most rapidly, we calculate

$$N''(t) = N \left(-\frac{b}{N} \right) + a - b \ln N = (a - b) - b \ln N$$

Thus, $N'(t)$ is increasing for $N < e^{a/b-1}$, is decreasing for $N > e^{a/b-1}$ and is therefore maximum when $N = e^{a/b-1}$. Therefore, the tumor increases most rapidly when $N = e^{a/b-1}$.

61. Find the maximum volume of a right-circular cone placed upside-down in a right-circular cone of radius $R = 3$ and height $H = 4$ as in Figure 3. A cone of radius r and height h has volume $\frac{1}{3}\pi r^2 h$.

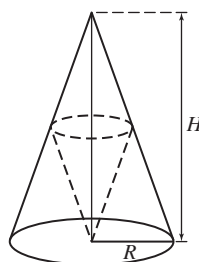


FIGURE 3

SOLUTION Let r denote the radius and h the height of the upside down cone. By similar triangles, we obtain the relation

$$\frac{4-h}{r} = \frac{4}{3} \quad \text{so} \quad h = 4\left(1 - \frac{r}{3}\right)$$

and the volume of the upside down cone is

$$V(r) = \frac{1}{3}\pi r^2 h = \frac{4}{3}\pi \left(r^2 - \frac{r^3}{3}\right)$$

for $0 \leq r \leq 3$. Thus,

$$\frac{dV}{dr} = \frac{4}{3}\pi(2r - r^2)$$

and the critical points are $r = 0$ and $r = 2$. Because $V(0) = V(3) = 0$ and

$$V(2) = \frac{4}{3}\pi\left(4 - \frac{8}{3}\right) = \frac{16}{9}\pi$$

the maximum volume of a right-circular cone placed upside down in a right-circular cone of radius 3 and height 4 is

$$\frac{16}{9}\pi$$

63. Show that the maximum area of a parallelogram $ADEF$ that is inscribed in a triangle ABC , as in Figure 4, is equal to one-half the area of $\triangle ABC$.

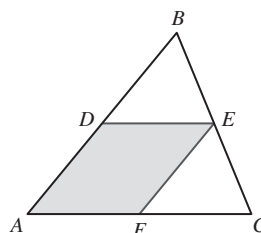


FIGURE 4

SOLUTION Let θ denote the measure of angle BAC . Then the area of the parallelogram is given by $\overline{AD} \cdot \overline{AF} \sin \theta$. Now, suppose that

$$\overline{BE}/\overline{BC} = x$$

Then, by similar triangles, $\overline{AD} = (1-x)\overline{AB}$, $\overline{AF} = \overline{DE} = x\overline{AC}$, and the area of the parallelogram becomes $\overline{AB} \cdot \overline{AC} x(1-x) \sin \theta$. The function $x(1-x)$ achieves its maximum value of $\frac{1}{4}$ when $x = \frac{1}{2}$. Thus, the maximum area of a parallelogram inscribed in a triangle $\triangle ABC$ is

$$\frac{1}{4}\overline{AB} \cdot \overline{AC} \sin \theta = \frac{1}{2} \left(\frac{1}{2}\overline{AB} \cdot \overline{AC} \sin \theta \right) = \frac{1}{2} (\text{area of } \triangle ABC)$$

65. Let f be a function whose graph does not pass through the x -axis and let $Q = (a, 0)$. Let $P = (x_0, f(x_0))$ be the point on the graph closest to Q (Figure 5). Prove that $\overline{PQ}$ is perpendicular to the tangent line to the graph of x_0 . *Hint:* Find the minimum value of the square of the distance from $(x, f(x))$ to $(a, 0)$.

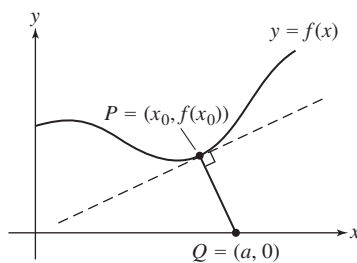


FIGURE 5

SOLUTION Let $P = (a, 0)$ and let $Q = (x_0, f(x_0))$ be the point on the graph of $y = f(x)$ closest to P . The slope of the segment joining P and Q is then

$$\frac{f(x_0)}{x_0 - a}$$

Now, let

$$q(x) = \sqrt{(x - a)^2 + (f(x))^2}$$

the distance from the arbitrary point $(x, f(x))$ on the graph of $y = f(x)$ to the point P . As $(x_0, f(x_0))$ is the point closest to P , we must have

$$q'(x_0) = \frac{2(x_0 - a) + 2f(x_0)f'(x_0)}{\sqrt{(x_0 - a)^2 + (f(x_0))^2}} = 0$$

Thus,

$$f'(x_0) = -\frac{x_0 - a}{f(x_0)} = -\left(\frac{f(x_0)}{x_0 - a}\right)^{-1}$$

In other words, the slope of the segment joining P and Q is the negative reciprocal of the slope of the line tangent to the graph of $y = f(x)$ at $x = x_0$; hence, the two lines are perpendicular.

67. Use Newton's Method to estimate $\sqrt[3]{25}$ to four decimal places.

SOLUTION Let $f(x) = x^3 - 25$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 25}{3x_n^2}$$

With $x_0 = 3$, we find

n	1	2	3
x_n	2.925925926	2.924018982	2.924017738

Thus, to four decimal places $\sqrt[3]{25} = 2.9240$.

69. Find the local extrema of $f(x) = \frac{e^{2x} + 1}{e^{x+1}}$.

SOLUTION To simplify the differentiation, we first rewrite $f(x) = \frac{e^{2x} + 1}{e^{x+1}}$ using the Laws of Exponents:

$$f(x) = \frac{e^{2x}}{e^{x+1}} + \frac{1}{e^{x+1}} = e^{2x-(x+1)} + e^{-(x+1)} = e^{x-1} + e^{-x-1}$$

Now,

$$f'(x) = e^{x-1} - e^{-x-1}$$

Setting the derivative equal to zero yields

$$e^{x-1} - e^{-x-1} = 0 \quad \text{or} \quad e^{x-1} = e^{-x-1}$$

Thus,

$$x - 1 = -x - 1 \quad \text{or} \quad x = 0$$

Next, we use the Second Derivative Test. With $f''(x) = e^{x-1} + e^{-x-1}$, it follows that

$$f''(0) = e^{-1} + e^{-1} = \frac{2}{e} > 0$$

Hence, $x = 0$ is a local minimum. Since $f(0) = e^{-1} + e^{-1} = \frac{2}{e}$, we conclude that the point $(0, \frac{2}{e})$ is a local minimum.

In Exercises 71–74, find the local extrema and points of inflection, and sketch the graph. Use L'Hôpital's Rule to determine the limits as $x \rightarrow 0+$ or $x \rightarrow \pm\infty$ if necessary.

71. $y = x \ln x$ ($x > 0$)

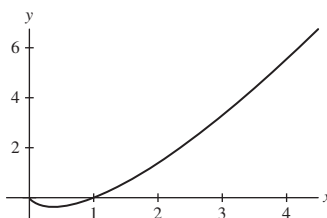
SOLUTION Let $y = x \ln x$. Then

$$y' = \ln x + x \left(\frac{1}{x} \right) = 1 + \ln x$$

and $y'' = \frac{1}{x}$. Solving $y' = 0$ yields the critical point $x = e^{-1}$. Since $y''(e^{-1}) = e > 0$, the function has a local minimum at $x = e^{-1}$. y'' is positive for $x > 0$, hence the function is concave up for $x > 0$ and there are no points of inflection. As $x \rightarrow 0+$ and as $x \rightarrow \infty$, we find

$$\begin{aligned} \lim_{x \rightarrow 0+} x \ln x &= \lim_{x \rightarrow 0+} \frac{\ln x}{x^{-1}} = \lim_{x \rightarrow 0+} \frac{x^{-1}}{-x^{-2}} = \lim_{x \rightarrow 0+} (-x) = 0; \\ \lim_{x \rightarrow \infty} x \ln x &= \infty \end{aligned}$$

The graph is shown below:



73. $y = x(\ln x)^2$ ($x > 0$)

SOLUTION Let $y = x(\ln x)^2$. Then

$$y' = x \frac{2 \ln x}{x} + (\ln x)^2 = 2 \ln x + (\ln x)^2 = \ln x(2 + \ln x)$$

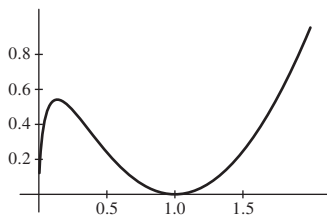
and

$$y'' = \frac{2}{x} + \frac{2 \ln x}{x} = \frac{2}{x}(1 + \ln x)$$

Solving $y' = 0$ yields the critical points $x = e^{-2}$ and $x = 1$. Since $y''(e^{-2}) = -2e^2 < 0$ and $y''(1) = 2 > 0$, the function has a local maximum at $x = e^{-2}$ and a local minimum at $x = 1$. $y'' < 0$ and the function is concave down for $x < e^{-1}$, whereas $y'' > 0$ and the function is concave up for $x > e^{-1}$; hence, there is a point of inflection at $x = e^{-1}$. As $x \rightarrow 0+$ and as $x \rightarrow \infty$, we find

$$\begin{aligned} \lim_{x \rightarrow 0+} x(\ln x)^2 &= \lim_{x \rightarrow 0+} \frac{(\ln x)^2}{x^{-1}} = \lim_{x \rightarrow 0+} \frac{2 \ln x \cdot x^{-1}}{-x^{-2}} = \lim_{x \rightarrow 0+} \frac{2 \ln x}{-x^{-1}} = \lim_{x \rightarrow 0+} \frac{2x^{-1}}{x^{-2}} = \lim_{x \rightarrow 0+} 2x = 0; \\ \lim_{x \rightarrow \infty} x(\ln x)^2 &= \infty \end{aligned}$$

The graph is shown below:



75.  Explain why L'Hôpital's Rule gives no information about $\lim_{x \rightarrow \infty} \frac{2x - \sin x}{3x + \cos 2x}$. Evaluate the limit by another method.

SOLUTION As $x \rightarrow \infty$, both $2x - \sin x$ and $3x + \cos 2x$ tend toward infinity, so L'Hôpital's Rule applies to $\lim_{x \rightarrow \infty} \frac{2x - \sin x}{3x + \cos 2x}$; however, the resulting limit, $\lim_{x \rightarrow \infty} \frac{2 - \cos x}{3 - 2 \sin 2x}$, does not exist due to the oscillation of $\sin x$ and $\cos x$. To evaluate the limit, we note

$$\lim_{x \rightarrow \infty} \frac{2x - \sin x}{3x + \cos 2x} = \lim_{x \rightarrow \infty} \frac{2 - \frac{\sin x}{x}}{3 + \frac{\cos 2x}{x}} = \frac{2}{3}$$

In Exercises 77–88, verify that L'Hôpital's Rule applies and evaluate the limit.

77. $\lim_{x \rightarrow 3} \frac{4x - 12}{x^2 - 5x + 6}$

SOLUTION The given expression is an indeterminate form of type $\frac{0}{0}$; therefore L'Hôpital's Rule applies. We find

$$\lim_{x \rightarrow 3} \frac{4x - 12}{x^2 - 5x + 6} = \lim_{x \rightarrow 3} \frac{4}{2x - 5} = 4$$

79. $\lim_{x \rightarrow 0^+} x^{1/2} \ln x$

SOLUTION First rewrite

$$x^{1/2} \ln x \quad \text{as} \quad \frac{\ln x}{x^{-1/2}}$$

The rewritten expression is an indeterminate form of type $\frac{\infty}{\infty}$; therefore L'Hôpital's Rule applies. We find

$$\lim_{x \rightarrow 0^+} x^{1/2} \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-1/2}} = \lim_{x \rightarrow 0^+} \frac{1/x}{-\frac{1}{2}x^{-3/2}} = \lim_{x \rightarrow 0^+} -\frac{x^{1/2}}{2} = 0$$

81. $\lim_{\theta \rightarrow 0} \frac{2 \sin \theta - \sin 2\theta}{\sin \theta - \theta \cos \theta}$

SOLUTION The given expression is an indeterminate form of type $\frac{0}{0}$; hence, we may apply L'Hôpital's Rule. We find

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{2 \sin \theta - \sin 2\theta}{\sin \theta - \theta \cos \theta} &= \lim_{\theta \rightarrow 0} \frac{2 \cos \theta - 2 \cos 2\theta}{\cos \theta - (\cos \theta - \theta \sin \theta)} = \lim_{\theta \rightarrow 0} \frac{2 \cos \theta - 2 \cos 2\theta}{\theta \sin \theta} \\ &= \lim_{\theta \rightarrow 0} \frac{-2 \sin \theta + 4 \sin 2\theta}{\sin \theta + \theta \cos \theta} = \lim_{\theta \rightarrow 0} \frac{-2 \cos \theta + 8 \cos 2\theta}{\cos \theta + \cos \theta - \theta \sin \theta} = \frac{-2 + 8}{1 + 1 - 0} = 3 \end{aligned}$$

83. $\lim_{t \rightarrow \infty} \frac{\ln(t+2)}{\log_2 t}$

SOLUTION The limit is an indeterminate form of type $\frac{\infty}{\infty}$; hence, we may apply L'Hôpital's Rule. We find

$$\lim_{t \rightarrow \infty} \frac{\ln(t+2)}{\log_2 t} = \lim_{t \rightarrow \infty} \frac{\frac{1}{t+2}}{\frac{1}{t \ln 2}} = \lim_{t \rightarrow \infty} \frac{t \ln 2}{t+2} = \lim_{t \rightarrow \infty} \frac{\ln 2}{1} = \ln 2$$

85. $\lim_{y \rightarrow 0} \frac{\sin^{-1} y - y}{y^3}$

SOLUTION The limit is an indeterminate form of type $\frac{0}{0}$; hence, we may apply L'Hôpital's Rule. We find

$$\lim_{y \rightarrow 0} \frac{\sin^{-1} y - y}{y^3} = \lim_{y \rightarrow 0} \frac{\frac{1}{\sqrt{1-y^2}} - 1}{3y^2} = \lim_{y \rightarrow 0} \frac{y(1-y^2)^{-3/2}}{6y} = \lim_{y \rightarrow 0} \frac{(1-y^2)^{-3/2}}{6} = \frac{1}{6}$$

87. $\lim_{x \rightarrow 0} \frac{\sinh(x^2)}{\cosh x - 1}$

SOLUTION The limit is an indeterminate form of type $\frac{0}{0}$; hence, we may apply L'Hôpital's Rule. We find

$$\lim_{x \rightarrow 0} \frac{\sinh(x^2)}{\cosh x - 1} = \lim_{x \rightarrow 0} \frac{2x \cosh(x^2)}{\sinh x} = \lim_{x \rightarrow 0} \frac{2 \cosh(x^2) + 4x^2 \sinh(x^2)}{\cosh x} = \frac{2 + 0}{1} = 2$$

89. Let $f(x) = e^{-Ax^2/2}$, where $A > 0$ is a constant. Given any n numbers $a_1, a_2, \dots, a_n$, set

$$\Phi(x) = f(x - a_1)f(x - a_2) \cdots f(x - a_n)$$

(a) Assume $n = 2$ and prove that Φ attains its maximum value at the average $x = \frac{1}{2}(a_1 + a_2)$. *Hint:* Calculate $\Phi'(x)$ using logarithmic differentiation.

(b) Show that for any n , Φ attains its maximum value at $x = \frac{1}{n}(a_1 + a_2 + \cdots + a_n)$. This fact is related to the role of $f(x)$ (whose graph is a bell-shaped curve) in statistics.

SOLUTION

(a) For $n = 2$ we have,

$$\Phi(x) = f(x - a_1) f(x - a_2) = e^{-\frac{A}{2}(x-a_1)^2} \cdot e^{-\frac{A}{2}(x-a_2)^2} = e^{-\frac{A}{2}((x-a_1)^2 + (x-a_2)^2)}$$

Since $e^{-\frac{A}{2}y}$ is a decreasing function of y , it attains its maximum value where y is minimum. Therefore, we must find the minimum value of

$$y = (x - a_1)^2 + (x - a_2)^2 = 2x^2 - 2(a_1 + a_2)x + a_1^2 + a_2^2$$

Now, $y' = 4x - 2(a_1 + a_2) = 0$ when

$$x = \frac{a_1 + a_2}{2}$$

We conclude that $\Phi(x)$ attains a maximum value at this point.

(b) We have

$$\Phi(x) = e^{-\frac{A}{2}(x-a_1)^2} \cdot e^{-\frac{A}{2}(x-a_2)^2} \cdots e^{-\frac{A}{2}(x-a_n)^2} = e^{-\frac{A}{2}((x-a_1)^2 + \cdots + (x-a_n)^2)}$$

Since the function $e^{-\frac{A}{2}y}$ is a decreasing function of y , it attains a maximum value where y is minimum. Therefore we must minimize the function

$$y = (x - a_1)^2 + (x - a_2)^2 + \cdots + (x - a_n)^2$$

We find the critical points by solving:

$$y' = 2(x - a_1) + 2(x - a_2) + \cdots + 2(x - a_n) = 0$$

$$2nx = 2(a_1 + a_2 + \cdots + a_n)$$

$$x = \frac{a_1 + \cdots + a_n}{n}$$

We verify that this point corresponds the minimum value of y by examining the sign of y'' at this point: $y'' = 2n > 0$. We conclude that y attains a minimum value at the point $x = \frac{a_1 + \cdots + a_n}{n}$, hence $\Phi(x)$ attains a maximum value at this point.